# EART 22201 Igneous minerals and processes

## An introduction to the rock-forming minerals



Lesson 3


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# An introduction to the rock-forming minerals

**1.1 Cations, anions and co-ordination polyhedra**

1.2 The $SiO_4$ tetrahedron

1.3 Silicate mineral groups and their structures

1.4 Isomorphism and polymorphism

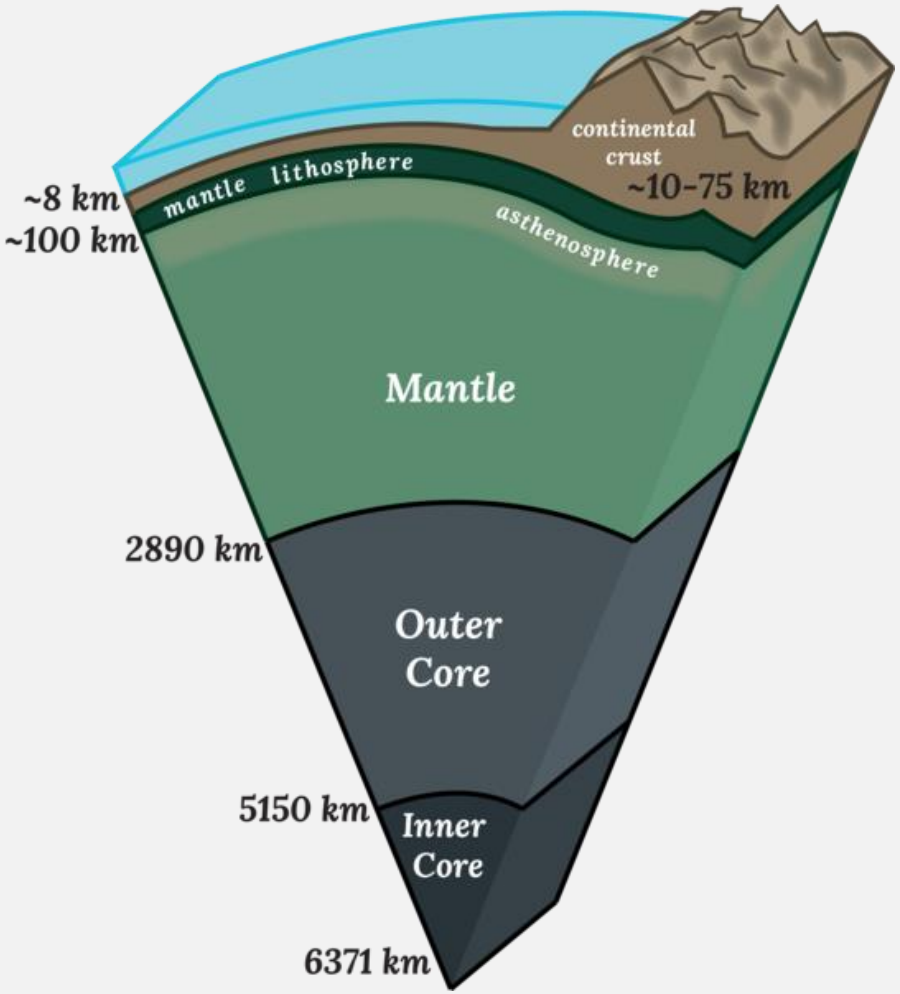




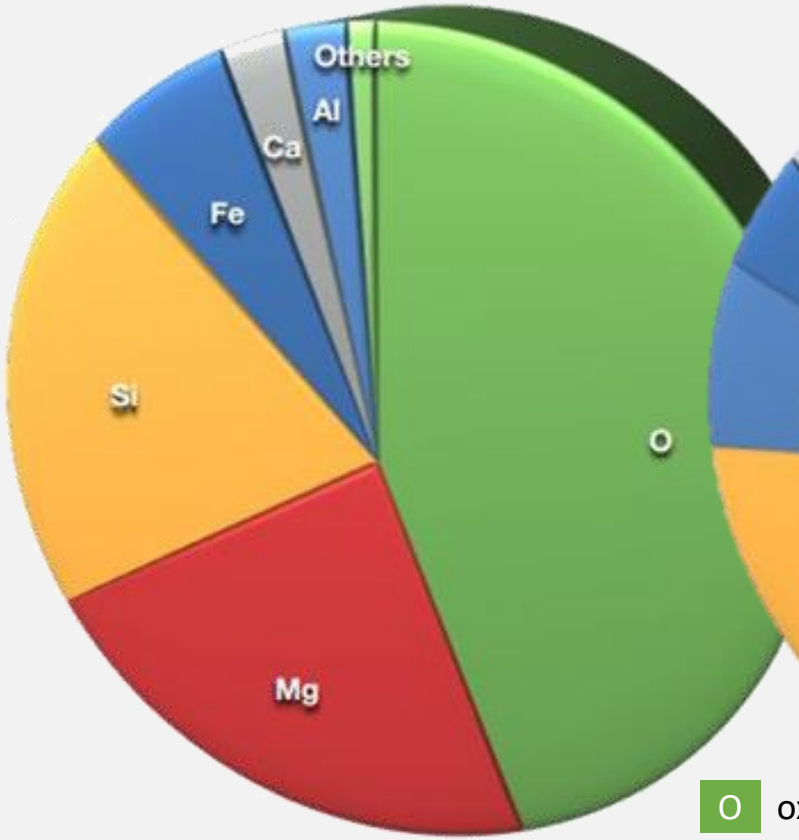
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# Which chemical elements are important in igneous petrology?

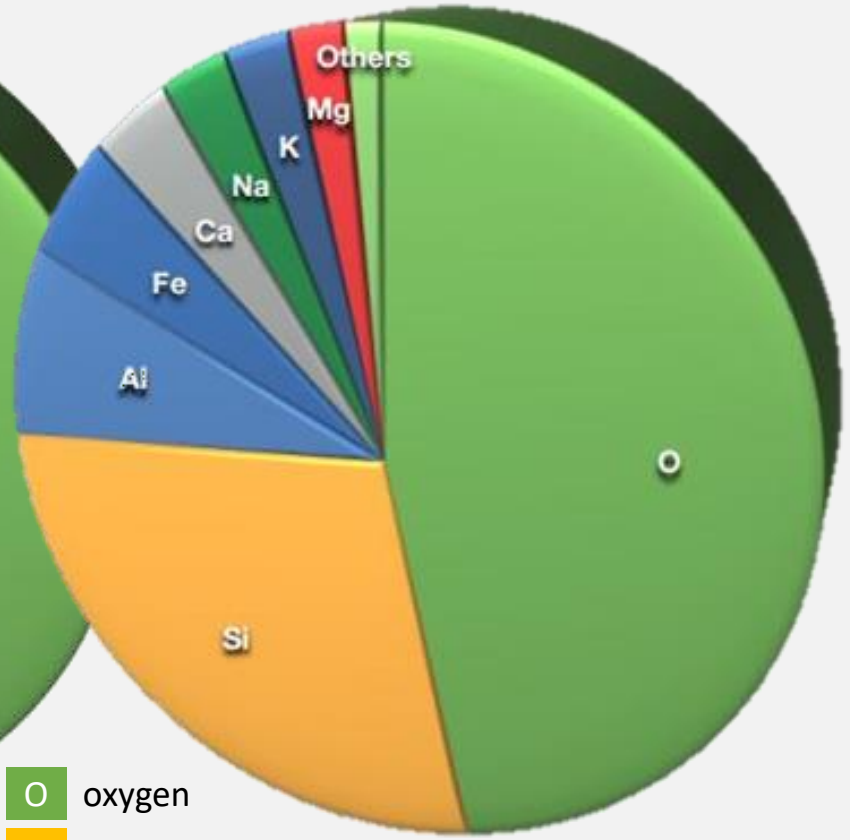
See also EART 26201 Principles of Geochemistry, Lecture 1




Bulk Silicate Earth




Crust




O oxygen
Si silicon
Mg magnesium
Fe iron


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# Cations and anions

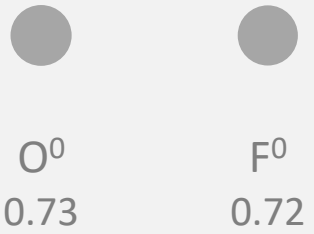
See also EART 26201 Principles of Geochemistry, Lecture 1



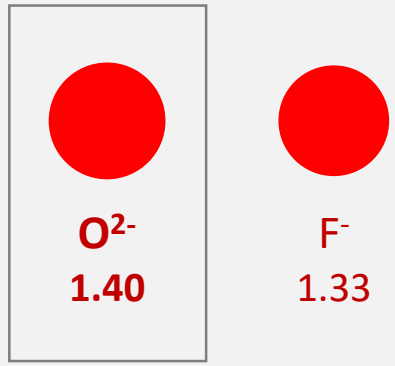

cations




neutral elements




anions



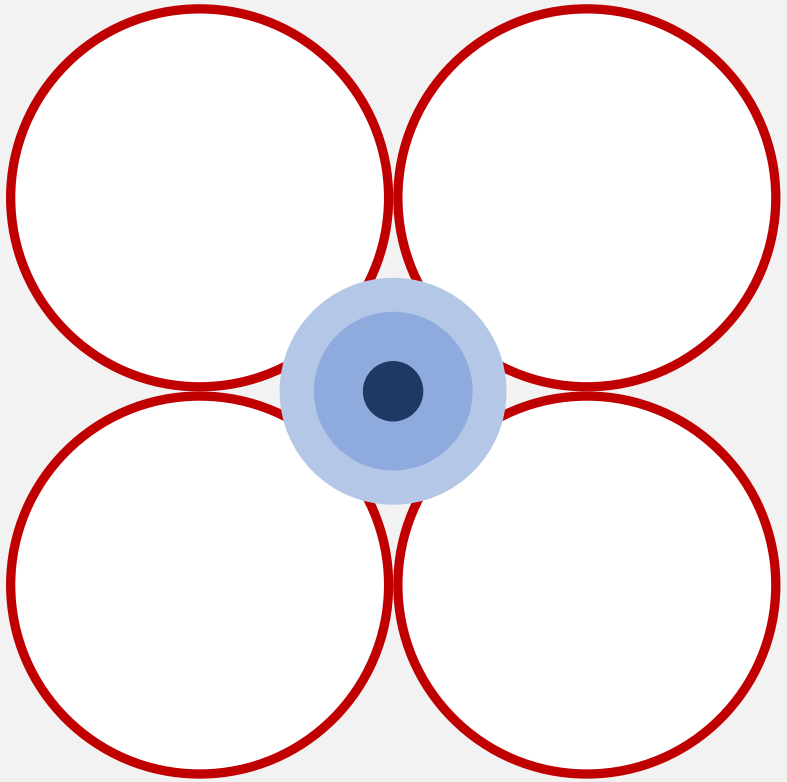

Atomic radii are given in angstroms (Å).

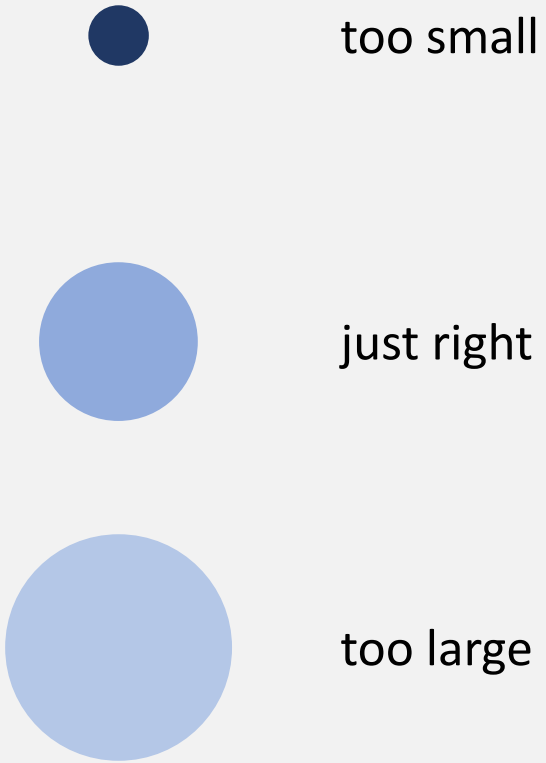


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# Cations and anions

See also EART 26201 Principles of Geochemistry, Lecture 1



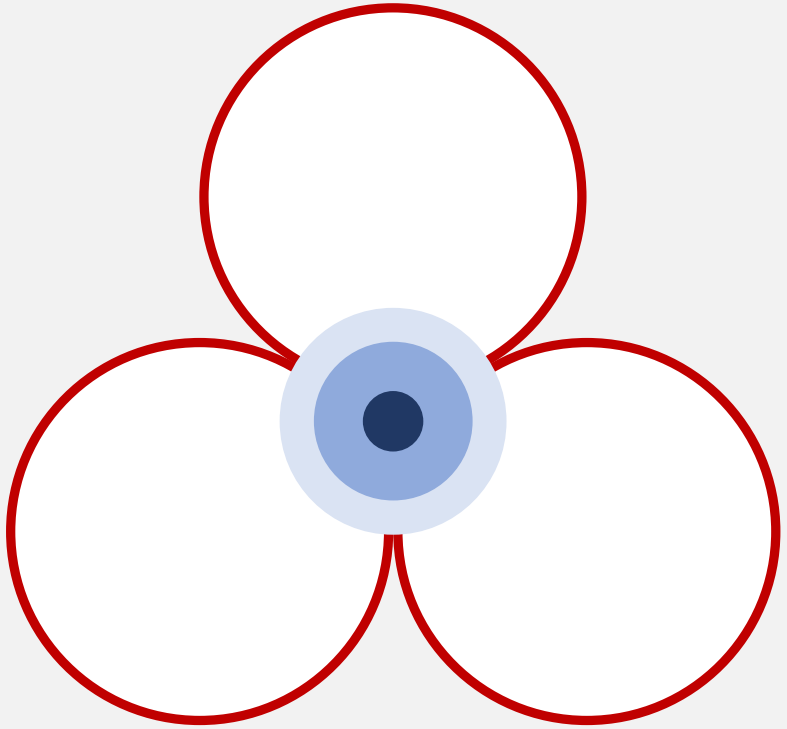





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# Cations and anions

See also EART 26201 Principles of Geochemistry, Lecture 1





just right



too large

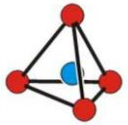
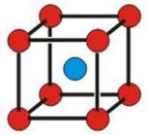
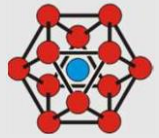


too large



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# Co-ordination polyhedra

| Radius ratio | Cation co-ordination | Co-ordination number | Polyhedron geometry |
|---|---|---|---|
| 0.16 – 0.23 | 3 anions at corners of a triangle | 3 |  |
| 0.23 – 0.41 | 4 anions at corners of a tetrahedron | 4 |  |
| 0.41 – 0.73 | 6 anions at corners of an octahedron | 6 |  |
| 0.73 – 1.0 | 8 anions at corners of a cube | 8 |  |
| > 1.0 | 12 anions at corners of a cuboctahedron | 12 |  |

$$\text{radius ratio} = \frac{\text{cation radius}}{\text{anion radius}}$$



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1.1 Cations, anions and co-ordination polyhedra

**1.2 The $SiO_4$ tetrahedron**

1.3 Silicate mineral groups and their structures

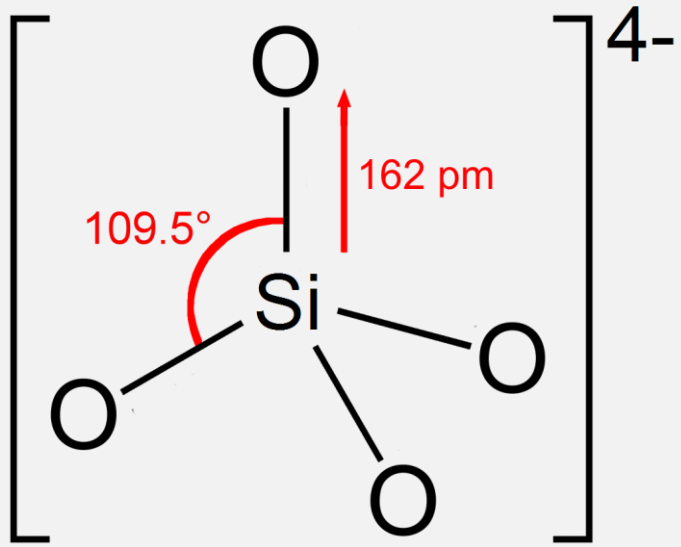
1.4 Isomorphism and polymorphism





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# The $[SiO_4]^{4-}$ tetrahedron

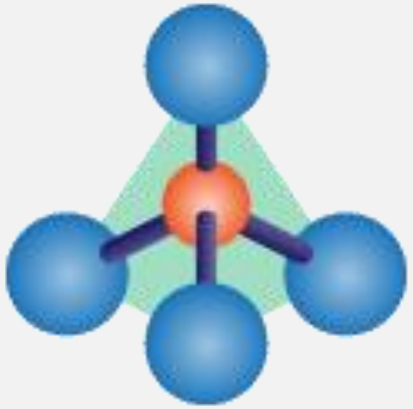


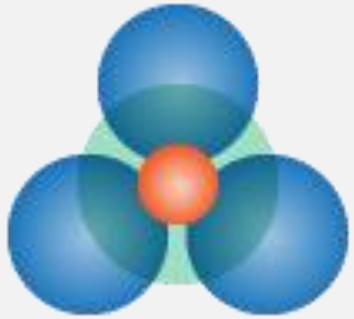

$[SiO_4]^{4-}$

$Si^{4+}$ cation radius in crystal = 0.4 Å

$O^{2-}$ anion radius in crystal = 1.4 Å

radius ratio = 0.3







$[SiO_4]^{4-}$ groups are linked together to make different types of silicate minerals



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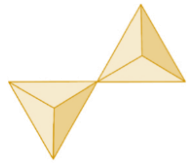
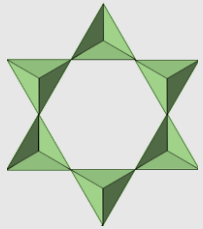
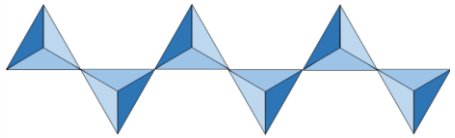
1.4 Isomorphism and polymorphism





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# Silicate mineral groups

| Mineral group | General formula | Shared oxygens per $[SiO_4]$ group | How are the silica tetrahedra arranged? | Example |
|---|---|---|---|---|
| Orthosilicates (island silicates) | $[SiO_4]^{4-}$ | 0 |  | olivine garnet |
| Double-island silicates | $[Si_2O_7]^{6-}$ | 1 |  | epidote zoisite |
| Ring silicates | $[Si_3O_9]^{6-}$ | 2 |  | beryl cordierite |
| Single chain silicates | $[Si_2O_6]^{4-}$ | 2 |  | pyroxenes |

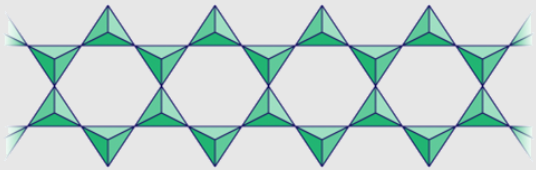
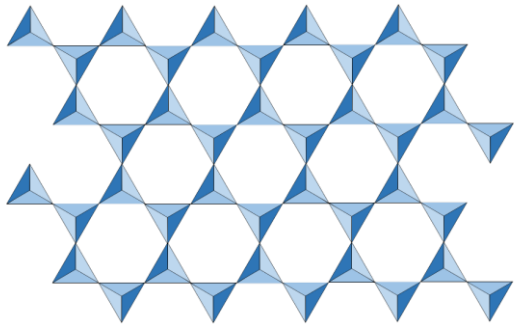
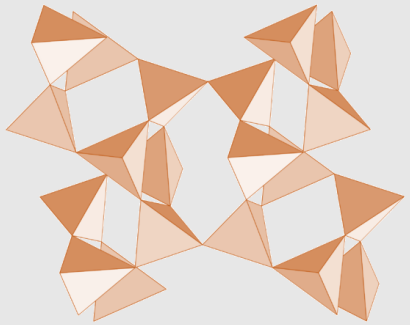
$[SiO_4]^{4-}$ tetrahedra can be polymerized in different configurations.

Silicate minerals are grouped according to how the $[SiO_4]^{4-}$ tetrahedra are polymerized in the crystal structure.



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# Silicate mineral groups

| Mineral group | General formula | Shared oxygens per $[SiO_4]$ group | How are the silica tetrahedra arranged? | Example |
|---|---|---|---|---|
| Double chain silicates | $[Si_4O_{11}]^{6-}$ | 2 and 3 |  | amphiboles |
| Sheet silicates | $[Si_4O_{10}]^{4-}$ | 3 |  | serpentine<br>chlorite<br>micas<br>clay minerals |
| Framework silicates | $[Si_3O_8]^{-}$<br>$[SiO_2]$ | 4 |  | feldspars<br>quartz |

$[SiO_4]^{4-}$ tetrahedra can be polymerized in different configurations.

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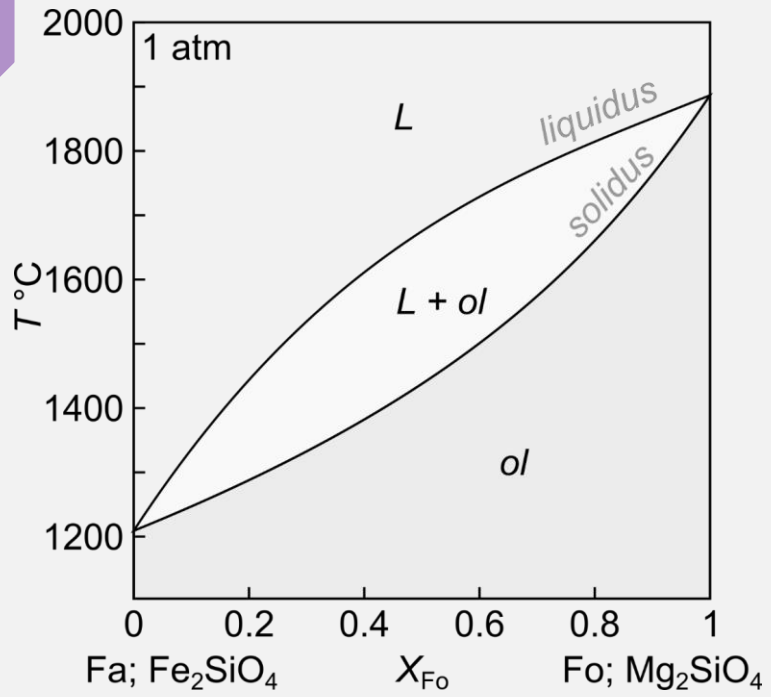
**1.4 Isomorphism and polymorphism**





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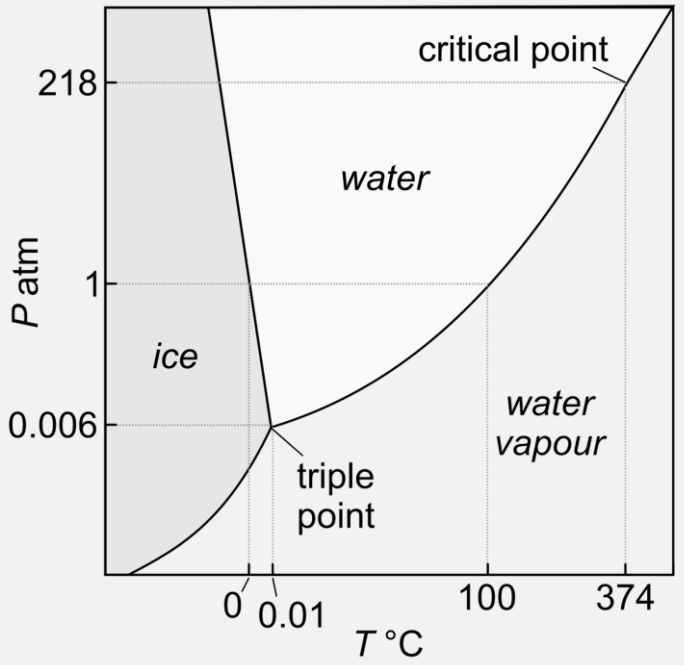
# Isomorphism




Same structure, different chemistry

Example: All the possible compositions of the olivine solid solution

# Polymorphism




Same chemistry, different structure

Example: Water, ice, water vapour

The stable polymorph is determined by the pressure and temperature conditions. Polymorphs found outside their stable P-T range are said to be **metastable**.

See also EART 26201 Principles of Geochemistry, Lecture 3


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# An introduction to the rock-forming minerals

You should now be able to:

- Explain that minerals are formed from cations and anions
- Recognise co-ordination polyhedra and use diagrams to count the cation co-ordination number
- Describe the shape, chemistry and charge of the silica tetrahedron
- Recognise and describe the structures of the principal silicate mineral groups
- Explain why different silicate structures accommodate different combinations of cations




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# Olivine

$(Mg,Fe)_2SiO_4$ orthosilicate orthorhombic





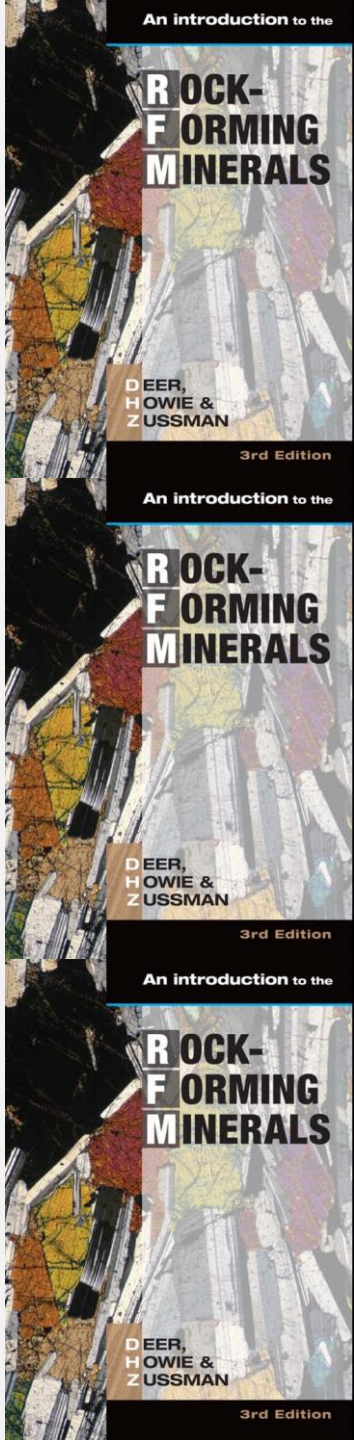




Found in:

Ultramafic igneous rocks
- Peridotite

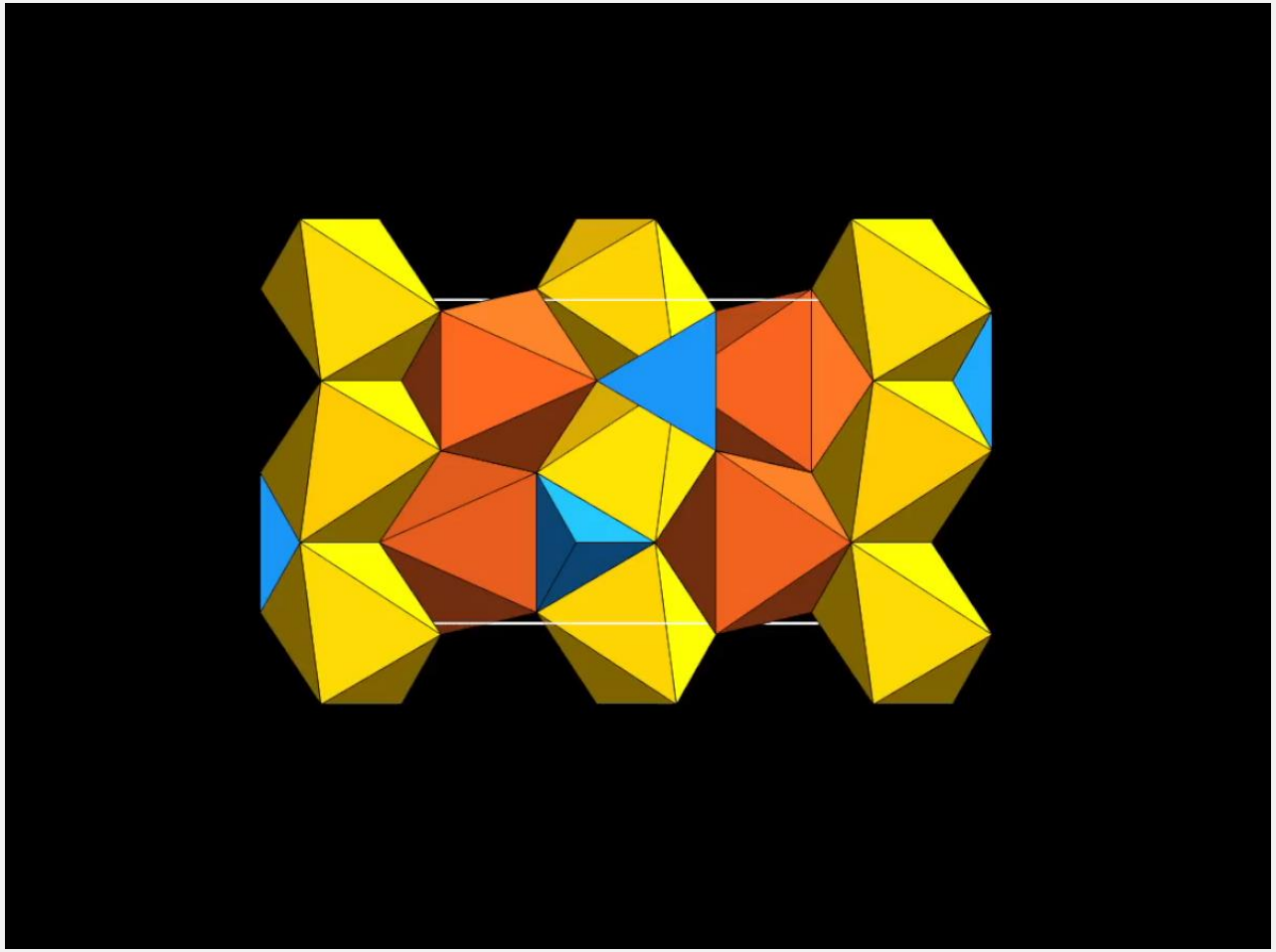
Mafic igneous rocks
- Basalt
- Gabbro

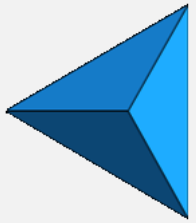




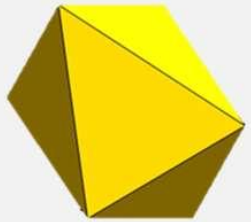

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# Olivine





silica tetrahedra



M1 cation site



M2 cation site

The M1 and M2 cation sites have similar sizes but slightly different shapes. Both sites take $M^{2+}$ cations.



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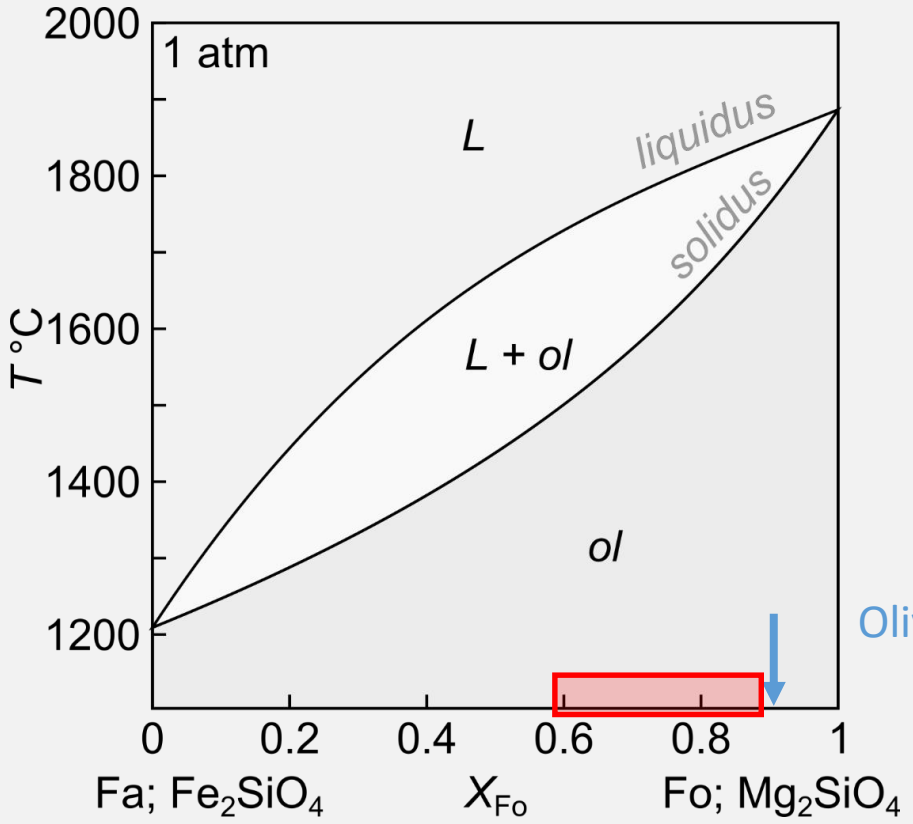
# Olivine composition, phase diagram and solid solution

Olivine phase diagram: See EART 26201 Principles of Geochemistry, Lecture 3

General formula: $X_2SiO_4$

Endmembers: $Mg_2SiO_4$ forsterite
$Fe_2SiO_4$ fayalite

Complete solid solution:
$(Mg,Fe)_2SiO_4$



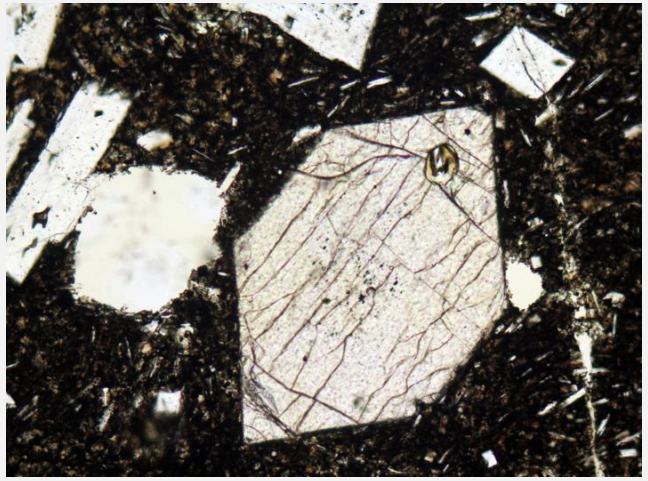

Olivine in the mantle: ~$Fo_{92}$

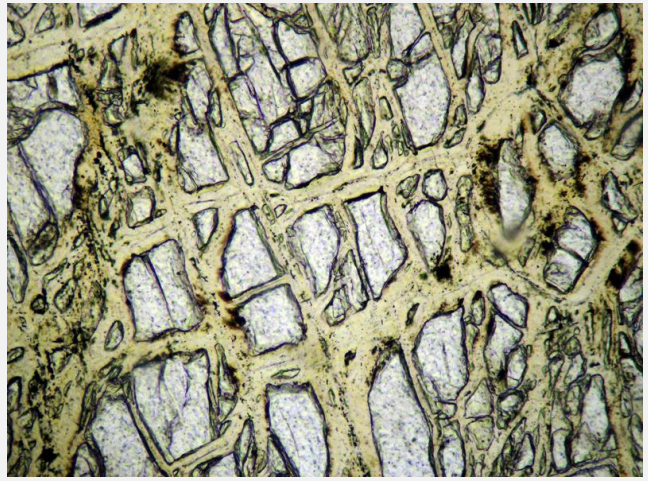
Olivine in most basalts and gabbros: $Fo_{60}$-$Fo_{90}$


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# Olivine

Distinguish from augite by its lack of colour, straight extinction, curving fractures with common alteration to serpentine

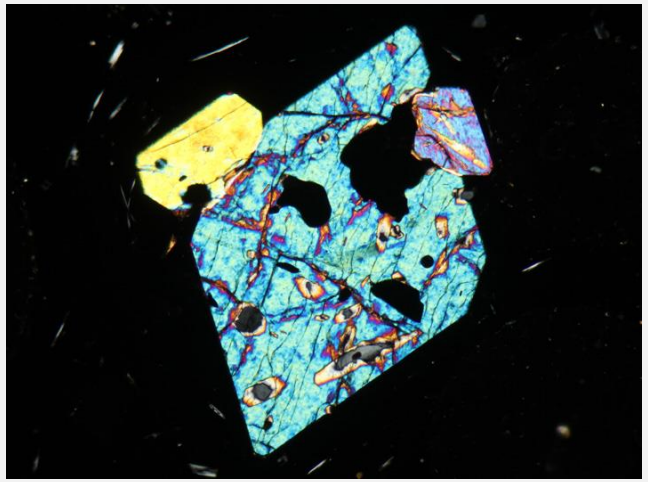




Key optical properties:

PPL

- Colourless
- High relief
- Curved fractures (common)
- Cleavages at 90° (rarely visible)
- Alteration to serpentine (common)





XP

- Interference colours up to high second order
- Straight extinction



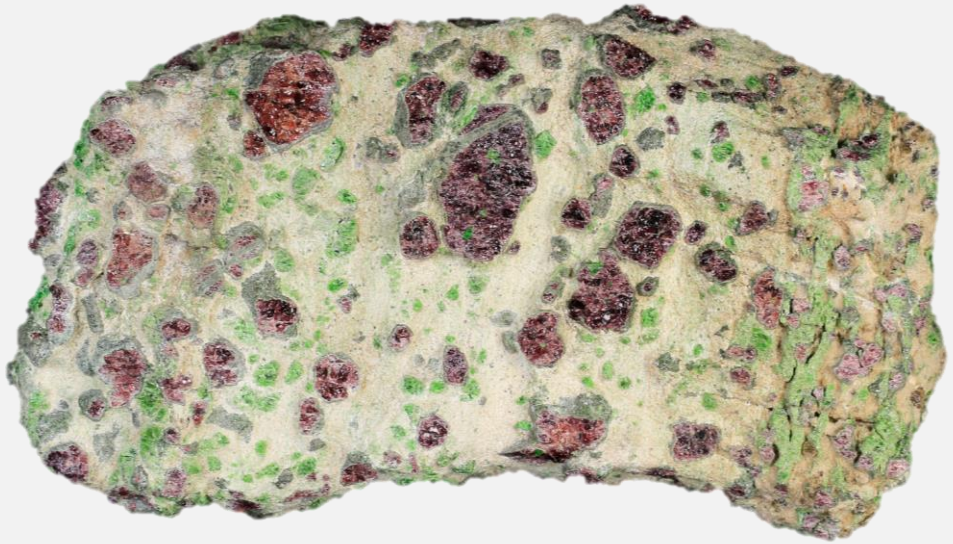
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# Garnet

$(A_3B_2)Si_3O_{12}$ orthosilicate cubic



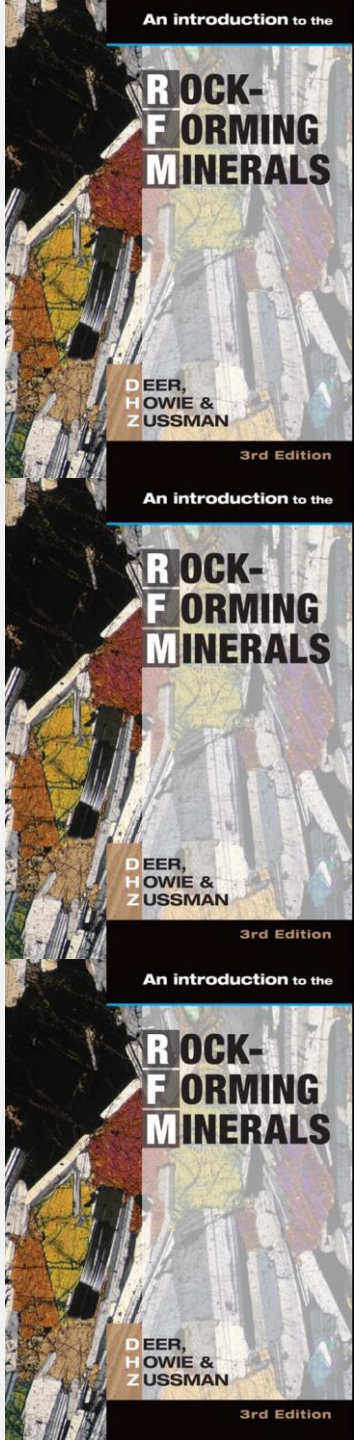






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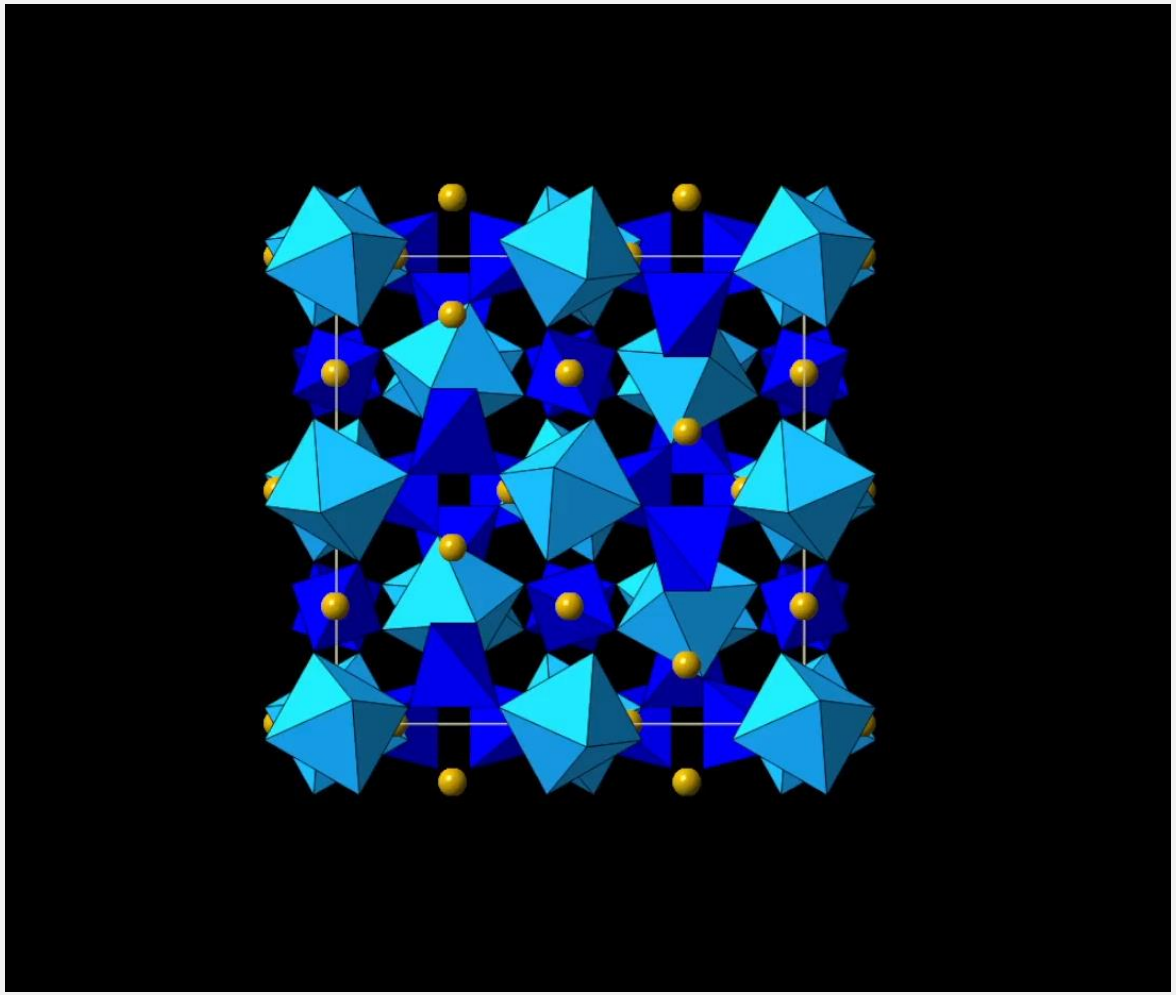
Ultramafic igneous rocks
- Peridotite






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# Garnet



General formula: $(A_3B_2)Si_3O_{12}$



**A** = Large $M^{2+}$ cation:
$Ca^{2+}$, $Mg^{2+}$, $Fe^{2+}$, $Mn^{2+}$
Distorted cubic [8] co-ordination



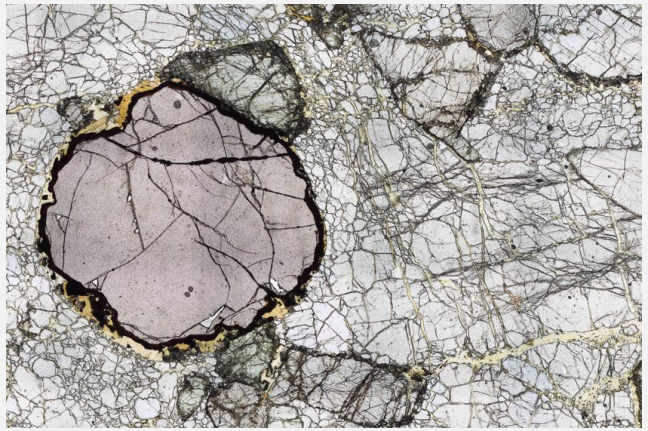
**B** = Small $M^{3+}$ cation:
$Al^{3+}$, $Fe^{3+}$, $Cr^{3+}$
Octahedral [6] co-ordination

Garnet in the mantle: $Mg_3Al_2Si_3O_{12}$


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# Garnet

Distinguish from vesicles and holes in the thin section by its high relief and pink colour in PPL

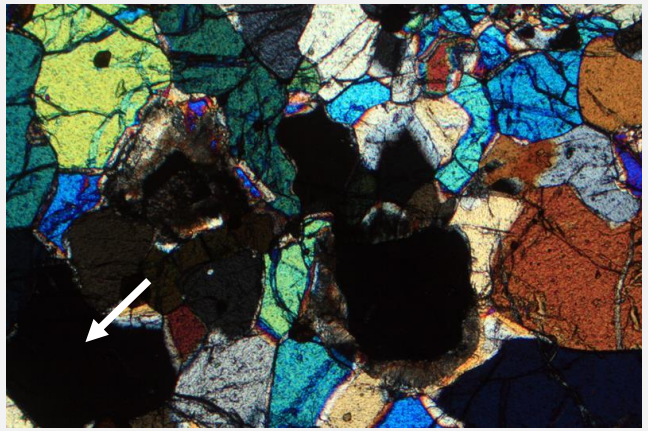


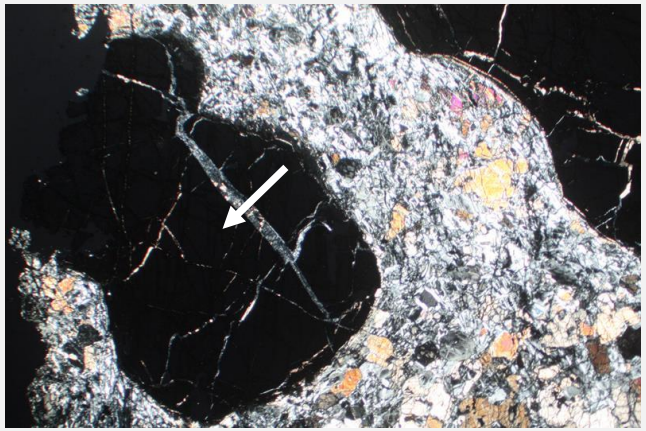


Key optical properties:

PPL

- Pinkish colour
- Very high relief
- Fractures (common)
- No cleavage





XP

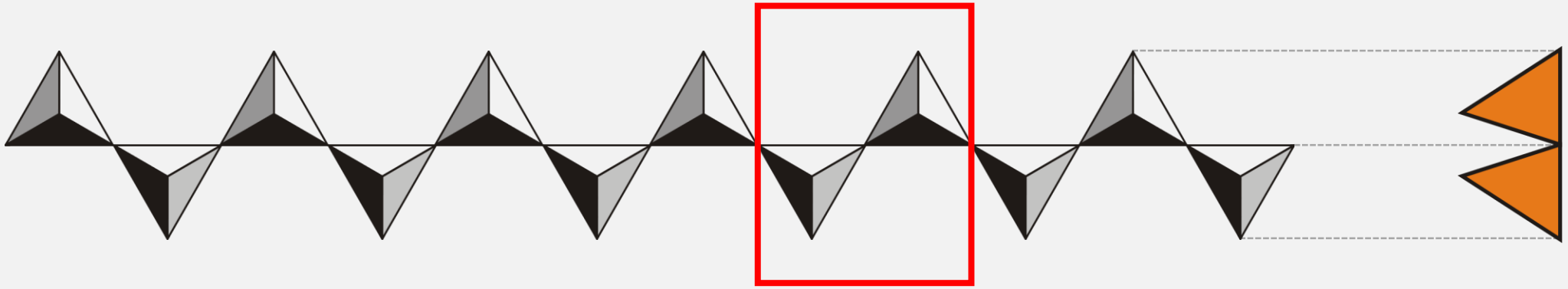
- Isotropic – always in extinction



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# Pyroxenes

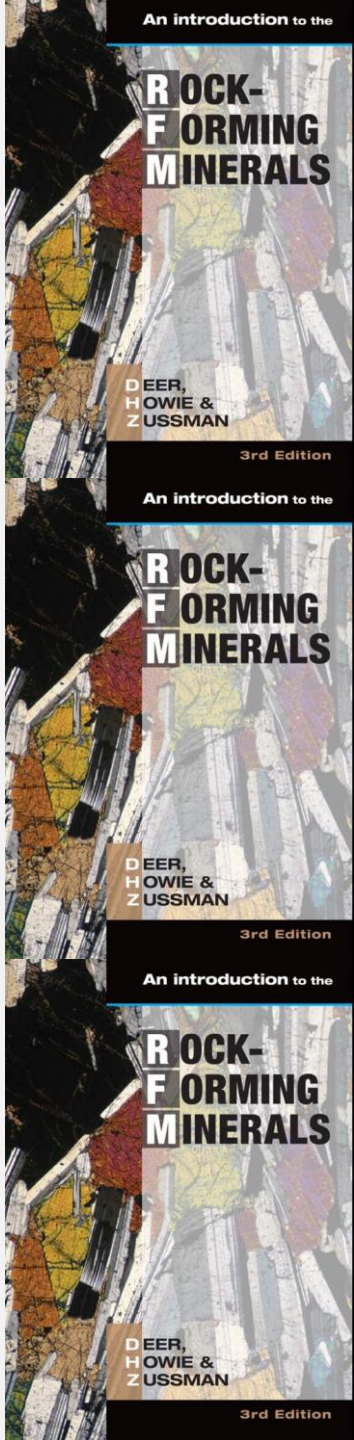
$(XY)Si_2O_6$ single chain silicates



$[Si_2O_6]^{4-}$

**Clinopyroxene** monoclinic

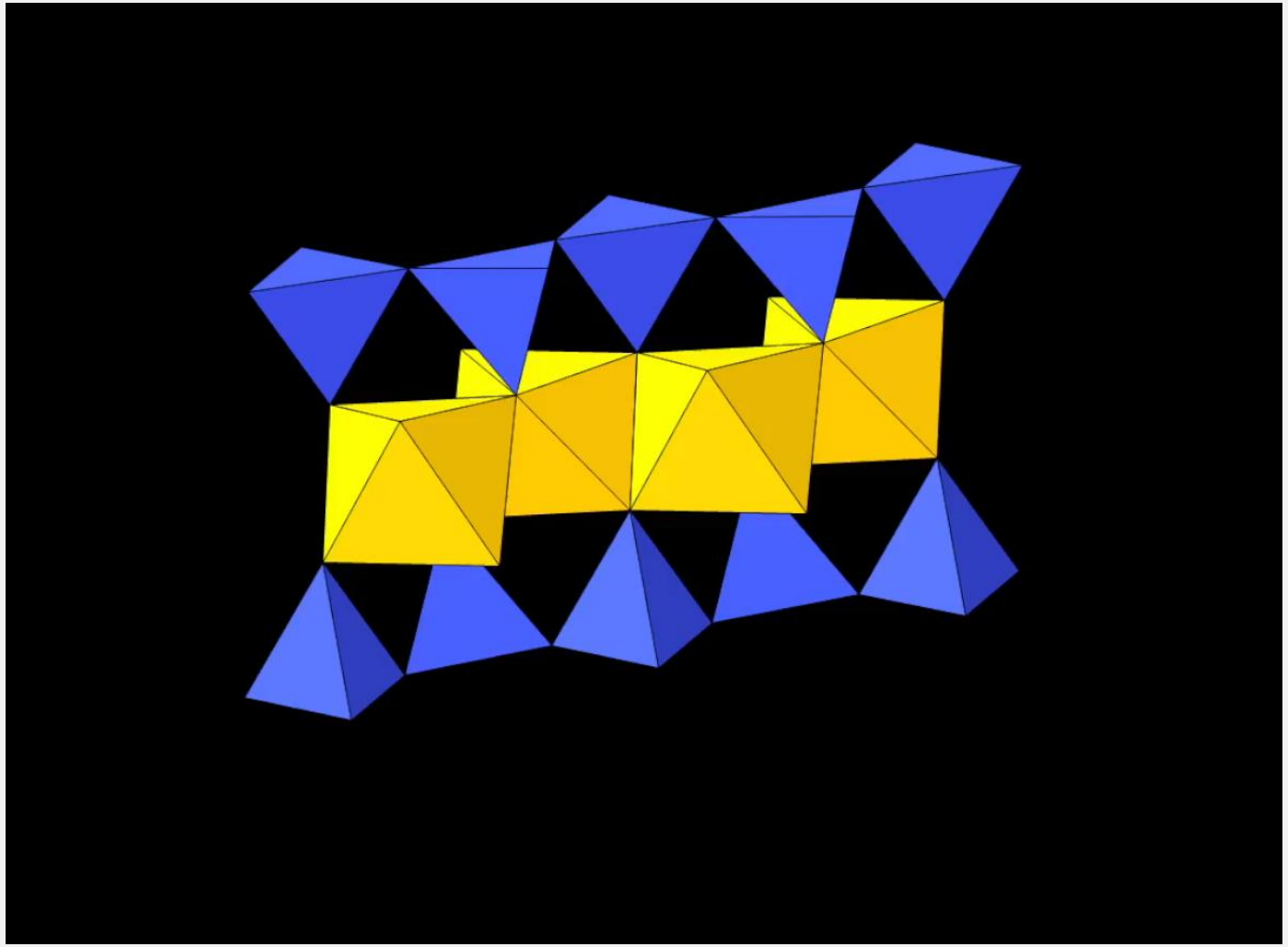
**Orthopyroxene** orthorhombic



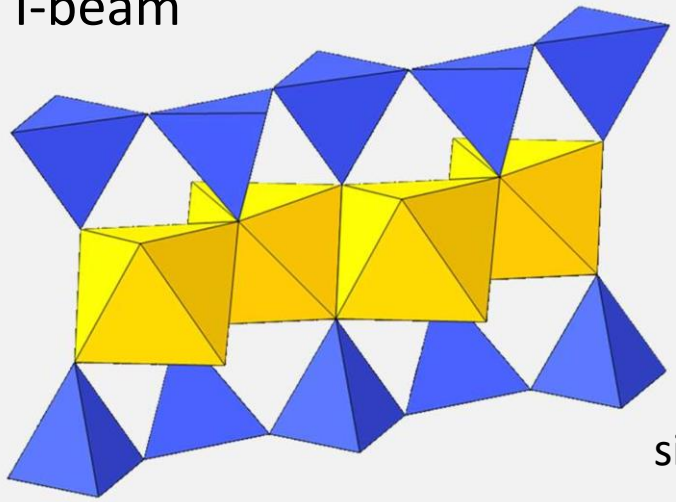



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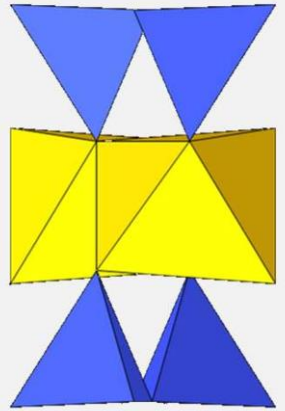
# Pyroxenes



I-beam



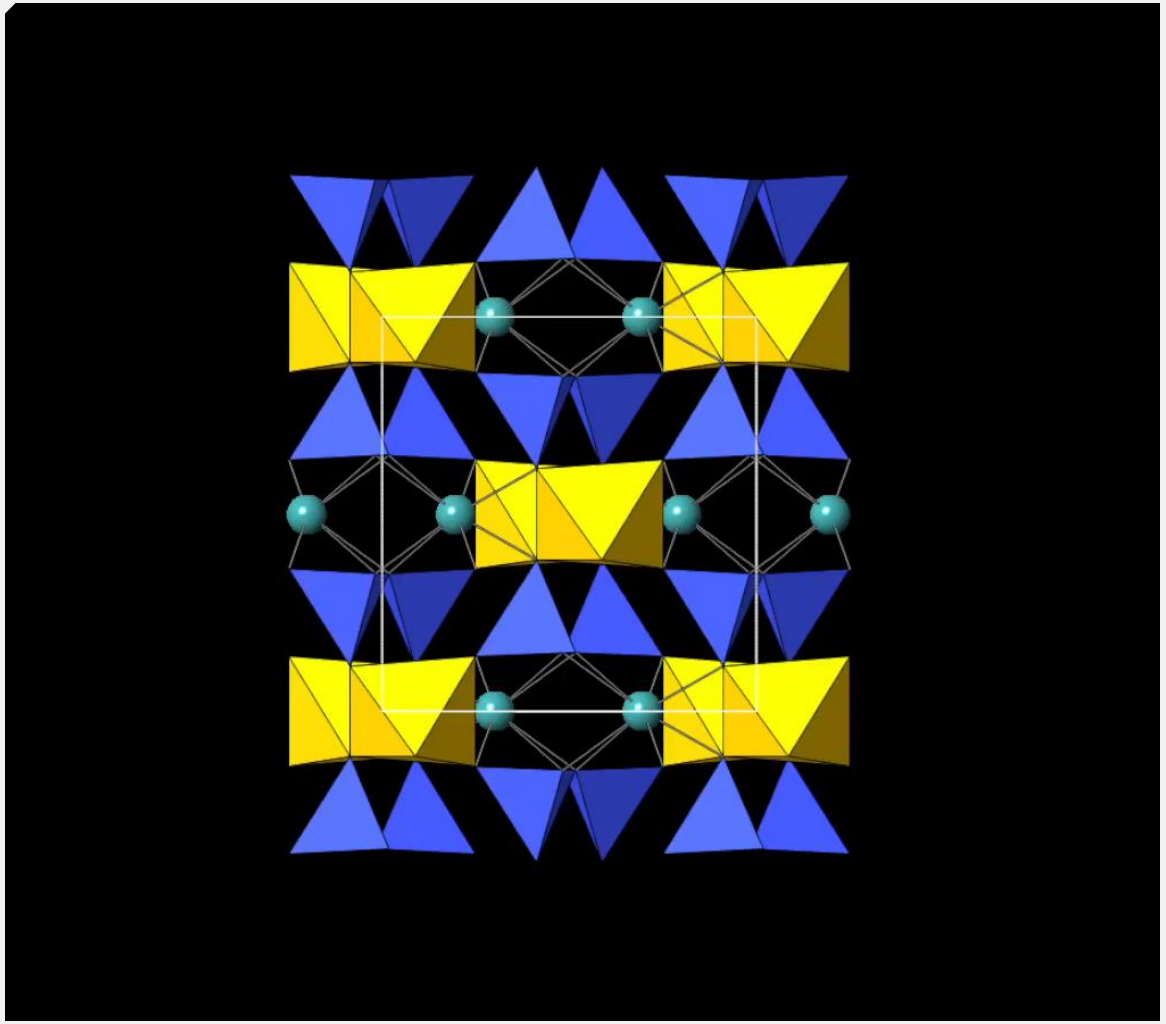
side view

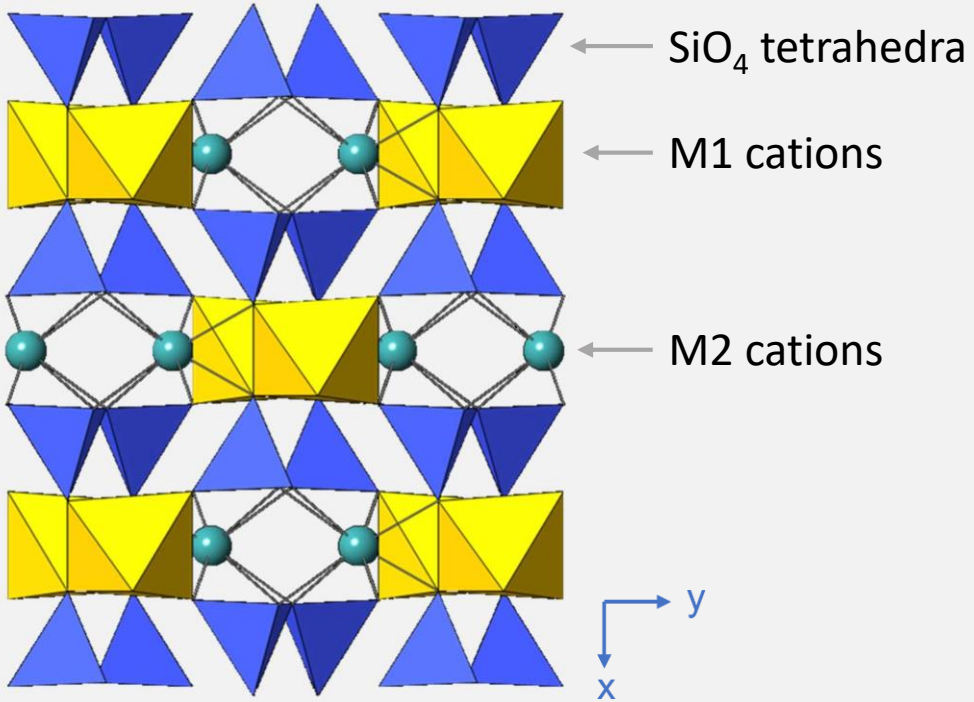


end-on view


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# Pyroxene structure:  clinopyroxene



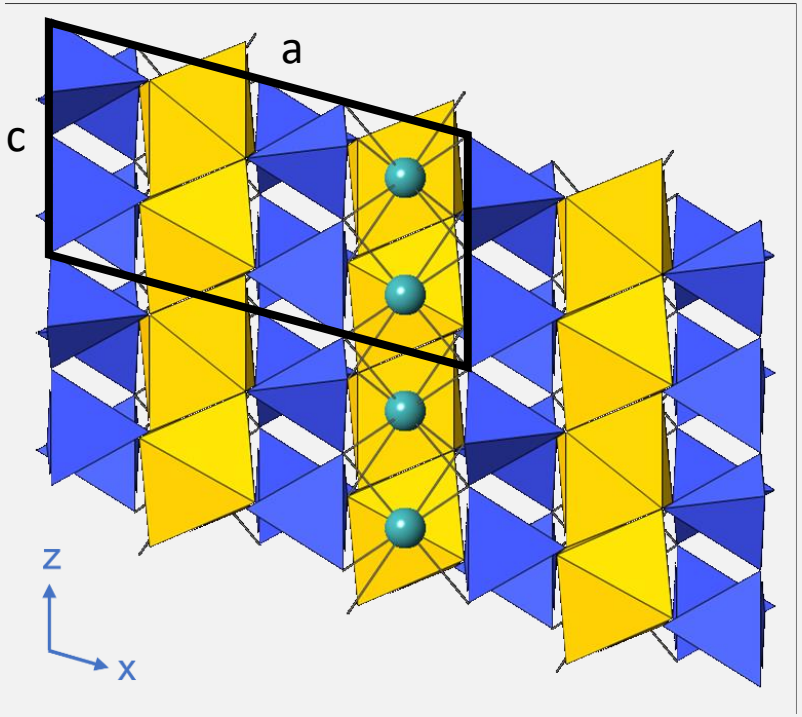




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## Clinopyroxene

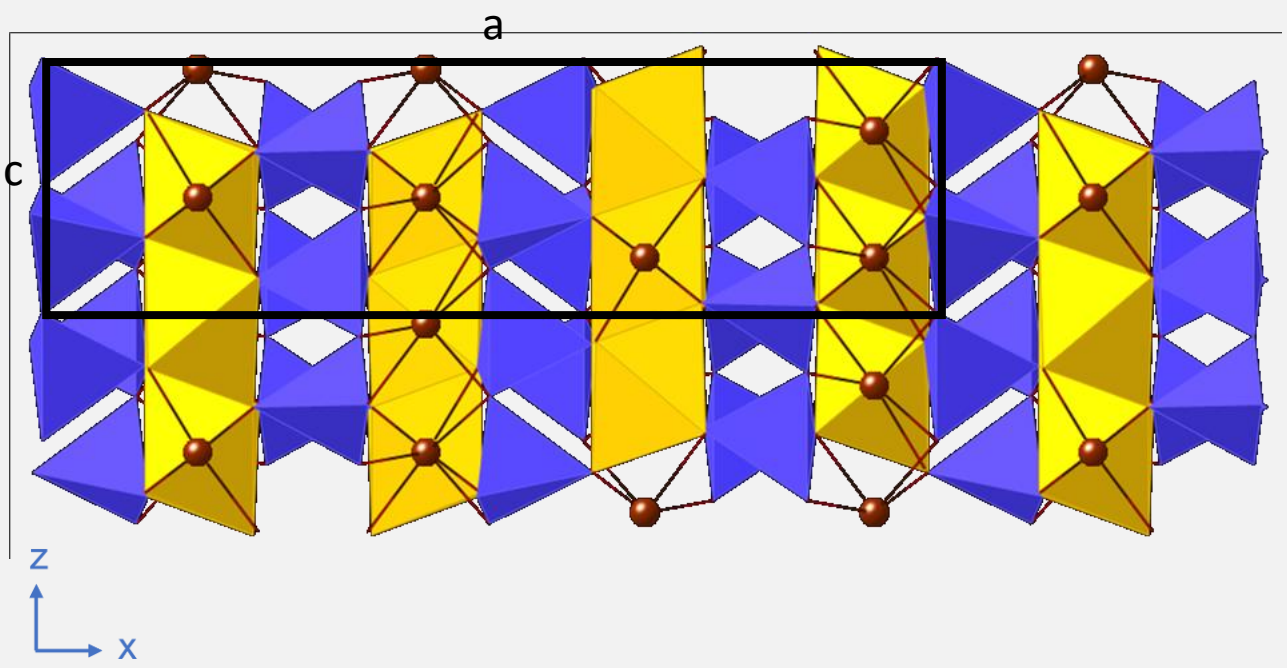
monoclinic




Cpx unit cell: two sets of stacked I-beams

## Orthopyroxene

orthorhombic

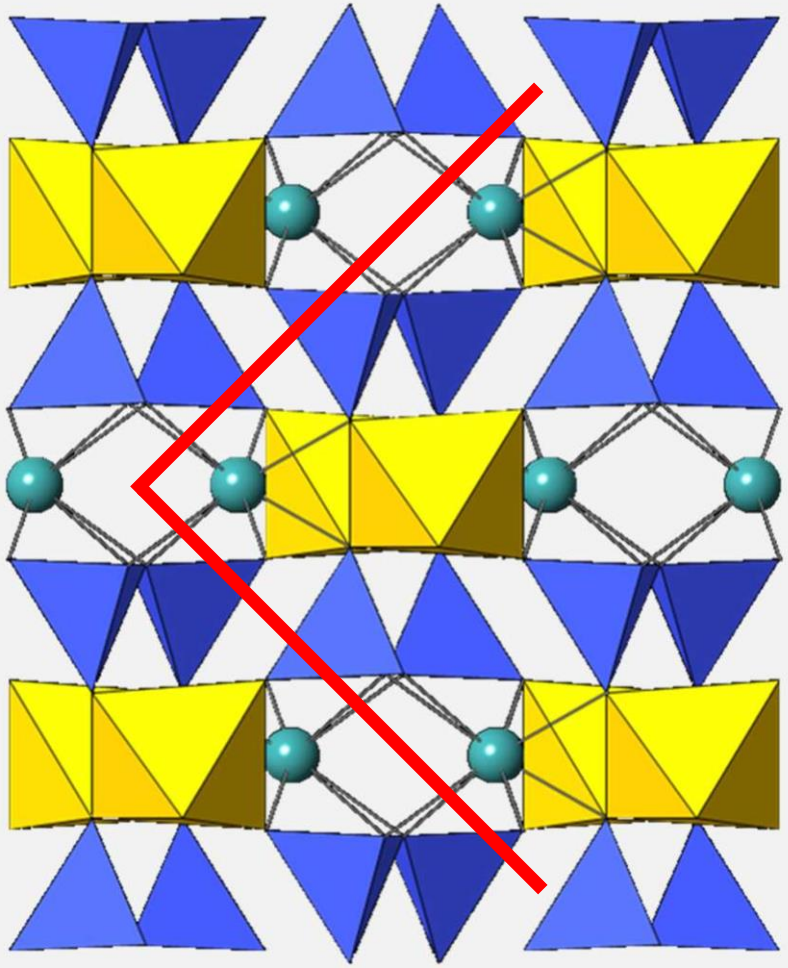



Opx unit cell: four sets of stacked I-beams

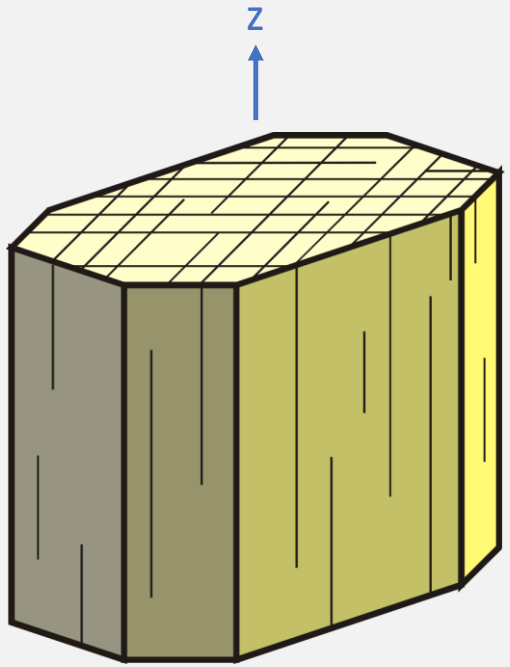
Clinopyroxene and orthopyroxene are both single chain silicates and have the same basic structure, but in detail their symmetry is different and their unit cell sizes are different.


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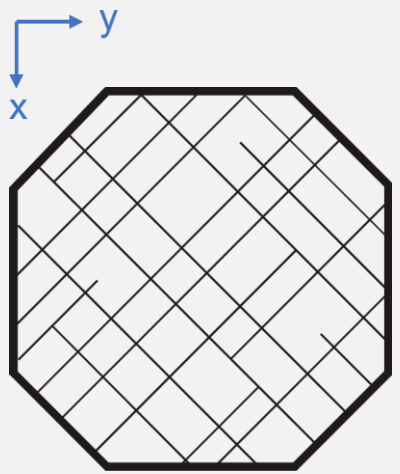
# Pyroxenes: cleavage



2 cleavages at ~90°



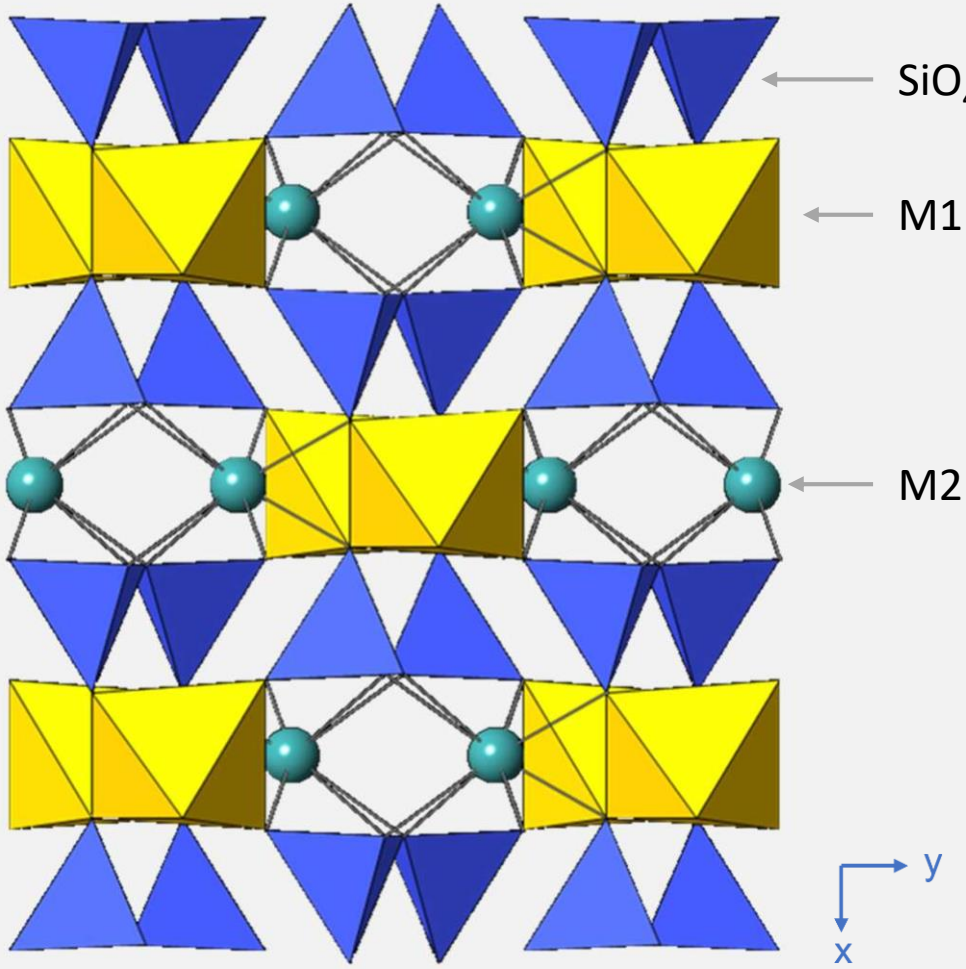

3D shape




2D cross-section


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# Pyroxene compositions and solid solutions




General formula:
**$XYSi_2O_6$**

**Y** = M1 (small)

In opx and cpx: usually $Mg^{2+}$ or $Fe^{2+}$

Other possible cations: $Mn^{2+}$, $Al^{3+}$, $Fe^{3+}$

**X** = M2 (large)

In opx: usually $Mg^{2+}$ or $Fe^{2+}$

In cpx: usually $Ca^{2+}$

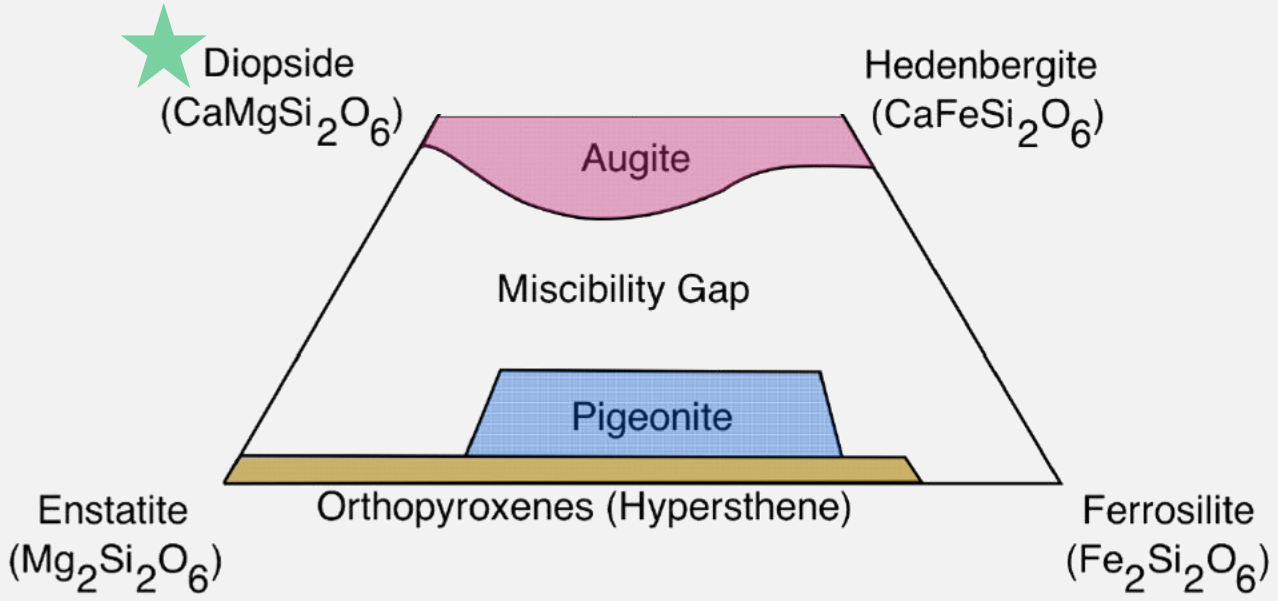
Other possible cations in cpx: $Na^{+}$


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# Pyroxene compositions and solid solutions

Exsolution in pyroxenes: see EART 26201 Principles of Geochemistry, Lecture 4

General formula:
$XYSi_2O_6$




Augite: high-Ca clinopyroxene
Very common in basalts and gabbros

Pigeonite: low-Ca clinopyroxene
Found in some gabbros

Orthopyroxene
Found in mantle rocks, and sometimes in mafic or intermediate igneous rocks

Chrome diopside: $Ca(Mg,Cr)Si_2O_6$ clinopyroxene
Found in some mantle rocks


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# Augite (clinopyroxene)

$Ca(Mg,Fe)Si_2O_6$ single chain silicate monoclinic



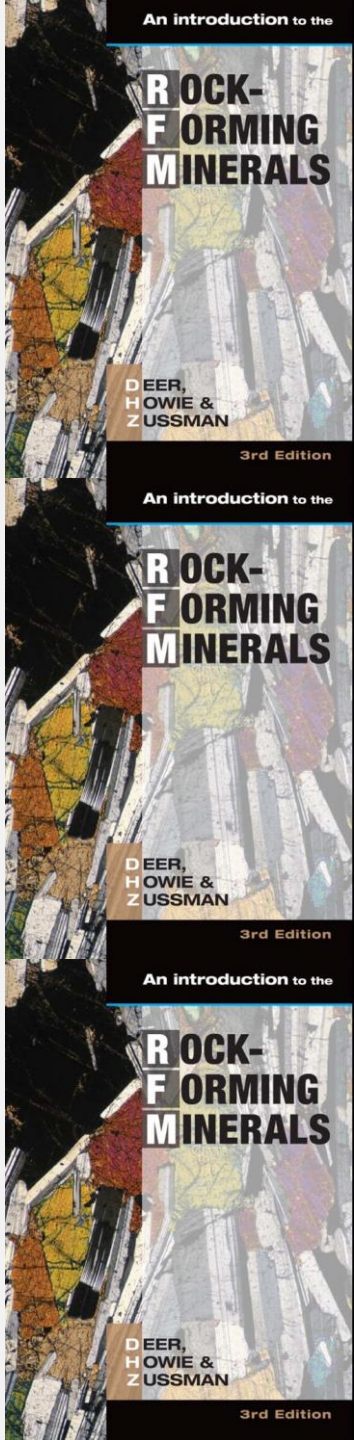




Found in:

Mafic igneous rocks
- Gabbro
- Basalt

Intermediate igneous rocks
- Andesite
- Diorite



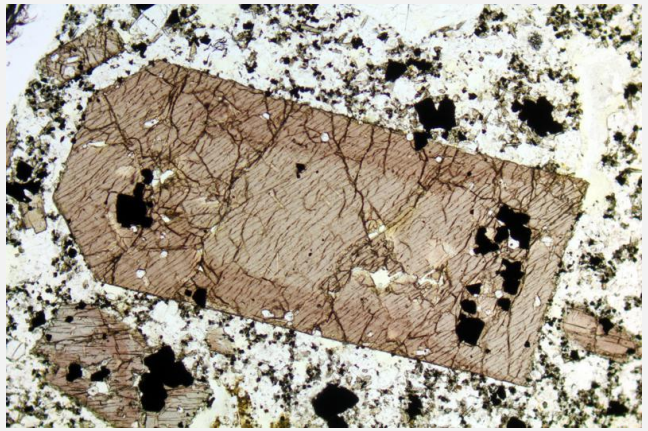


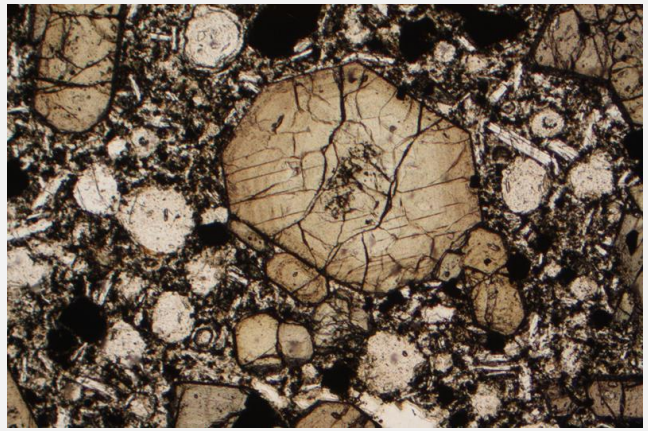

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# Clinopyroxene: augite

Distinguish from olivine by the slight coloration in PPL

Distinguish from opx by the bright interference colours, twinning (sometimes) and inclined extinction

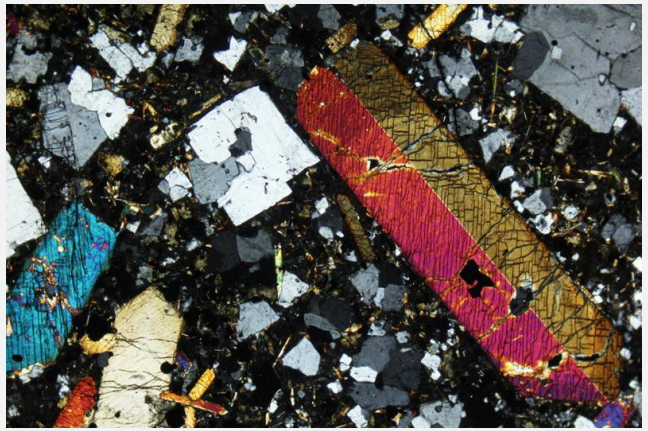


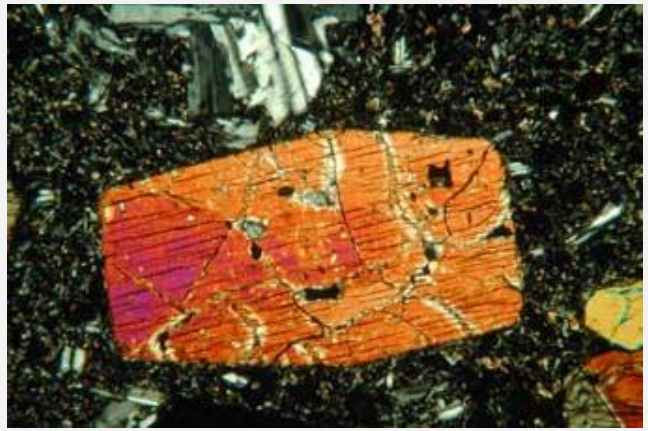


Key optical properties:

PPL

- High relief
- Usually pale grey-brown
- Two cleavages at 90°
- Most common to see only one of the cleavage traces





XP

- Interference colours up to mid second order: often blue or orange
- Most sections show inclined extinction with respect to the cleavage
- Extinction angle depends on composition. In augites, it is ~35-45°
- Sometimes twinned



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# Chrome diopside (clinopyroxene)

$Ca(Mg,Cr)Si_2O_6$ single chain silicate monoclinic

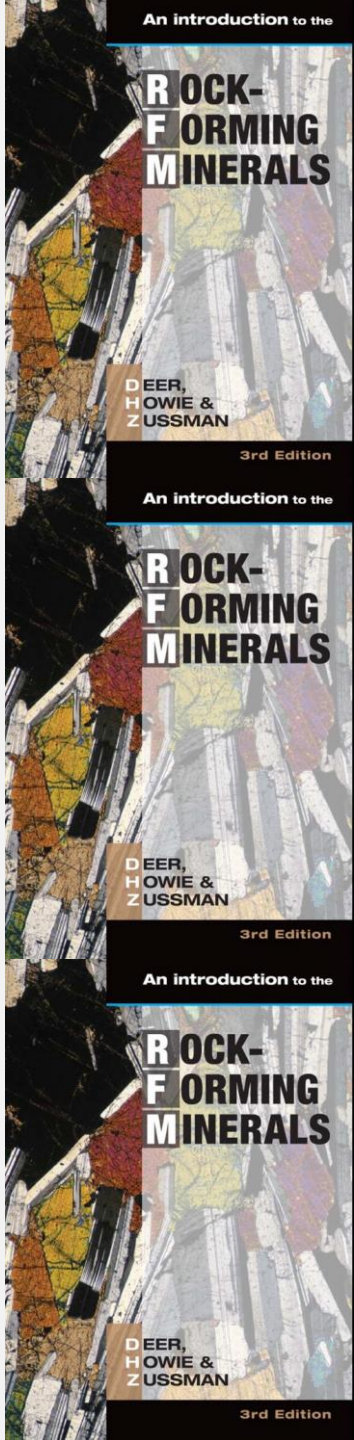






Found in:

Ultramafic igneous rocks
- Peridotite

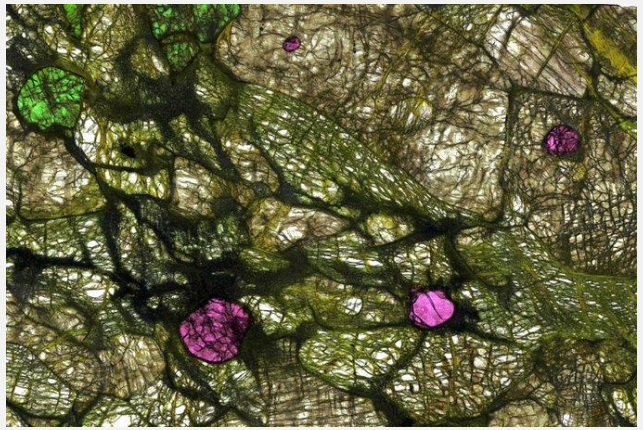


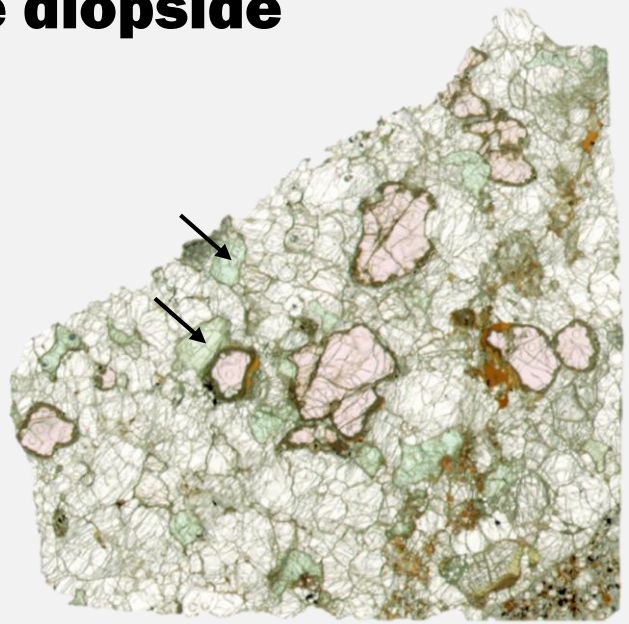



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# Clinopyroxene: chrome diopside

Distinguish from olivine and augite by the bright bottle green colour in hand specimen and in PPL

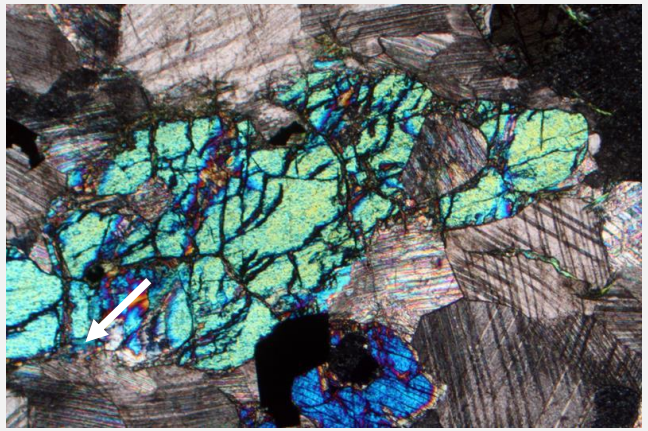


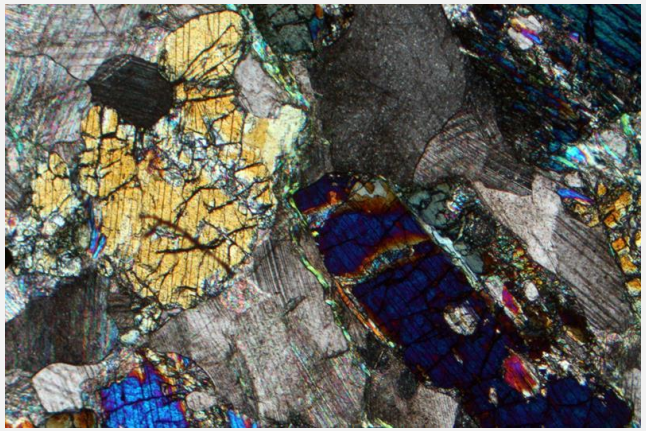


Key optical properties:

PPL

- High relief
- Bright bottle green
- Two cleavages at 90°
- Often fractured





XP

- Interference colours up to mid second order: often blue or orange
- Most sections show inclined extinction with respect to the cleavage
- Extinction angle is ~35-45°
- Sometimes twinned



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# Orthopyroxene

$(Mg,Fe)Si_2O_6$ single chain silicate orthorhombic



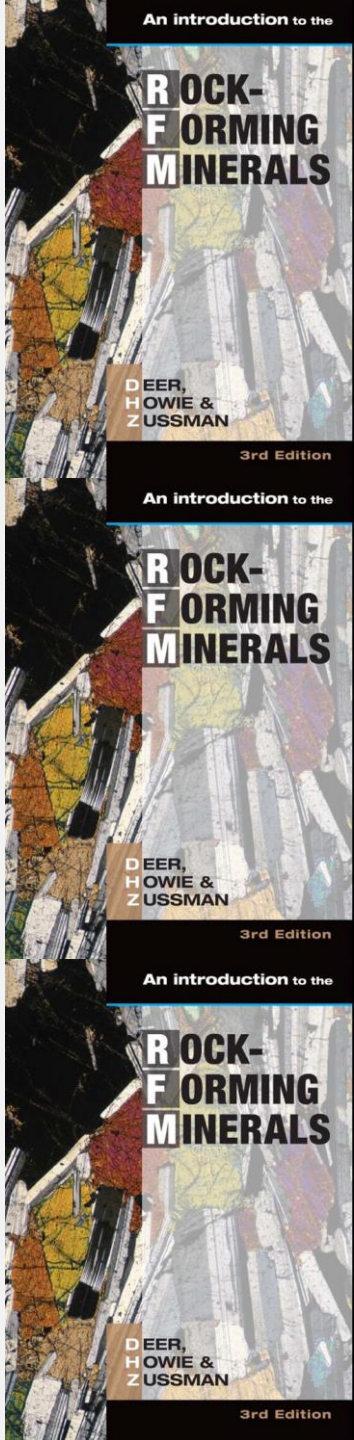




Found in:

Ultramafic igneous rocks
- Peridotite

Intermediate igneous rocks
- Andesite
- Diorite

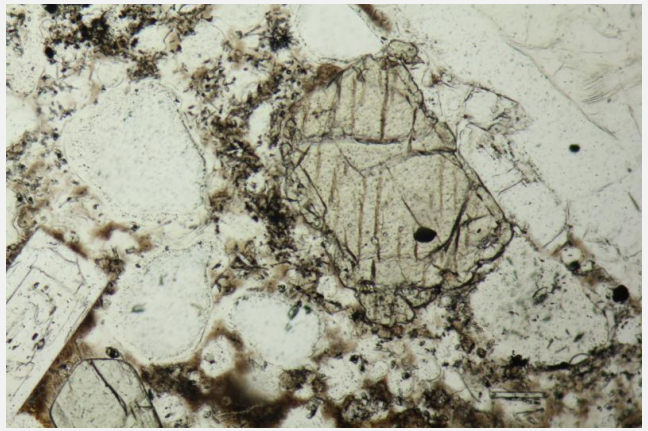


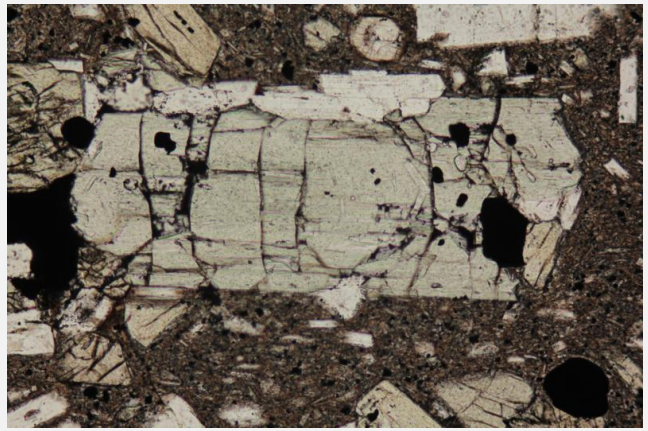



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# Orthopyroxene

Distinguish from augite by the low interference colours and straight extinction

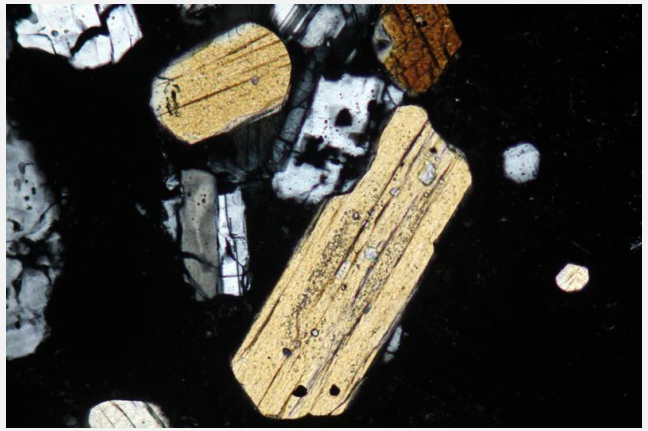


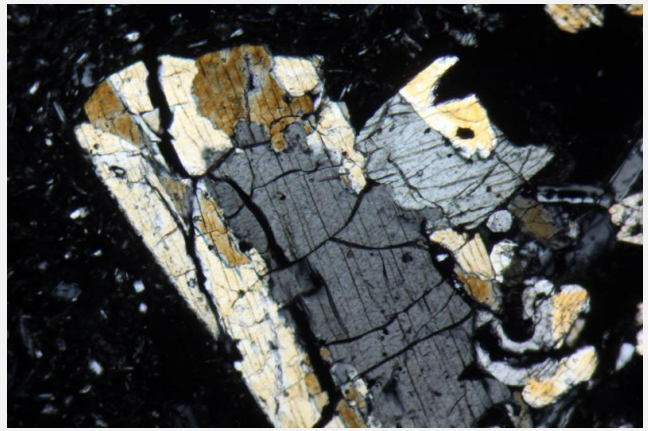


Key optical properties:

PPL

- Medium to high relief
- Usually colourless or very pale grey
- Two cleavages at 90°
- Most common to see only one of the cleavage traces





XP

- First-order interference colours: grey, yellow, white
- Straight extinction with respect to the cleavage

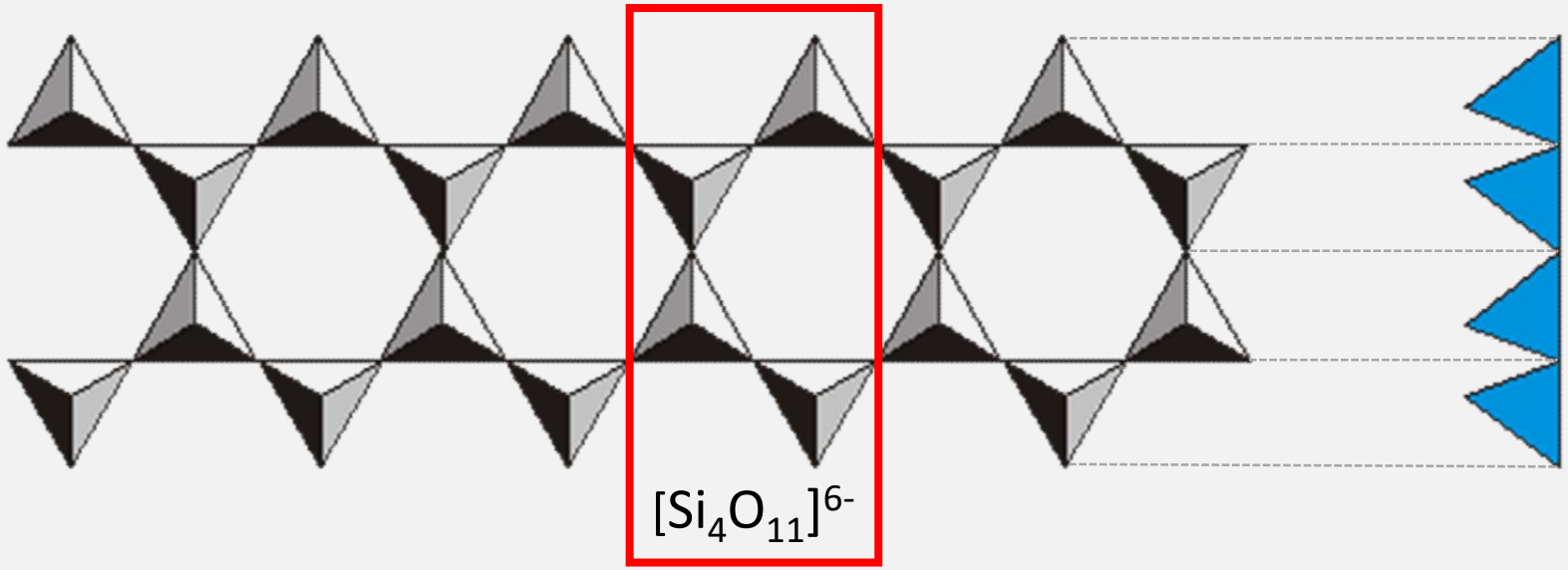


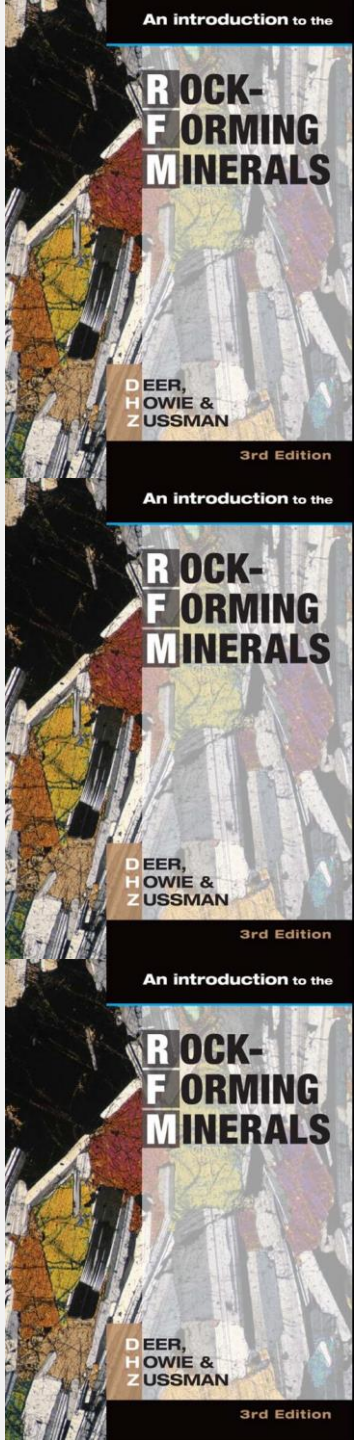
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# Amphiboles

$AB_2C_5Si_8O_{22}(OH)_2$ double chain silicates monoclinic

(with the exception of one or two orthorhombic amphiboles)

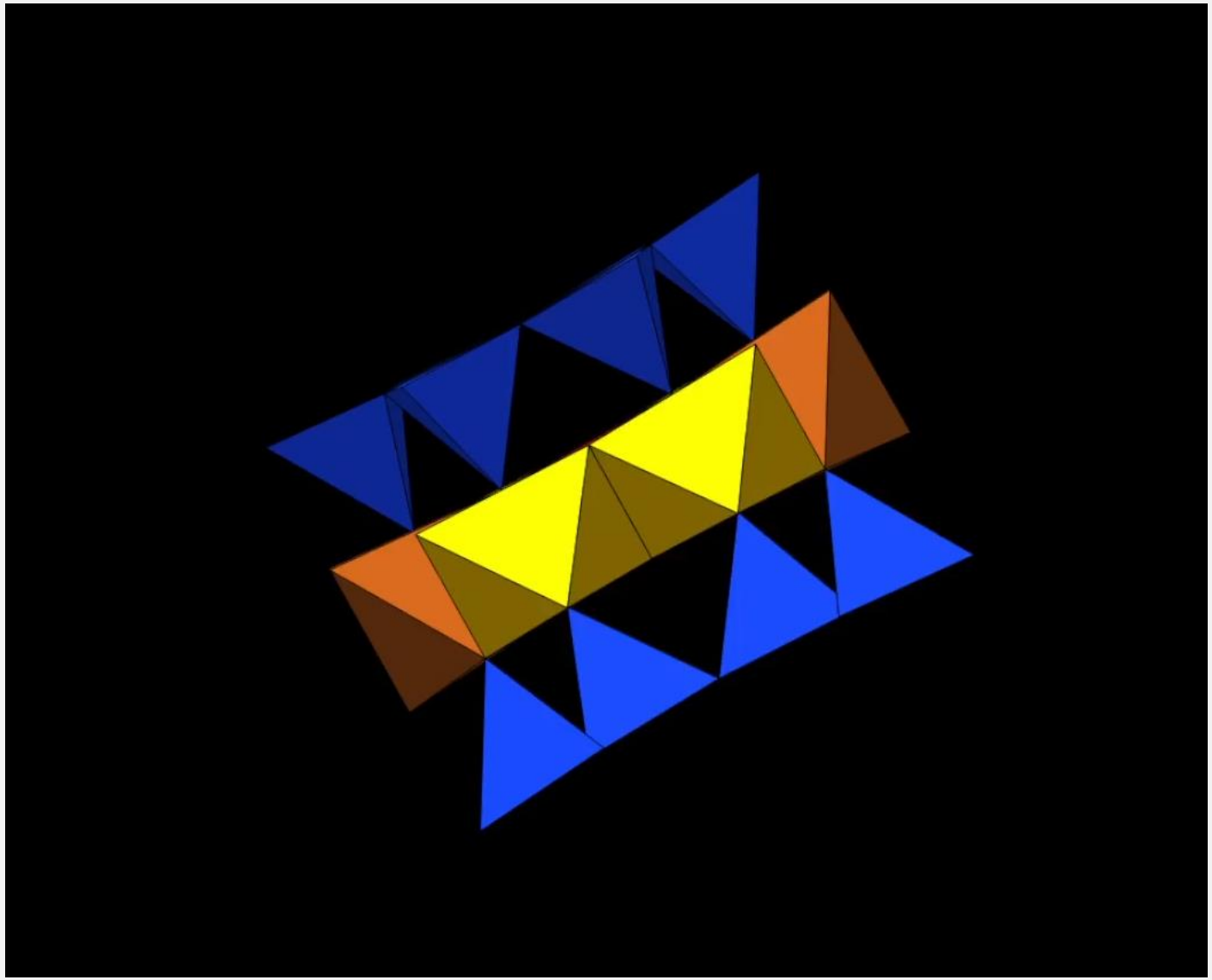




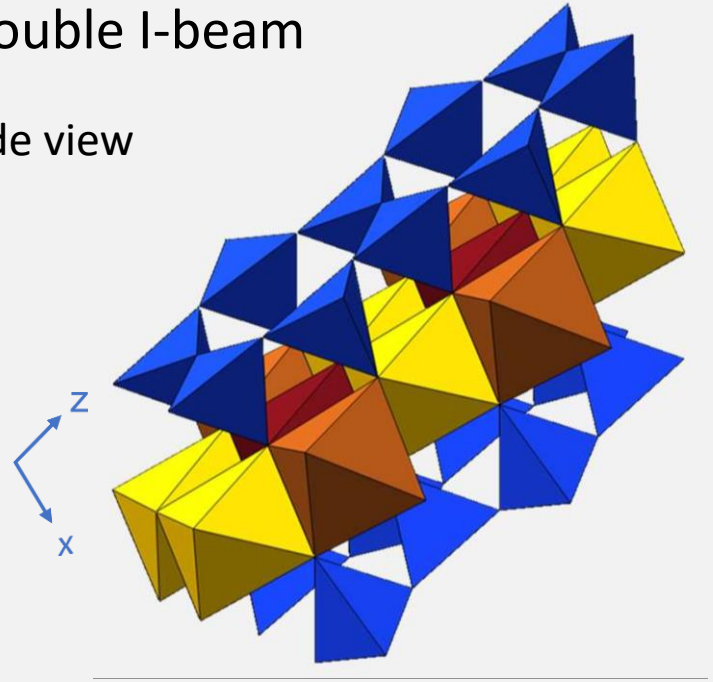

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# Amphiboles

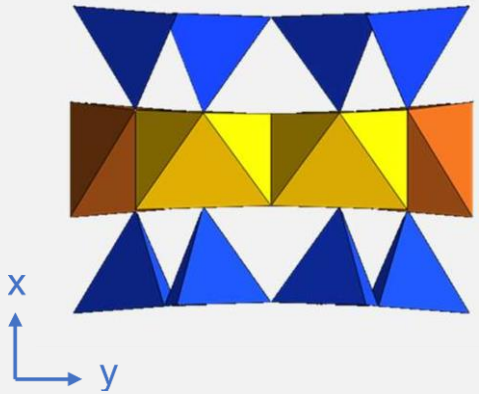


Double I-beam

side view



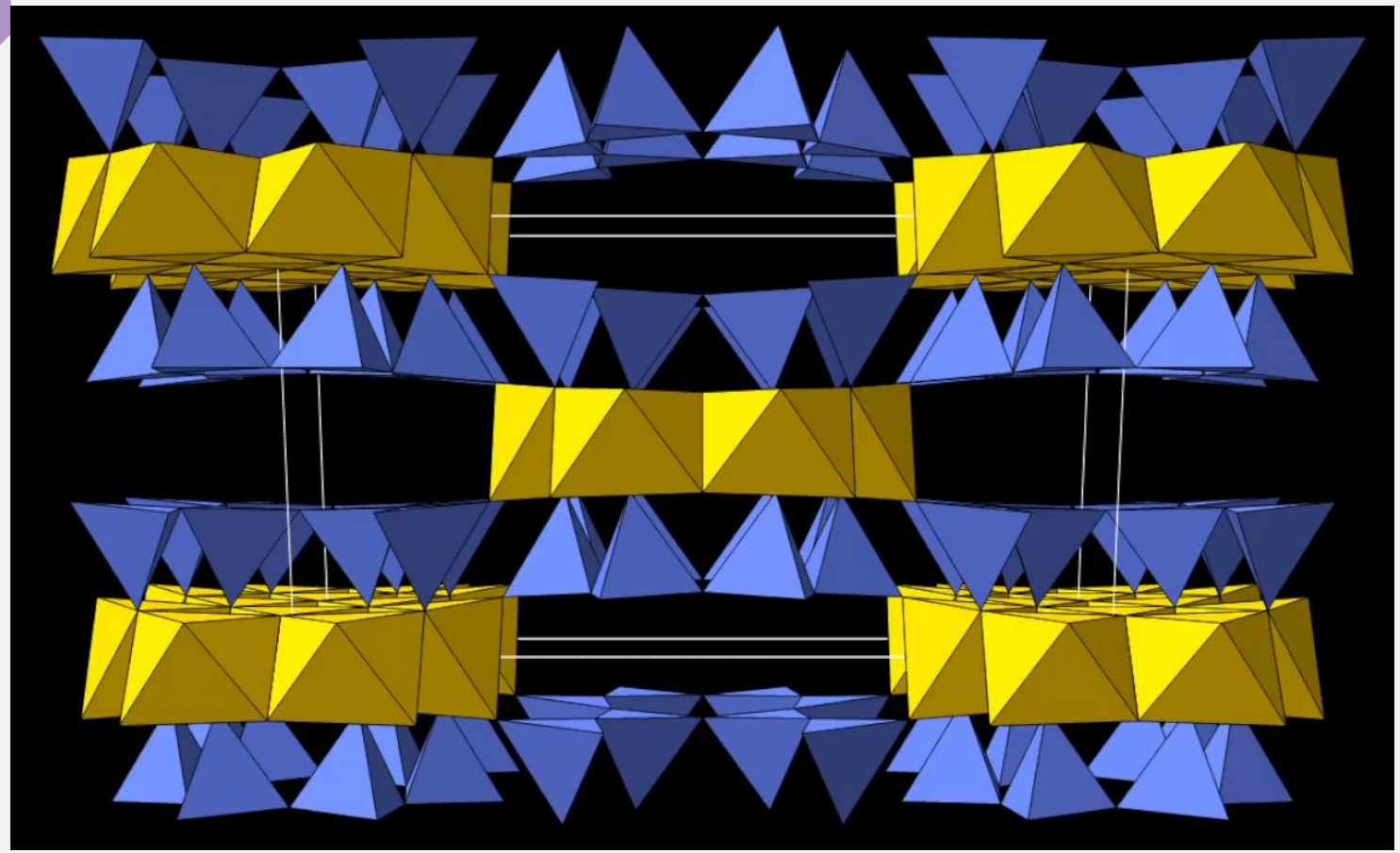

end-on view






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# Amphiboles



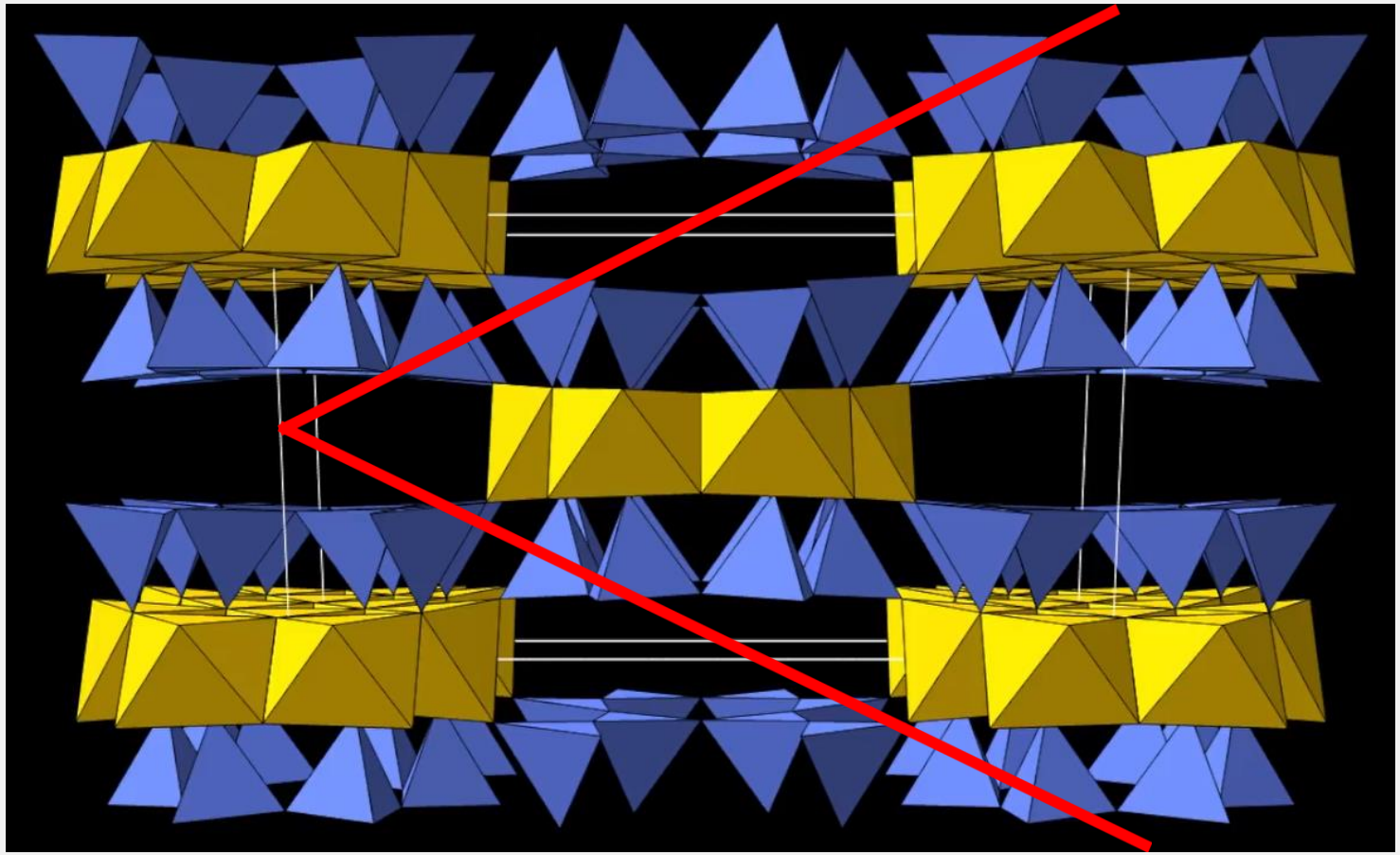
General formula:
$AB_2C_5Si_8O_{22}(OH)_2$

- **A** [12]: $Na^+$, $K^+$ …. or vacancy □
- **B** [8]: $Ca^{2+}$, $Na^+$, $Mg^{2+}$, $Fe^{2+}$ ….
- **C** [6]: $Mg^{2+}$, $Fe^{2+}$, $Al^{3+}$, $Fe^{3+}$ ….
- **$(OH)^-$**, sometimes $F^-$ or vacancy □

Multiple different cation sites with different sizes means that amphiboles as a mineral group have a wide range of compositions and solid solutions.

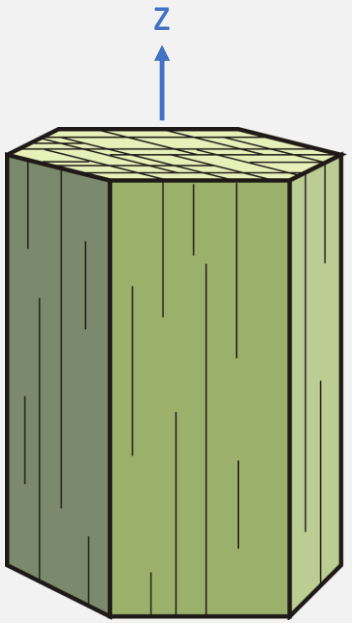

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# Amphiboles: cleavage

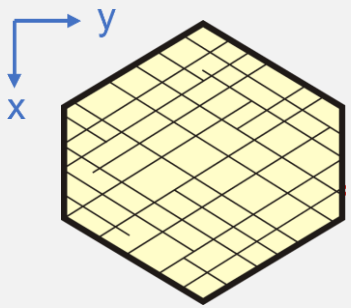


x
y

2 cleavages at ~60°




3D shape




2D cross-section


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# Hornblende

$Ca_2(Mg,Fe)_4Al[Si_7O_{22}](OH)_2$ double chain silicate monoclinic



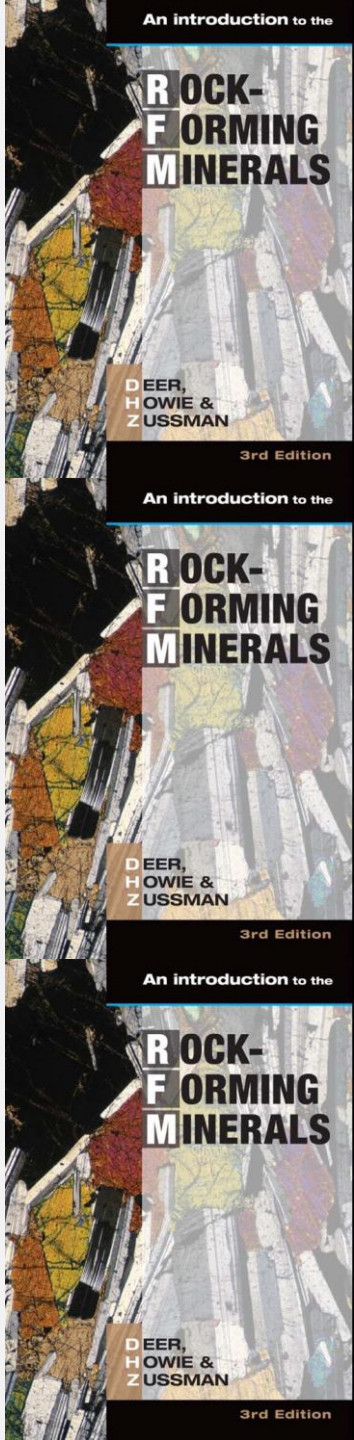




Found in:

Intermediate igneous rocks
- Andesite
- Diorite

Felsic igneous rocks
- Granite and granodiorite
- Rhyolite and dacite



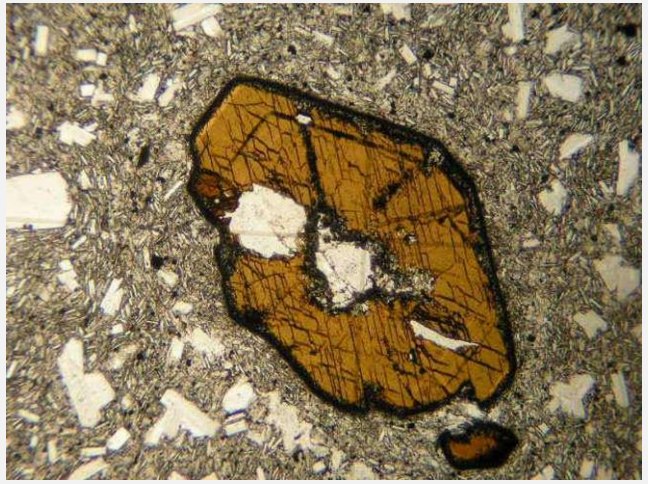


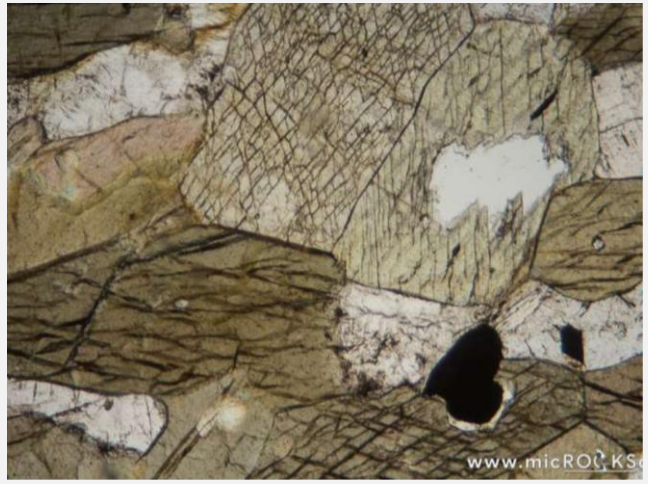

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# Hornblende

Distinguish from biotite by its two cleavages and inclined extinction

Distinguish from pyroxenes by its strong colour, pleochroism, 60° cleavage, ~20° extinction angle

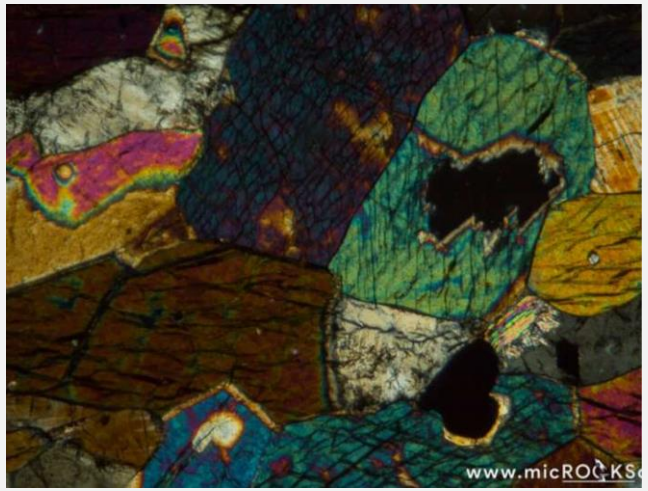





Key optical properties:

PPL

- Medium to high relief
- Various colours, usually brown or green
- Pleochroic
- Two cleavages at 60°
- In volcanic rocks, sometimes has a dark reaction rim






XP

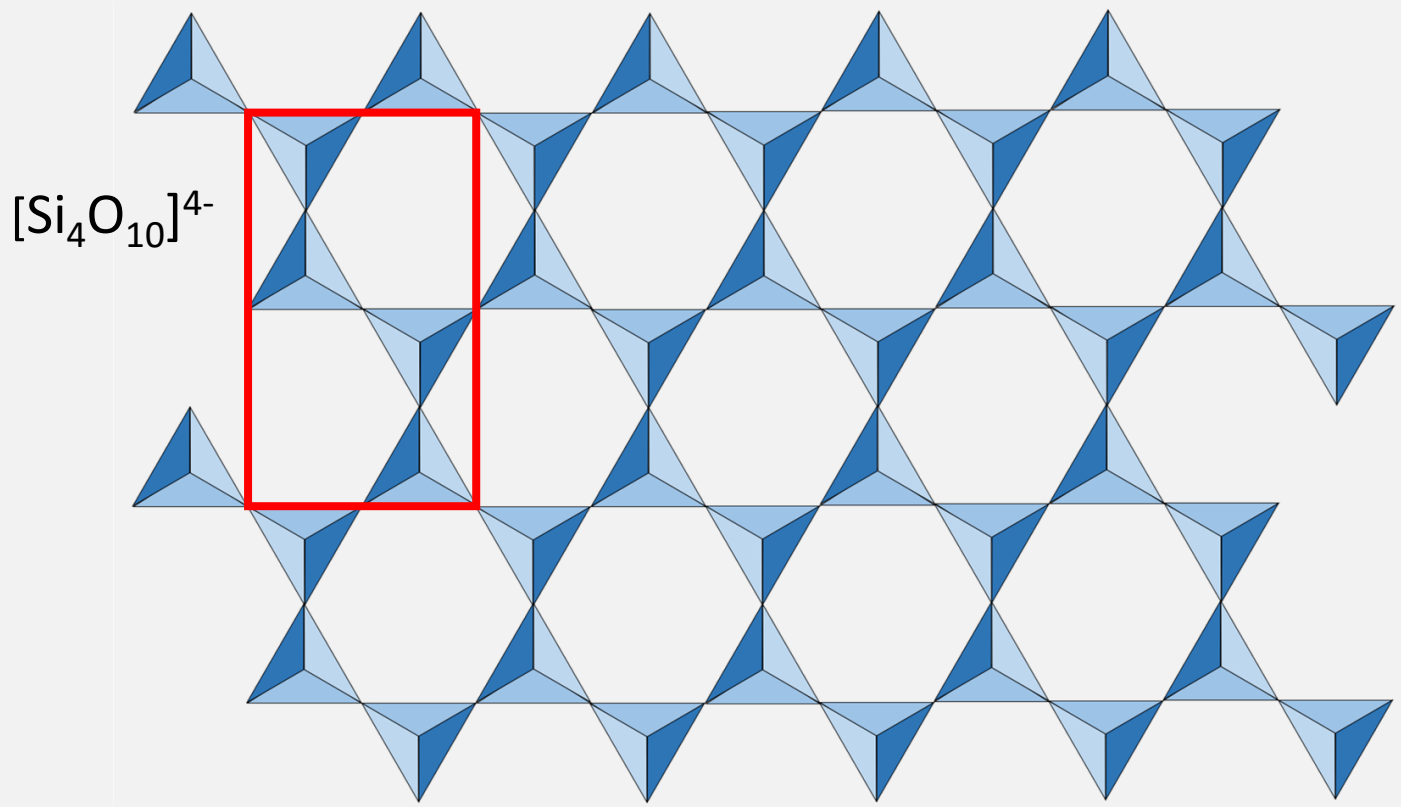
- Interference colours up to mid second order
- Elongate sections show inclined extinction with respect to the cleavage
- Extinction angle depends on composition and is usually ~15-25°
- Sometimes twinned

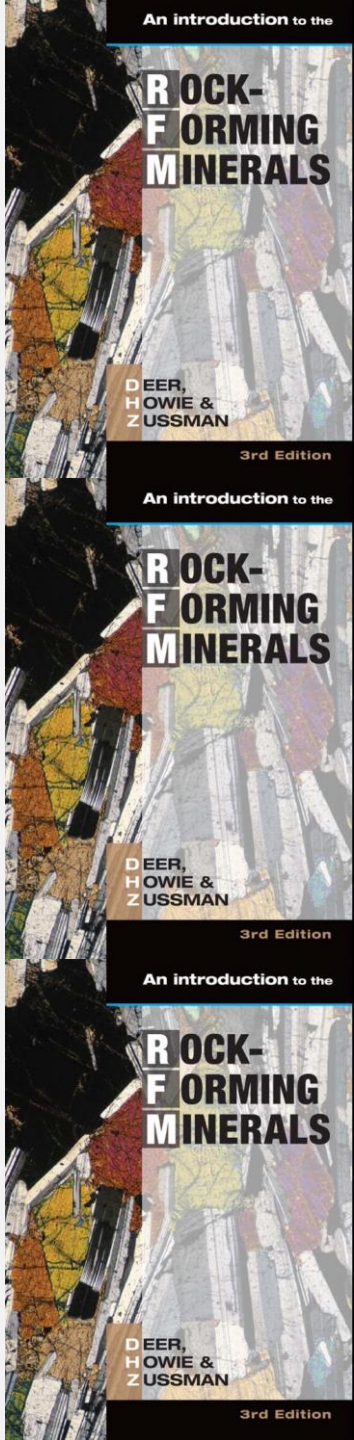


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# Micas

$AB_{2-3}Si_4O_{10}(OH,F)_2$ sheet silicates monoclinic or triclinic

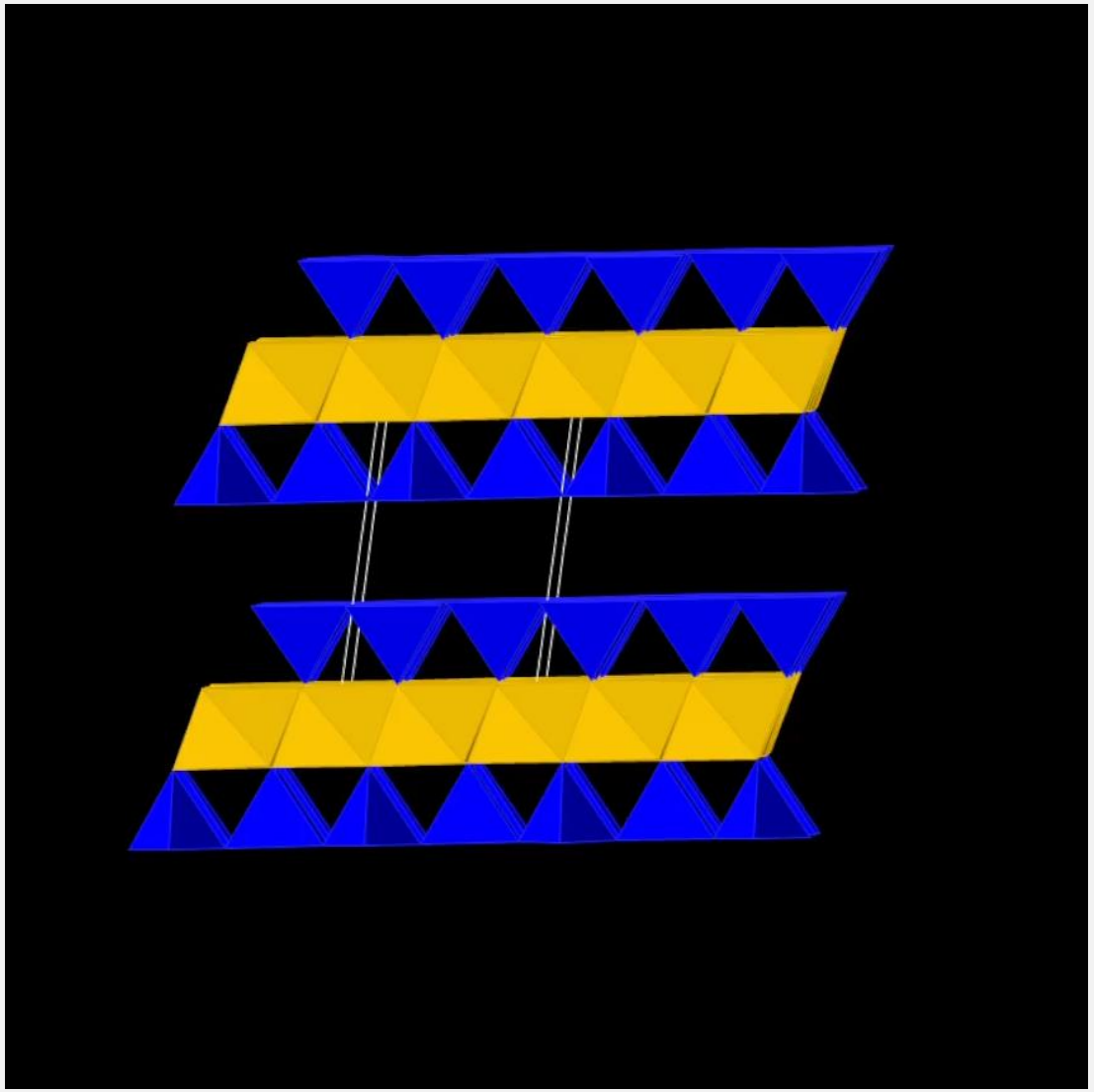


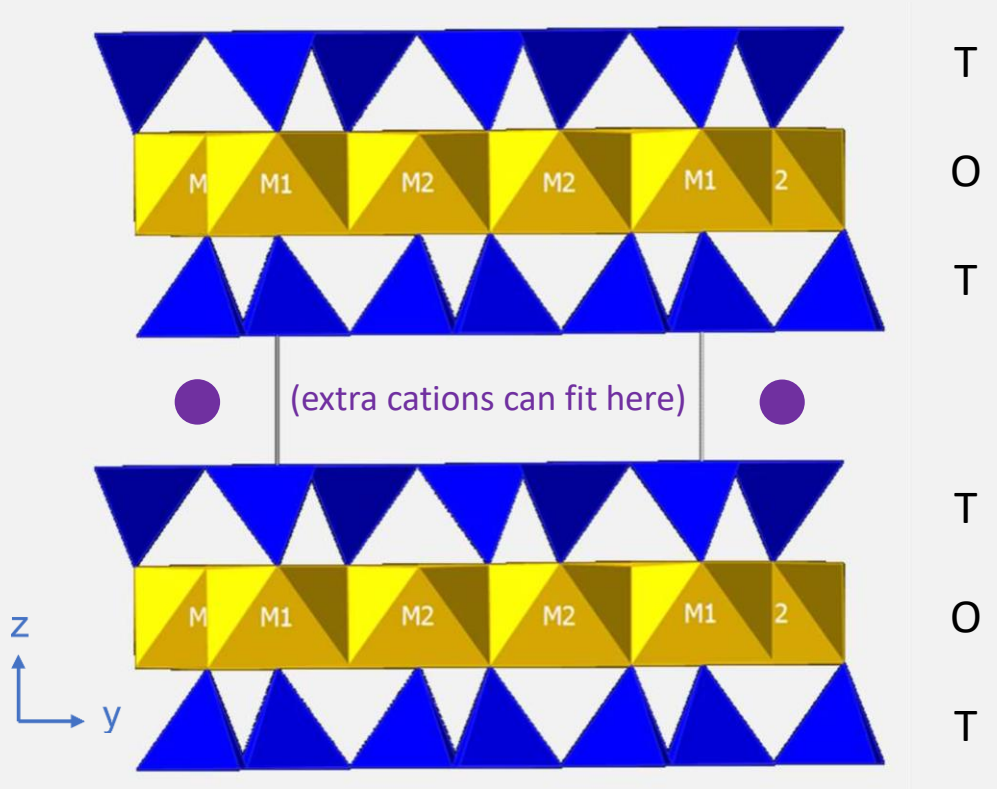






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# Micas

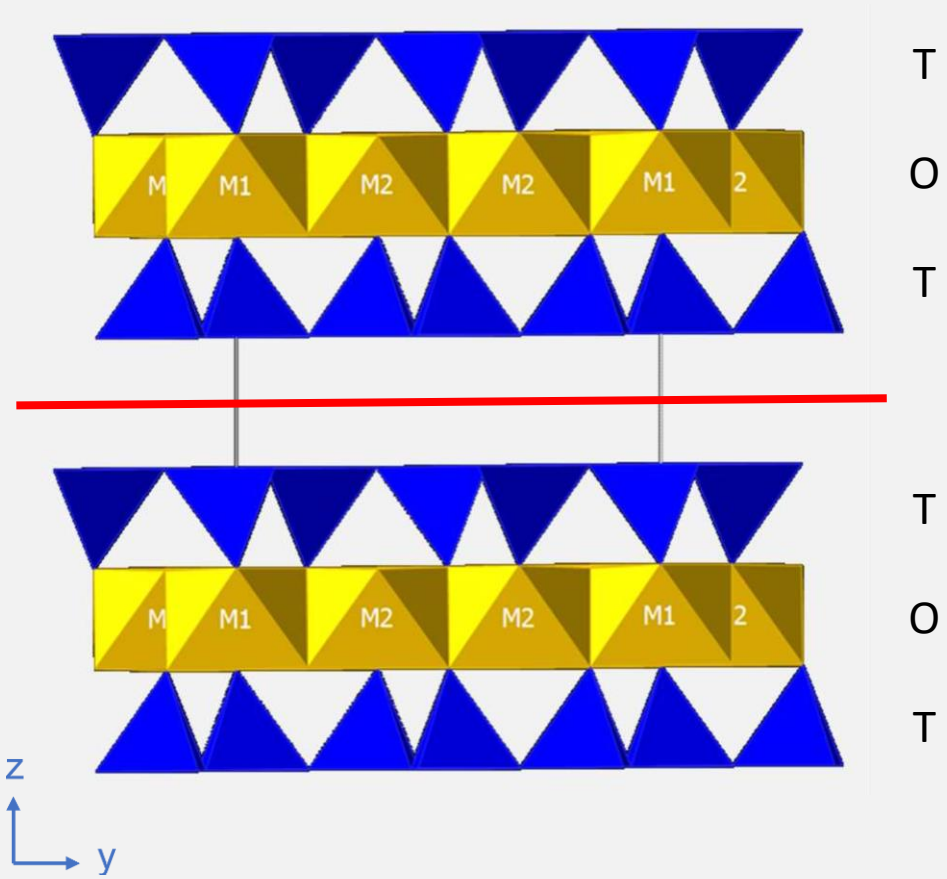




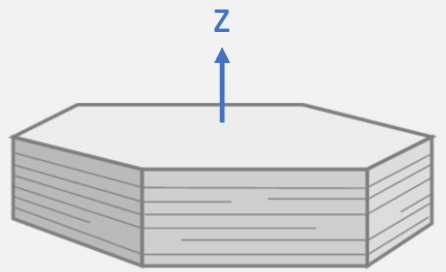

Micas and clay minerals are made by stacking T and O layers in different orders and combinations, with or without an extra layer of cations behind the T sheets


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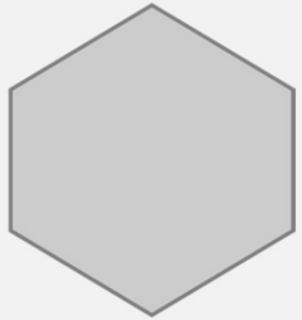
# Micas: cleavage




1 perfect basal cleavage




3D shape



2D cross-section ⊥ z



2D cross-section // z



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# Biotite (and phlogopite)

$K_2(Mg,Fe)_6[Al_2Si_6O_{20}](OH)_4$ sheet silicate monoclinic (near-hexagonal)





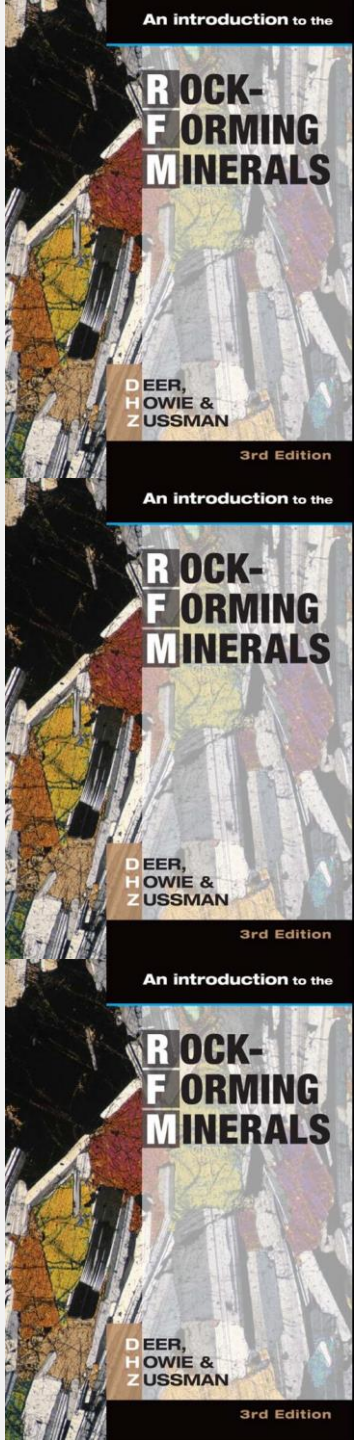
Biotite found in:

Intermediate igneous rocks
- Andesite
- Diorite

Felsic igneous rocks
- Granite and granodiorite
- Rhyolite and dacite

Phlogopite found in:

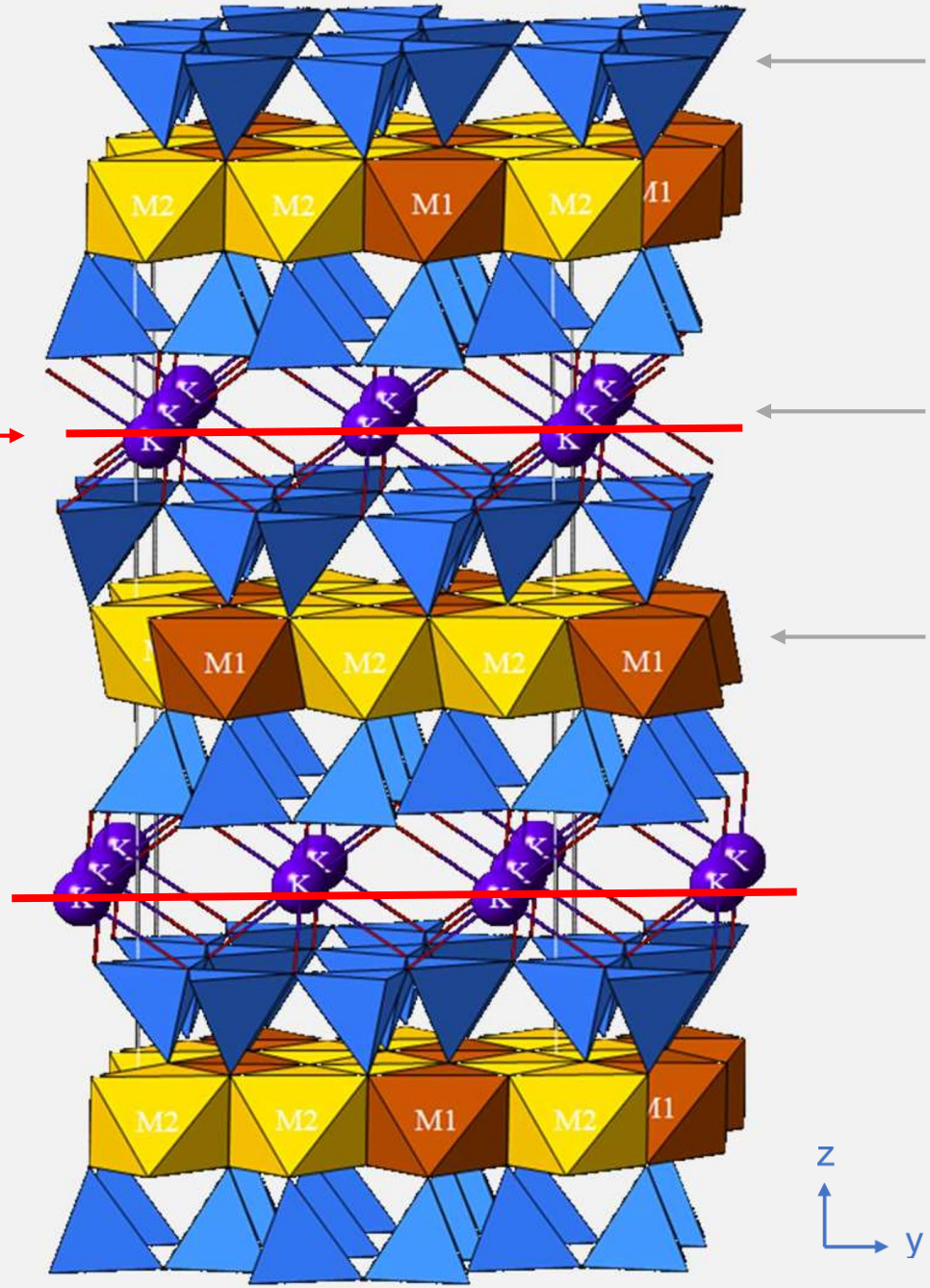
Mantle rocks
- Lherzolite






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# Biotite




cleavage plane

Layer of silica tetrahedra

Some $Si^{4+} \leftrightarrow Al^{3+}$ substitution

Layer of $K^+$ cations

Layer of $M^{2+}$ cations

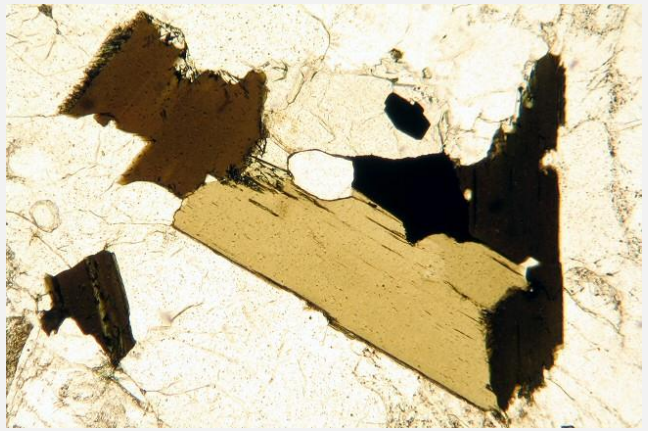
$Mg^{2+}$, $Fe^{2+}$

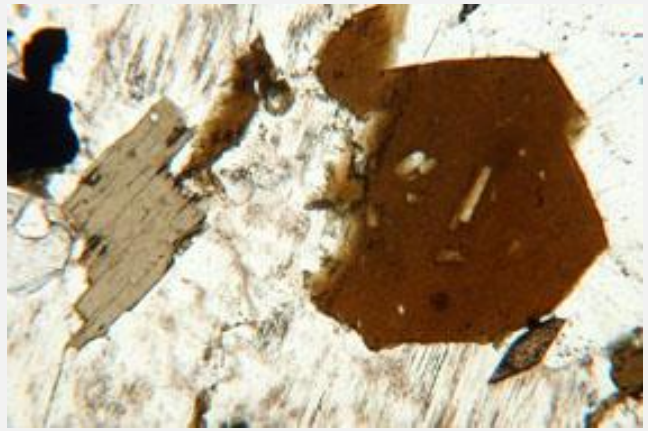


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# Biotite (and phlogopite)

Distinguish from hornblende by its distinctive brown-yellow pleochroism, one perfect cleavage, and straight extinction

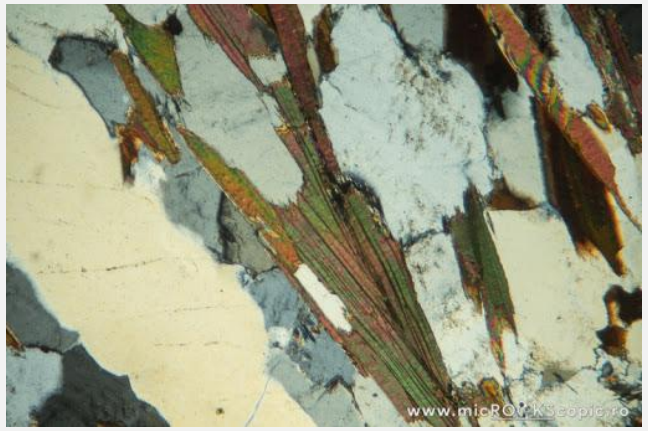


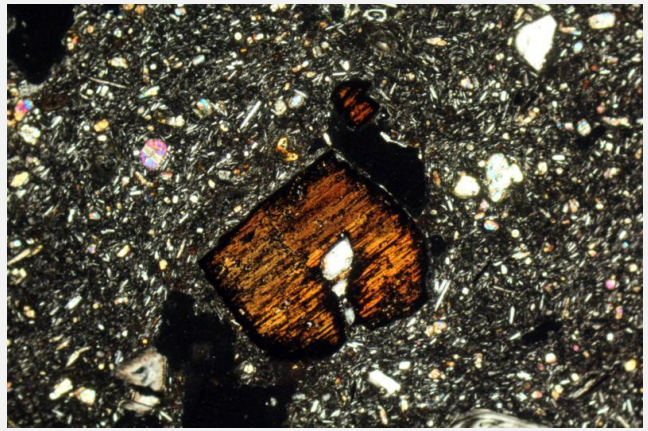


Key optical properties:

PPL

- Moderate relief
- Strongly pleochroic: dark brown to straw yellow
- One perfect cleavage
- In granites, may see pleochroic haloes
- May show alteration to chlorite



XP

- Interference colours up to mid third order: pale pinks and greens
- Straight, speckled extinction



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# Muscovite

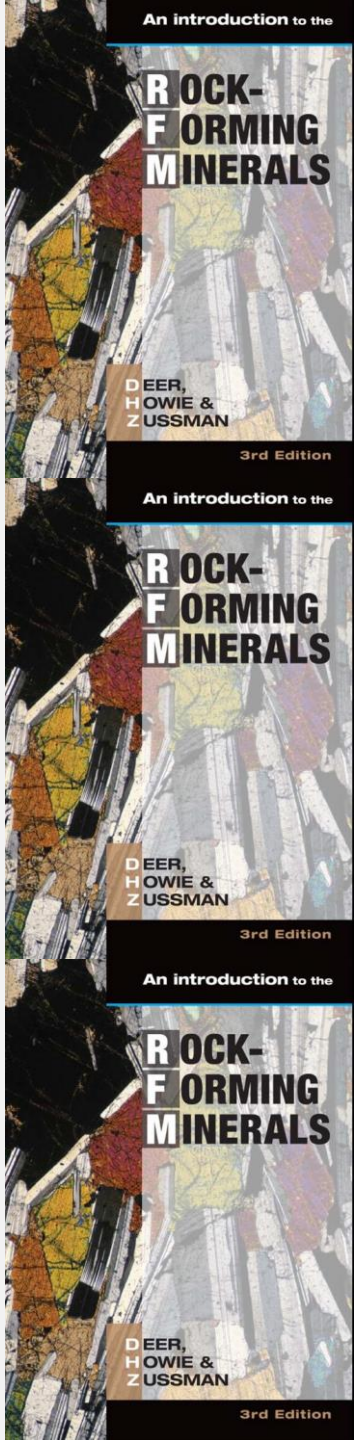
$K_4Al_4[Al_2Si_6O_{20}](OH,F)_4$ sheet silicate monoclinic







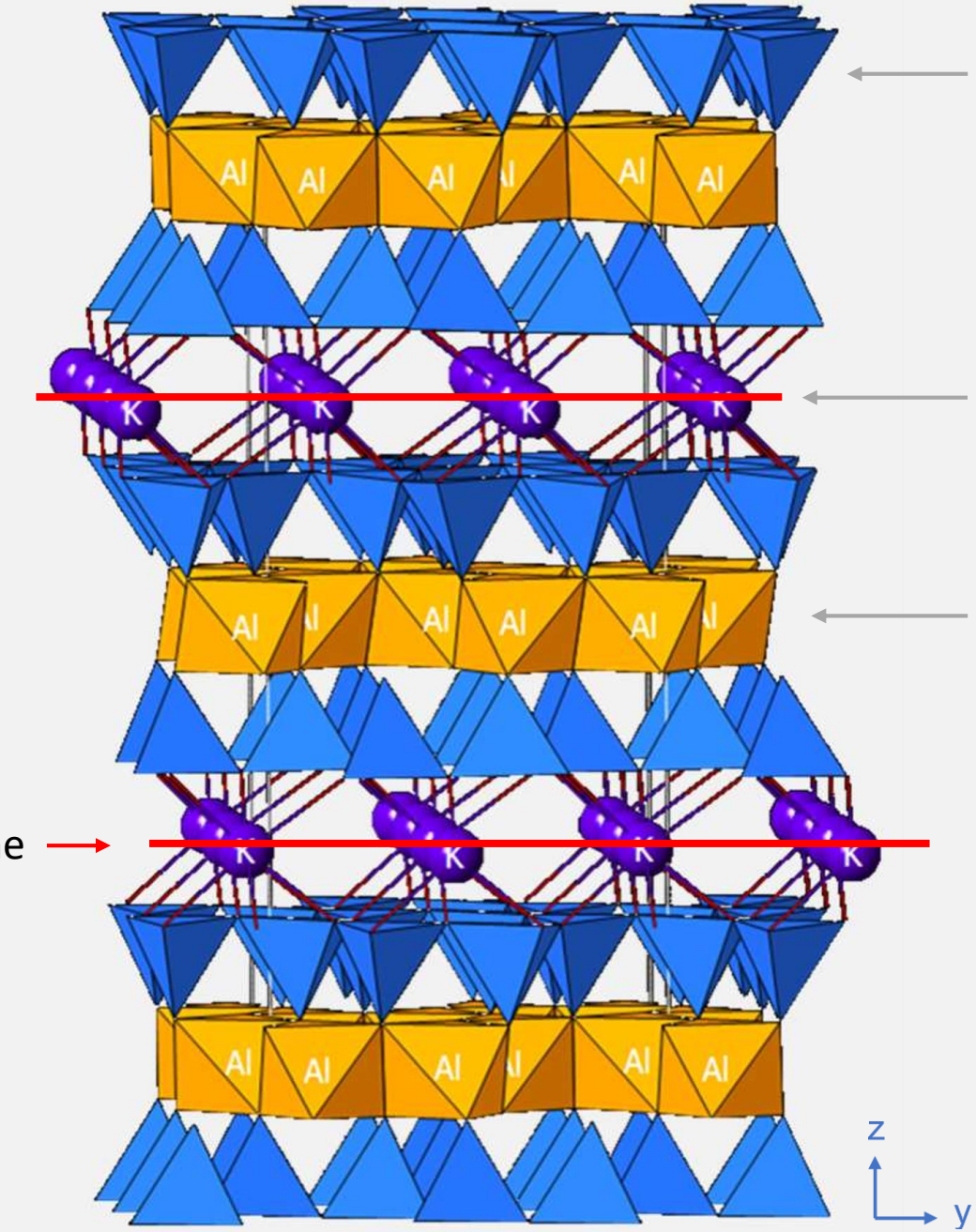
Felsic igneous rocks
- S-type granites and granodiorites






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# Muscovite




Layer of silica tetrahedra

Some $Si^{4+} \leftrightarrow Al^{3+}$ substitution

Layer of $K^+$ cations

Layer of $Al^{3+}$ cations

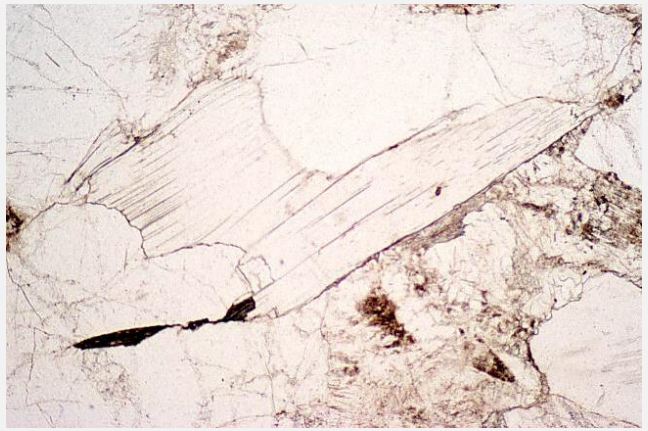
cleavage plane

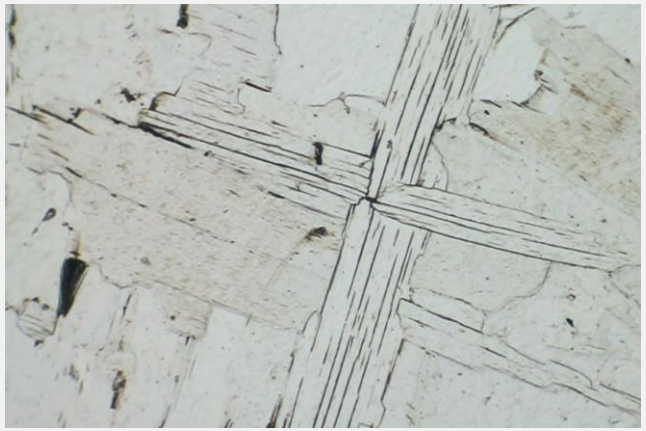


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# Muscovite

Distinguish from quartz and feldspars by its one perfect cleavage, straight extinction, and bright interference colours

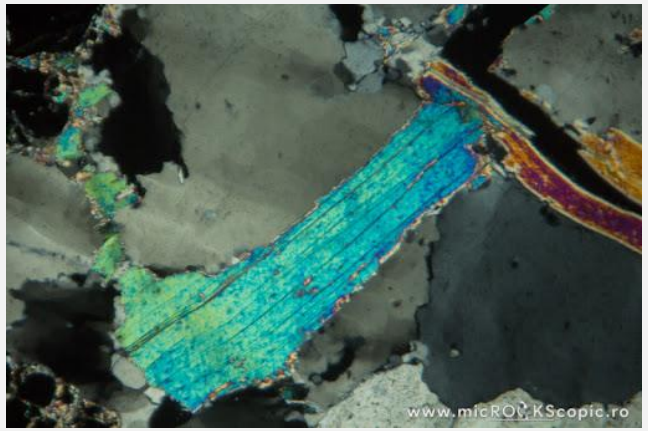


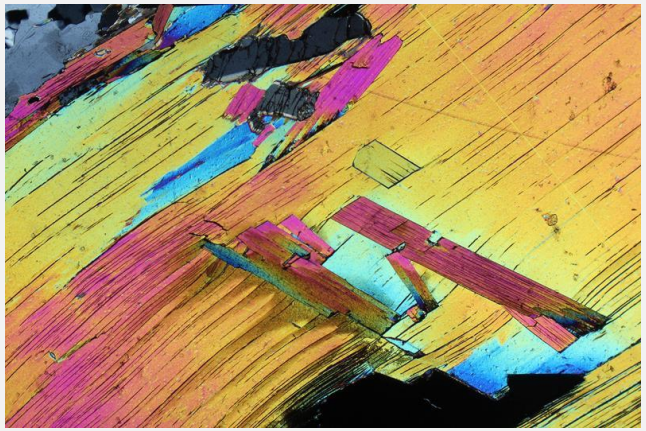


Key optical properties:

PPL

- Moderate relief
- Colourless
- One perfect cleavage



XP

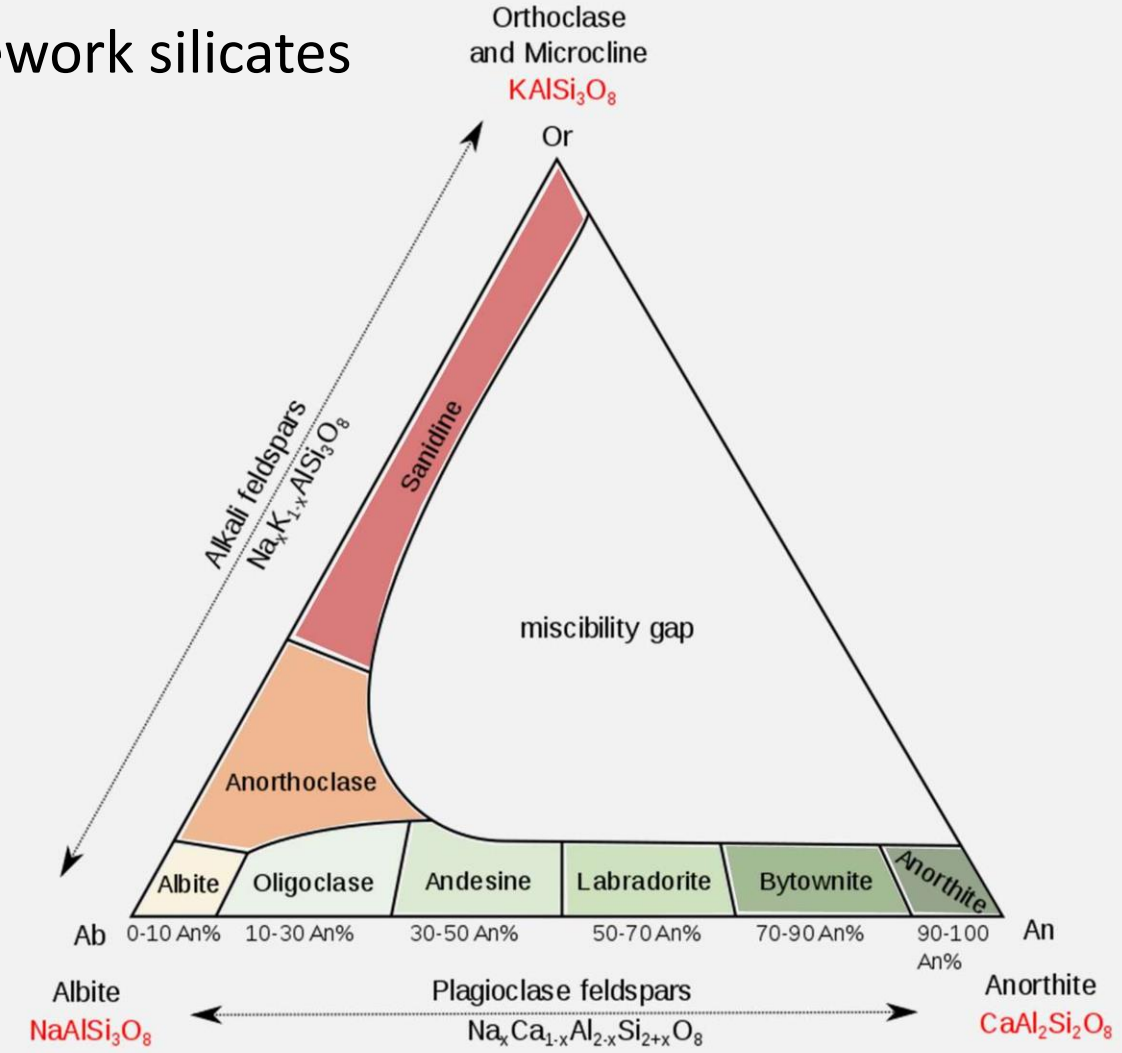
- Interference colours up to low third order: often blue or orange
- Straight extinction



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# Feldspars

framework silicates




monoclinic or triclinic

Plagioclase feldspars found in:

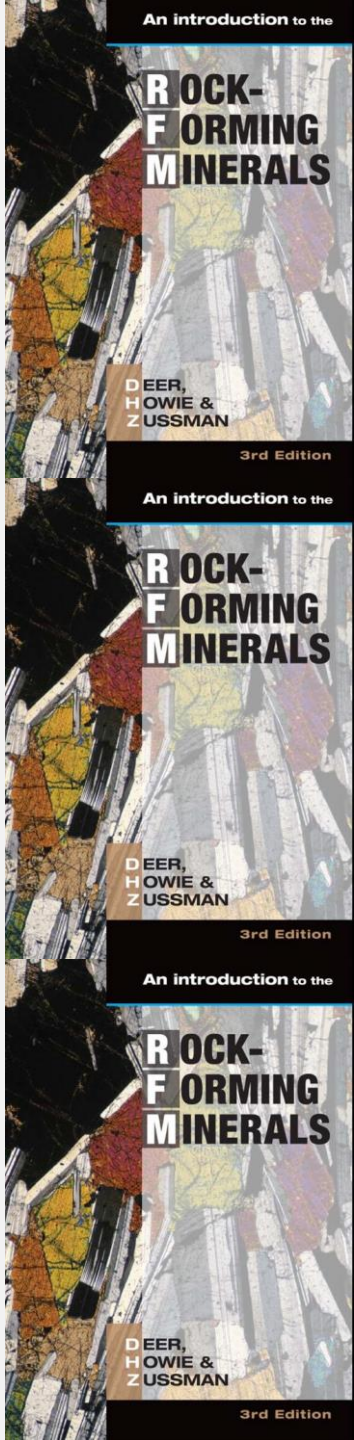
- All common mafic, intermediate and felsic igneous rocks

NOT found in mantle rocks

Alkali feldspars found in:

Felsic igneous rocks

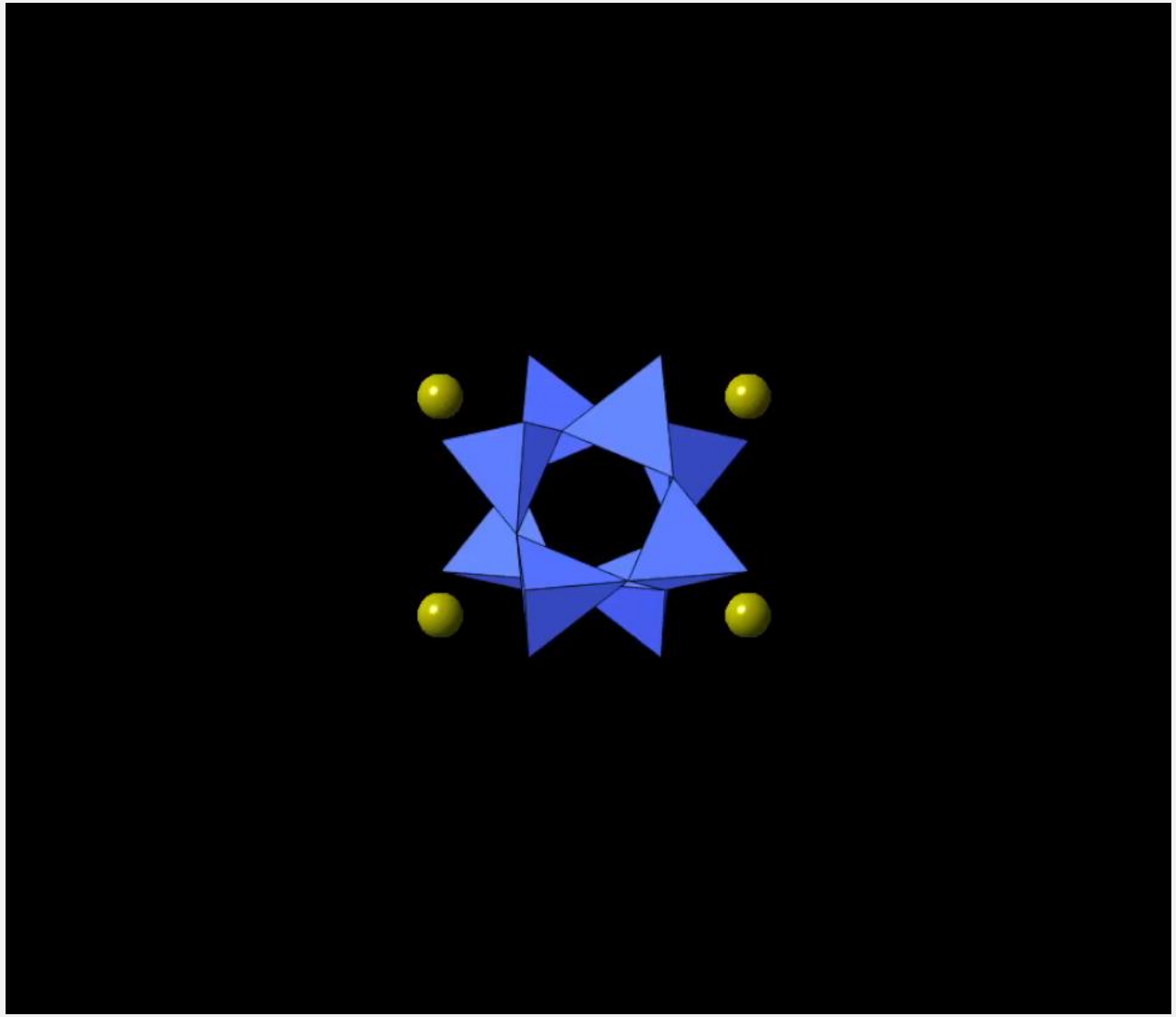
- Granite and granodiorite
- Rhyolite and dacite





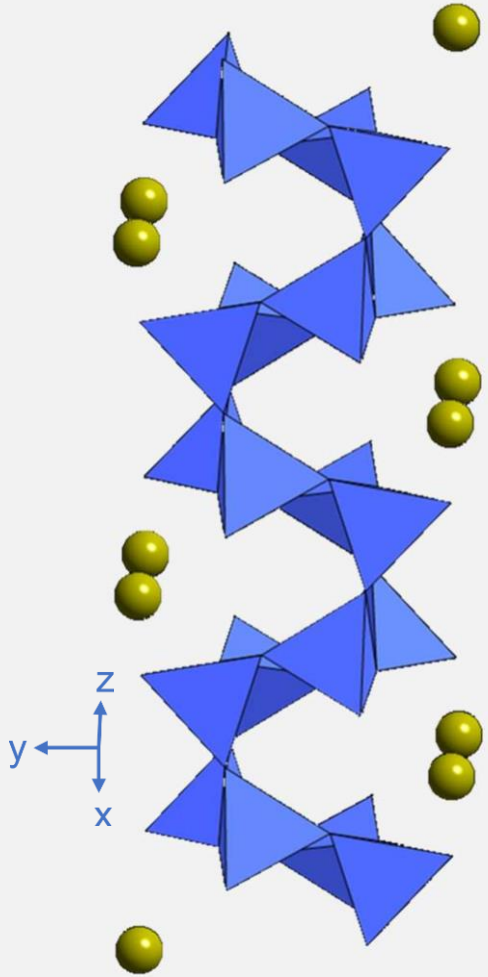

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# Feldspars



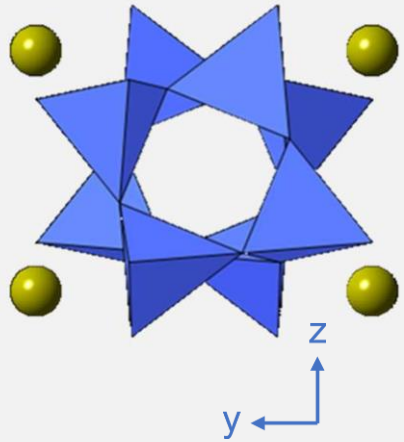
## Crankshaft chain

side view




Feldspar framework is made of linked chains of corner-sharing tetrahedra

end-on view






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# Alkali feldspar

$(K,Na)AlSi_3O_8$ framework silicate monoclinic or triclinic

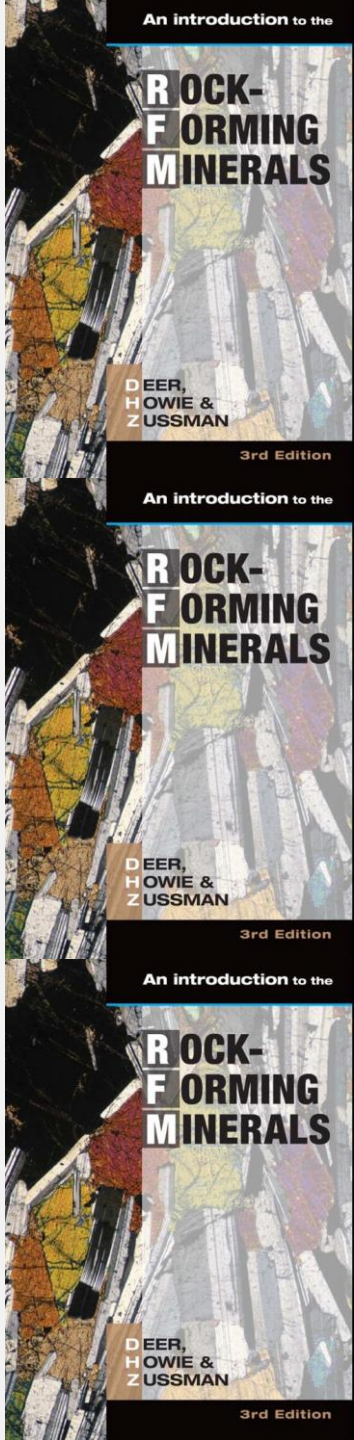






Found in:

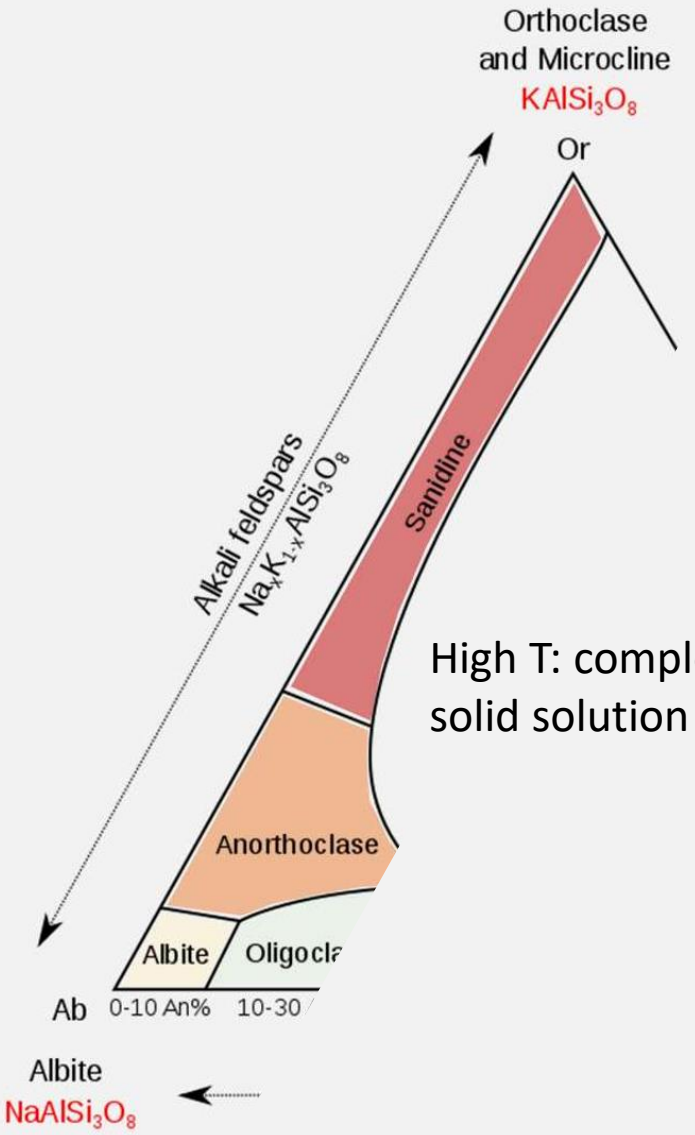
Felsic igneous rocks
- Granite and granodiorite
- Rhyolite and dacite



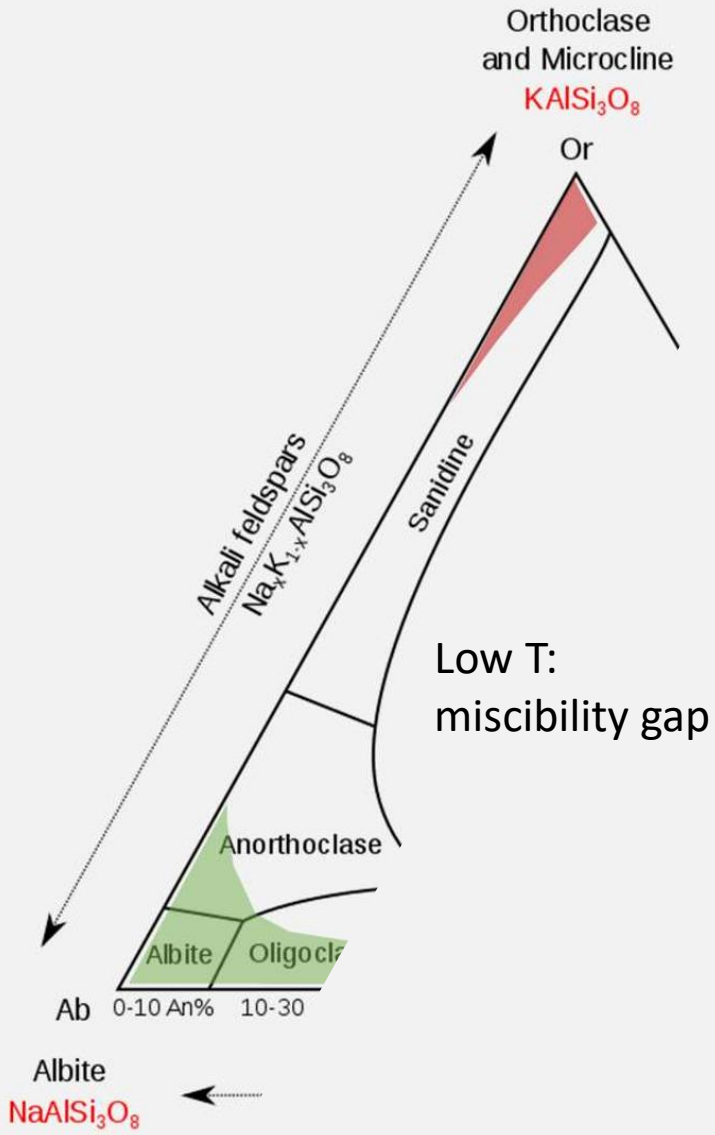



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# Alkali feldspar compositions and solid solution




High T: complete solid solution




Low T: miscibility gap

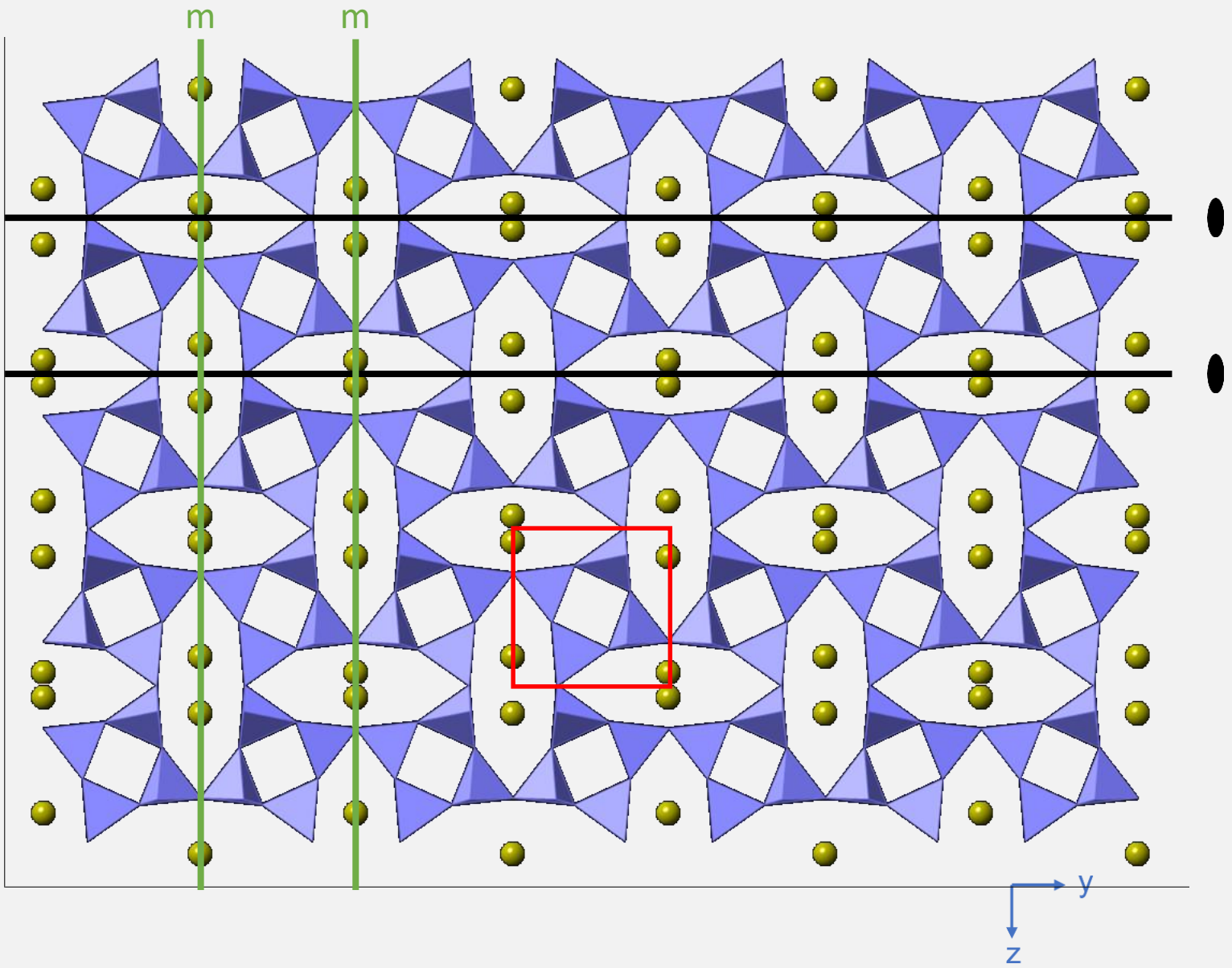
Miscibility gap exists at low temperature, due to the large size difference between $K^+$ and $Na^+$ cations

| | |
|---|---|
| $K^+$ | 0.155 nm |
| $Na^+$ | 0.132 nm |


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# Alkali feldspar structure at high T

# Sanidine ($KAlSi_3O_8$) structure at low T




Monoclinic symmetry: a mirror plane perpendicular to a diad

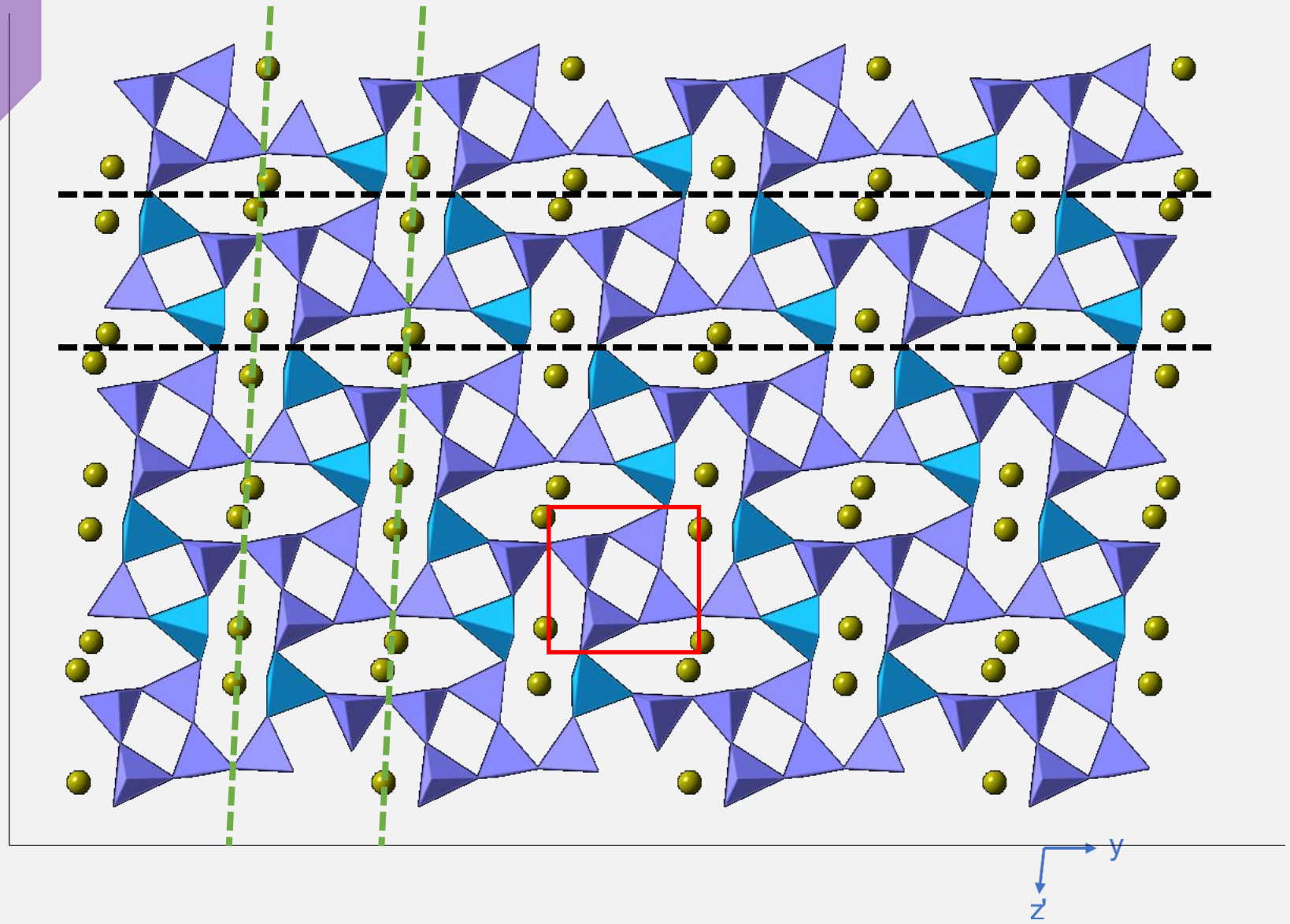
At high temperature: complete solid solution of Na and K on cation sites

If K-rich alkali feldspars are cooled quickly, they retain the monoclinic structure



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# Albite ($NaAlSi_3O_8$) structure at low T



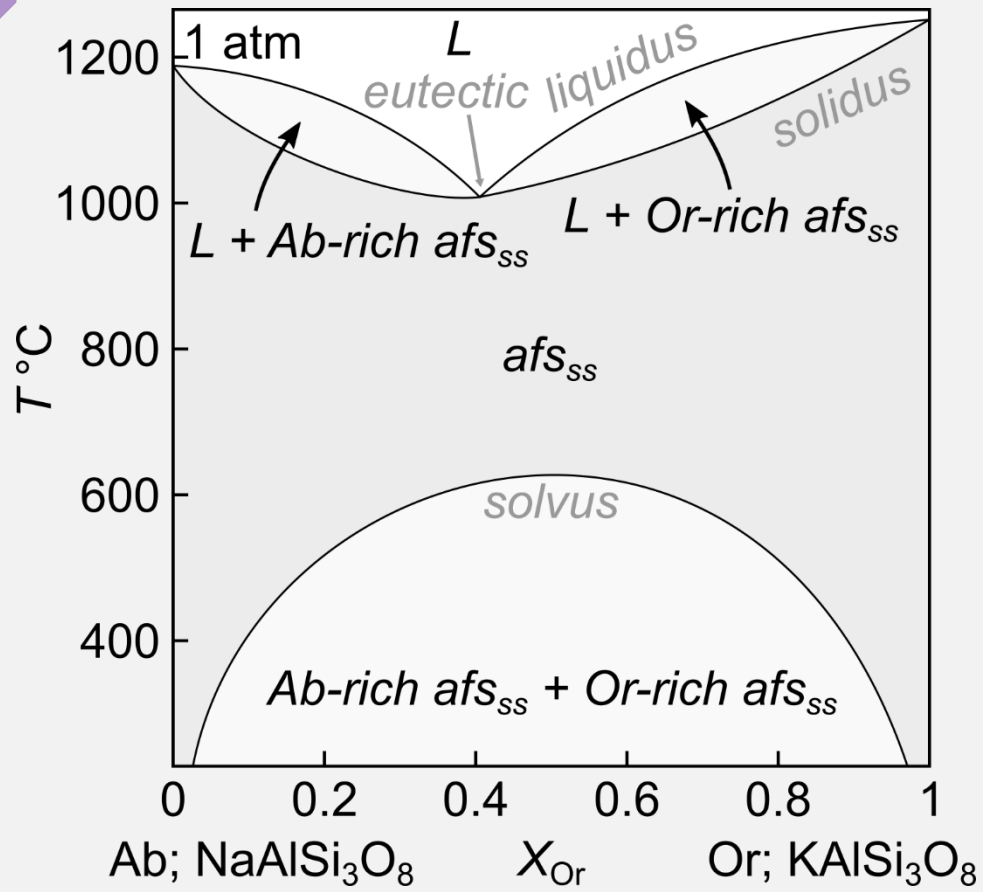

Triclinic symmetry: diad and mirror plane are lost. Cation site shapes are distorted.

In Na-rich alkali feldspars, the monoclinic structure collapses at low temperature

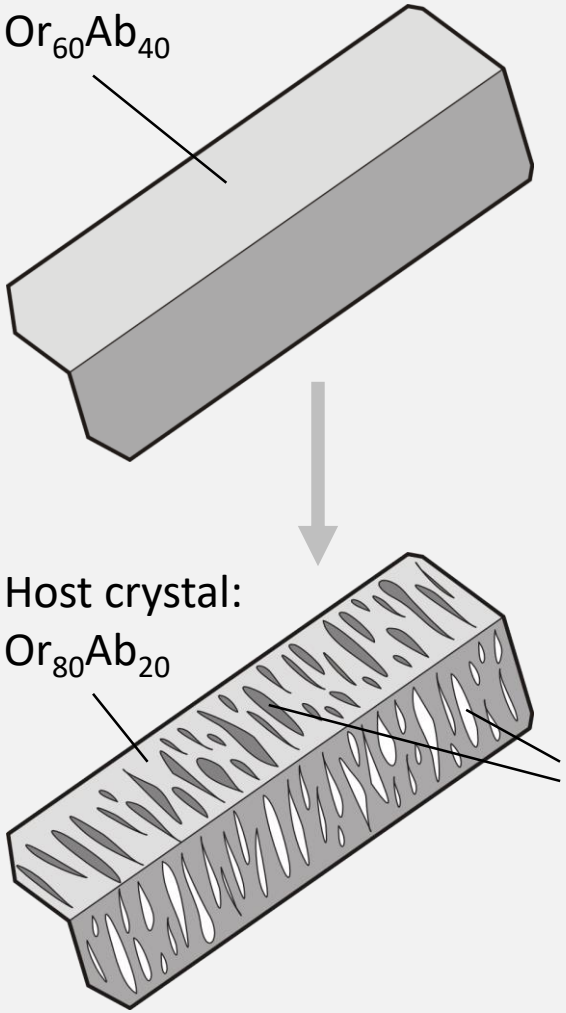

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# Alkali feldspars: exsolution

Exsolution in alkali feldspars: see EART 26201 Principles of Geochemistry, Lecture 4




$Na^+ \leftrightarrow K^+$

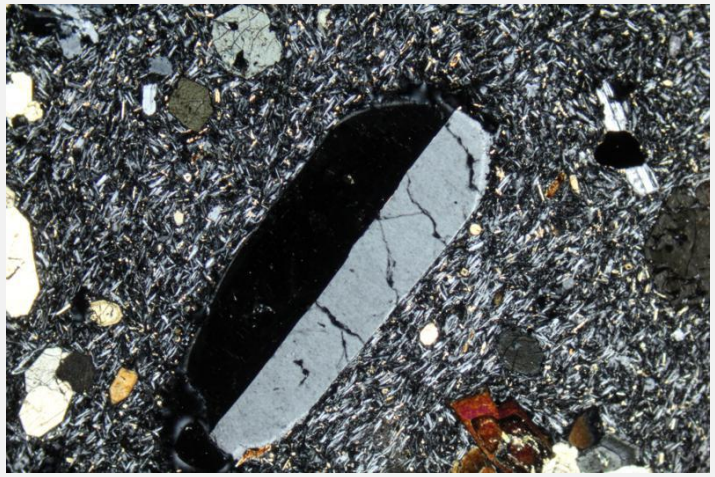




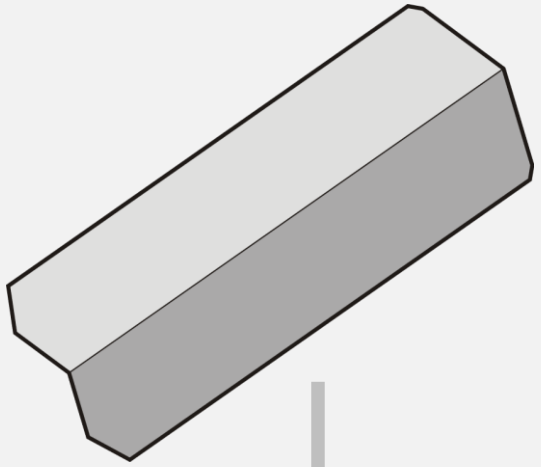
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# Alkali feldspars: exsolution

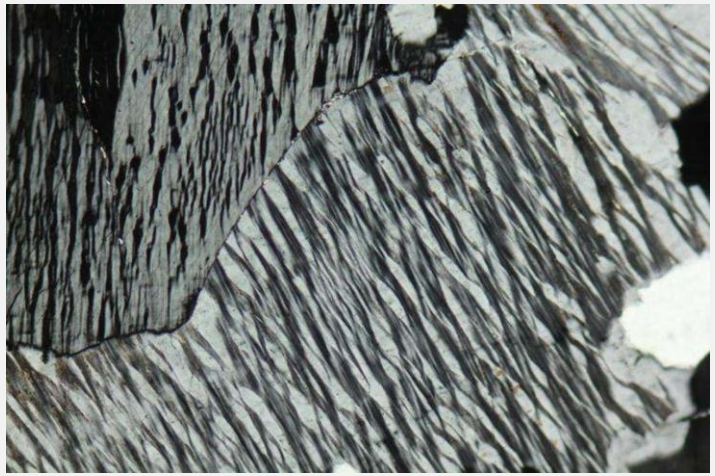
Exsolution in alkali feldspars: see EART 26201 Principles of Geochemistry, Lecture 4



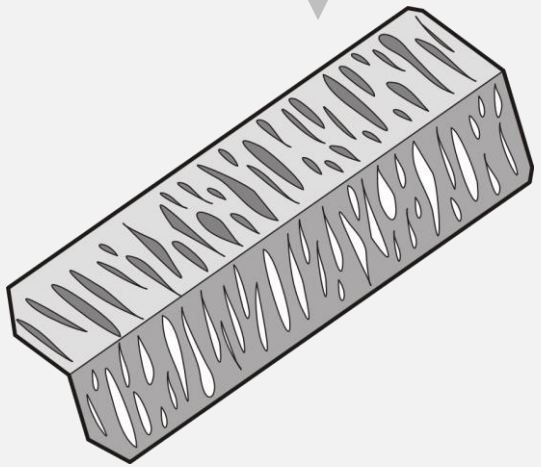
One simple twin, along the length of the crystal. Found in rapidly cooled volcanic rocks, such as rhyolites.



Sanidine



Perthitic texture. Common in plutonic rocks, such as granites. The thickness of the lamellae is proportional to the cooling rate.



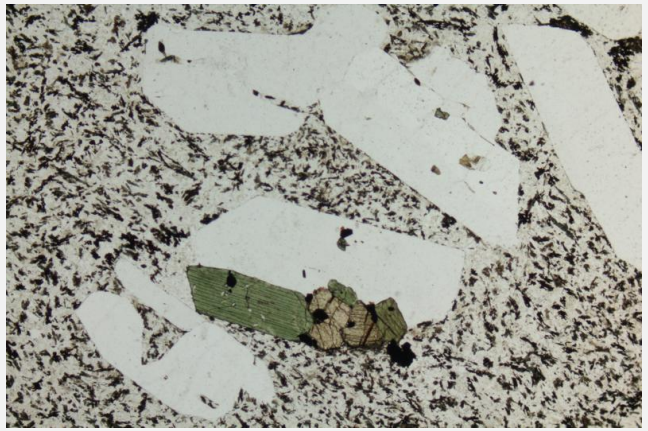
Orthoclase with perthitic texture

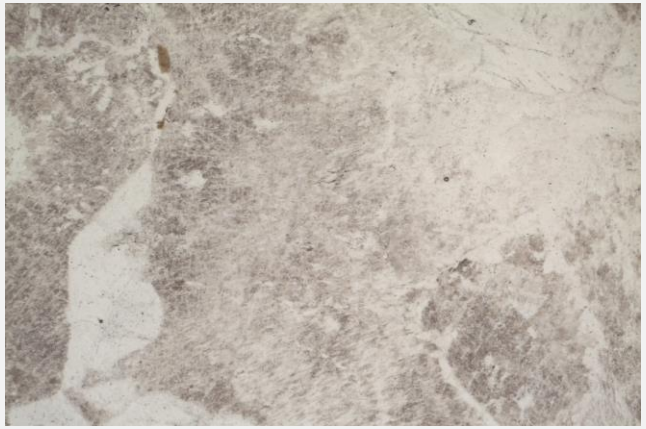


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# Alkali feldspars

Distinguish from plagioclase by either simple or cross-hatch twinning (not lamellar twinning), and/or perthitic texture (if visible)



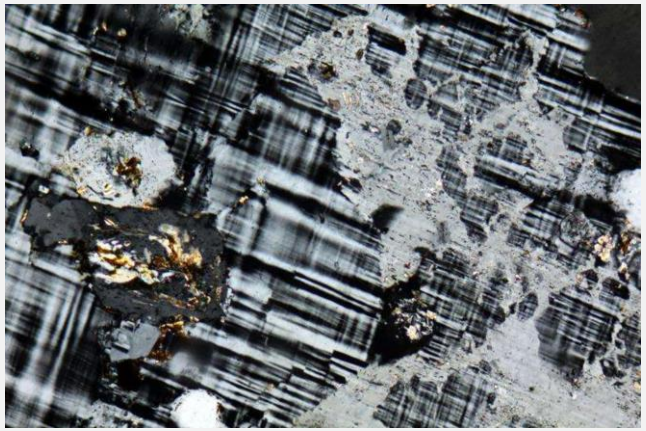


Key optical properties:

PPL

- Very low relief
- Colourless
- Cleavage traces very rarely visible
- May see cloudy alteration to clay minerals





XP

- Low first-order interference colours
- Twinning
  - Sanidine and orthoclase: simple twins
  - Microcline: multiple finely spaced cross-hatch twins at ~90°
- Often see perthitic texture in orthoclase and microcline

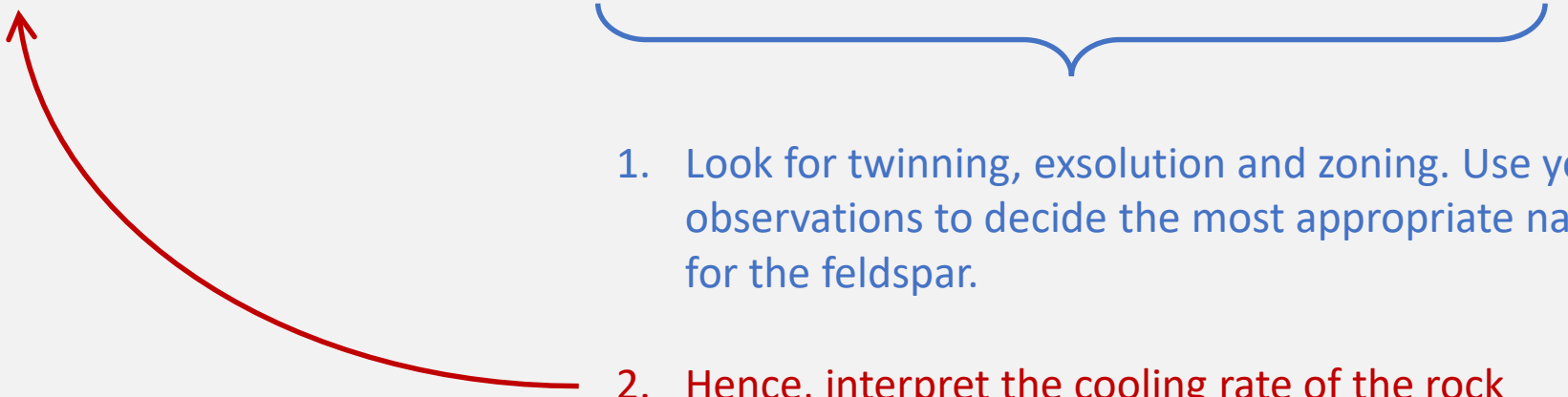


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# Which alkali feldspar is it? What should I call it in my rock description?

It is often possible to give a more detailed mineral identification than just “alkali feldspar” based on the type of twinning, presence/absence of perthitic exsolution, and presence/absence of zoning.

| | Cooling rate | Symmetry | Twinning | Perthite lamellae | Zoning |
|---|---|---|---|---|---|
| Sanidine | fast | monoclinic | simple | no | rare |
| Orthoclase | intermediate | monoclinic | simple | yes | possible |
| Microcline | very slow | triclinic | cross-hatch | yes | possible |



1. Look for twinning, exsolution and zoning. Use your observations to decide the most appropriate name for the feldspar.

2. Hence, interpret the cooling rate of the rock



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# Plagioclase

$(Na,Ca)(Al,Si)Si_2O_8$

framework silicate

triclinic



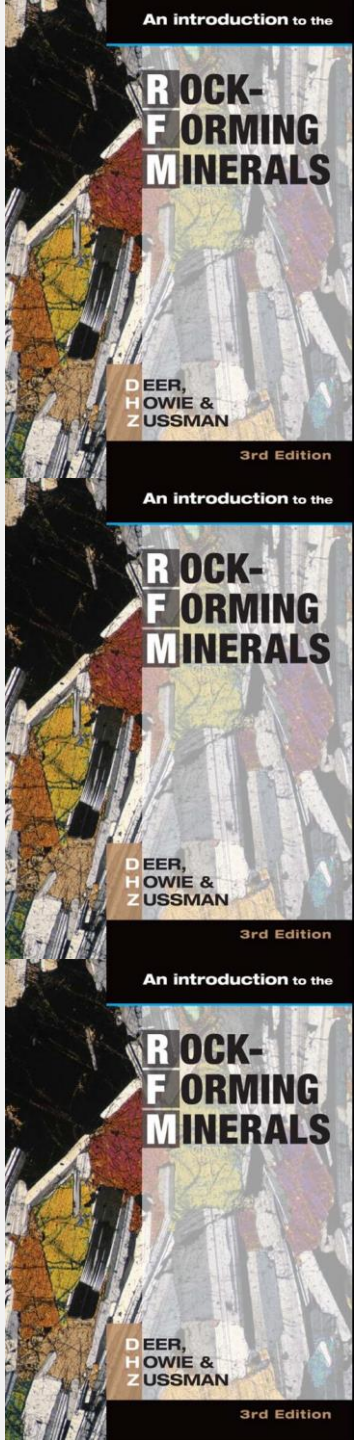






Common in all types of mafic, intermediate and felsic igneous rocks

NOT found in mantle rocks

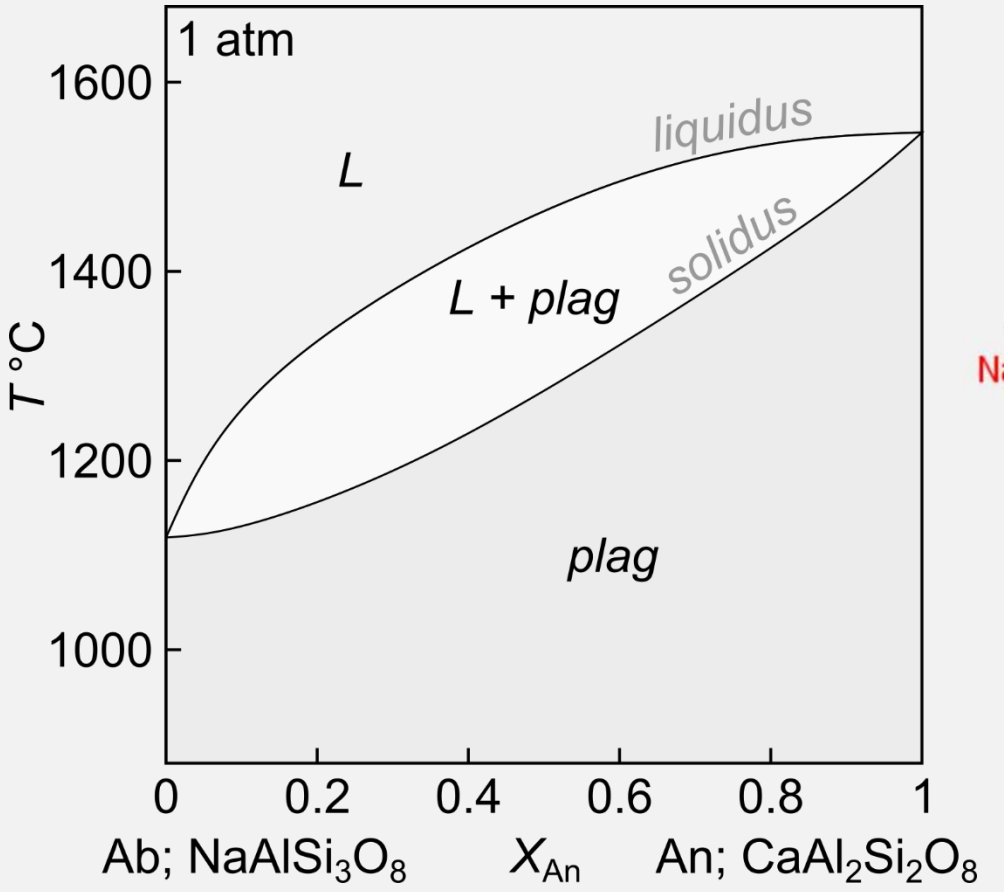


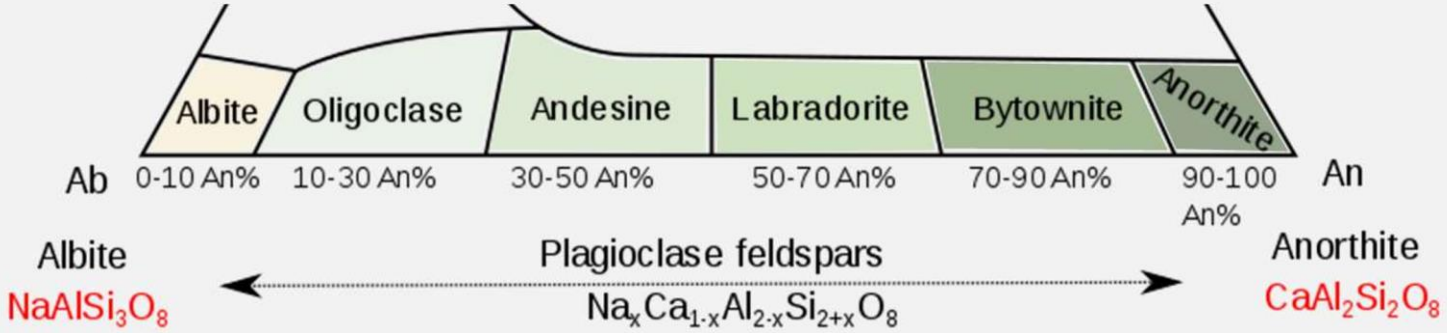



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# Plagioclase phase diagram and solid solution

Plagioclase phase diagram: See EART 26201 Principles of Geochemistry, Lecture 3




Complete solid solution from albite to anorthite.
Plagioclase never shows any exsolution textures in thin section.

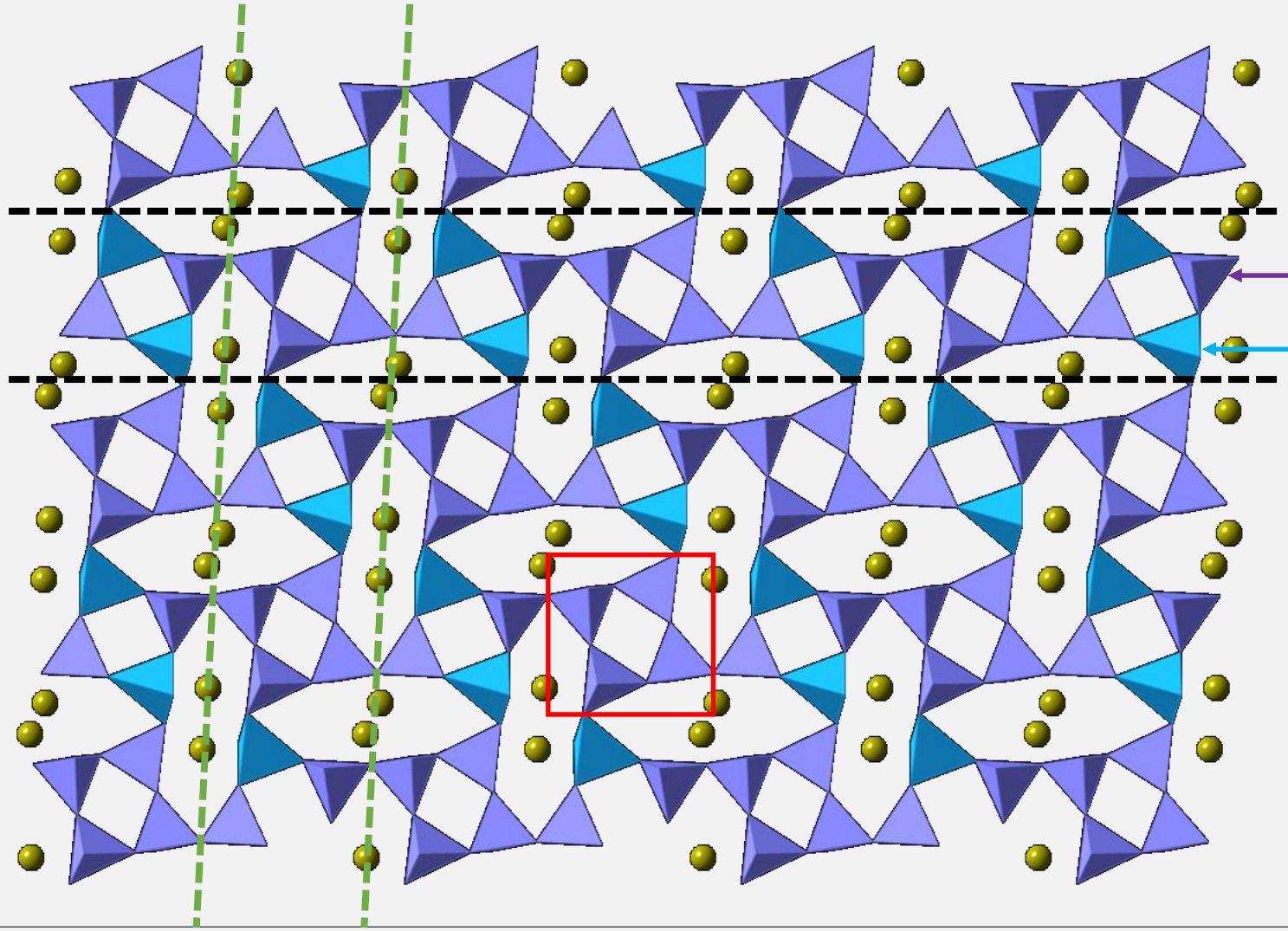
The composition of plagioclase can be measured from a thin section, using the extinction angle

$[NaSi]^{5+} \leftrightarrow [CaAl]^{5+}$

Measuring plagioclase composition in a thin section: see Week 6 and Practical Reference Handout


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# Albite (Na-feldspar) structure



Triclinic symmetry

silica tetrahedra

alumina tetrahedra

Albite formula: $NaAlSi_3O_8$

Ratio of Al:Si is 1:3

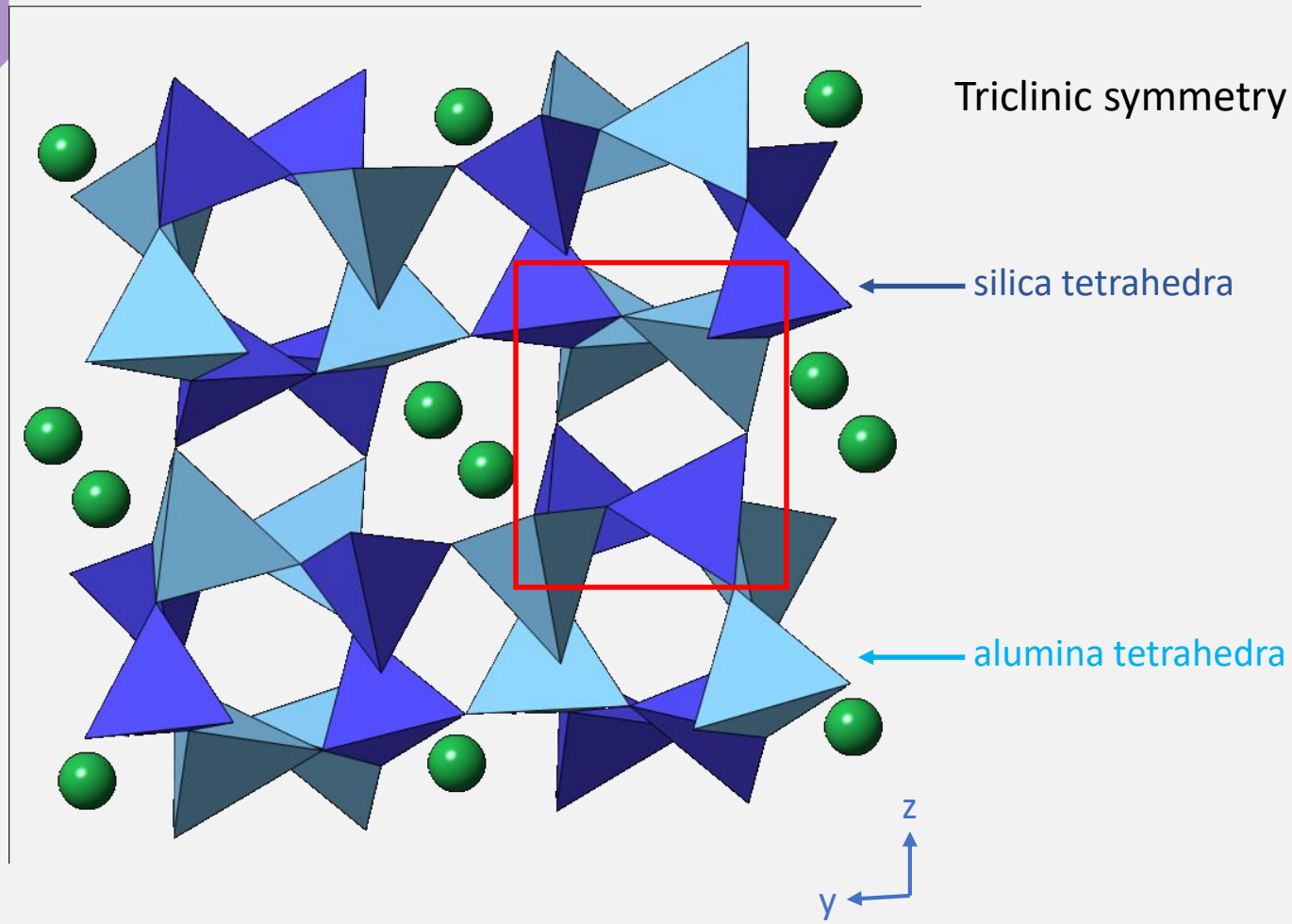
y

z



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# Anorthite (Ca-feldspar) structure




Triclinic symmetry

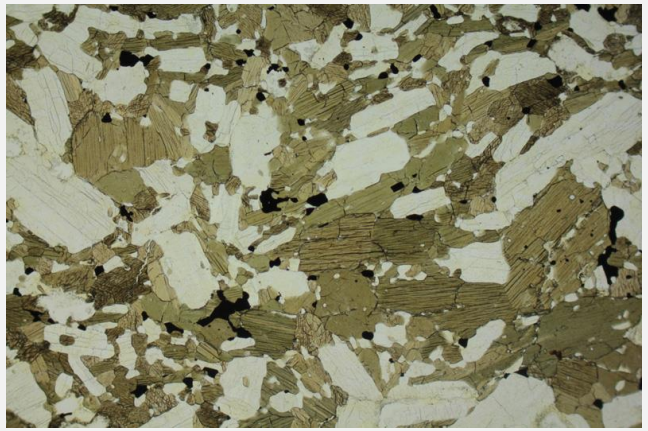
Anorthite formula: $CaAl_2Si_2O_8$

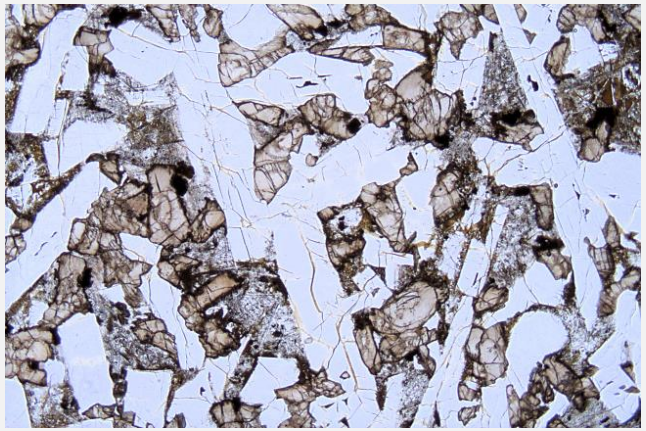
Ratio of Al:Si is 2:2


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# Plagioclase

Distinguish from alkali feldspar and quartz by the distinctive multiple lamellar twinning

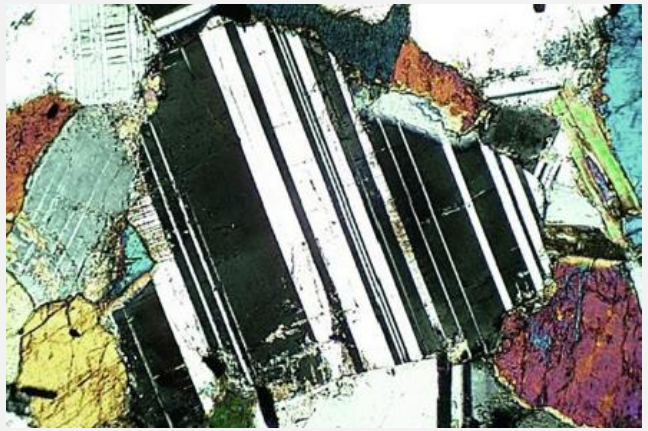


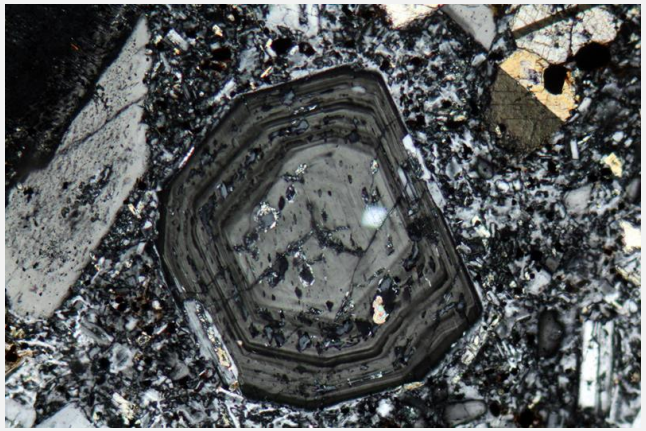


Key optical properties:

PPL

- Very low relief
- Colourless
- Often has tabular habit
- Cleavage traces very rarely visible
- May see cloudy alteration to sericite





XP

- Low first-order interference colours
- Multiple lamellar twinning
- Sometimes zoned
- Never shows exsolution textures – no perthite lamellae



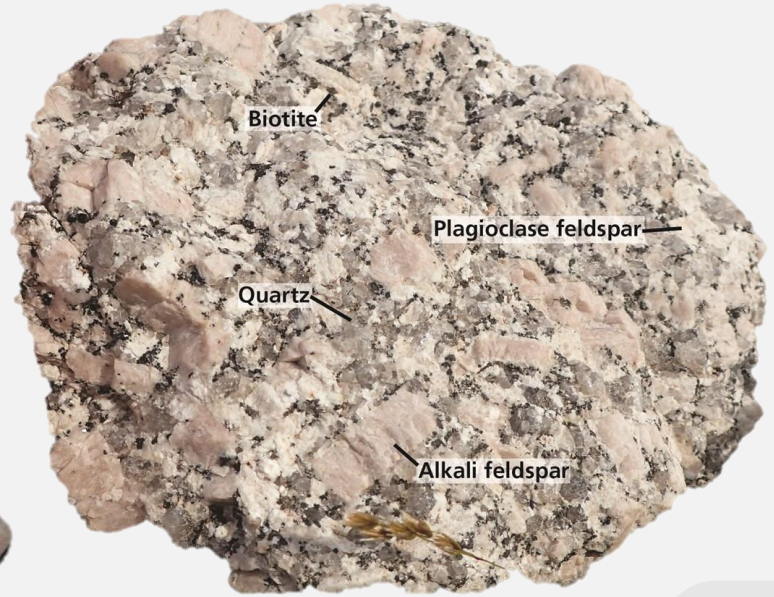
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# Quartz

$SiO_2$ framework silicate trigonal

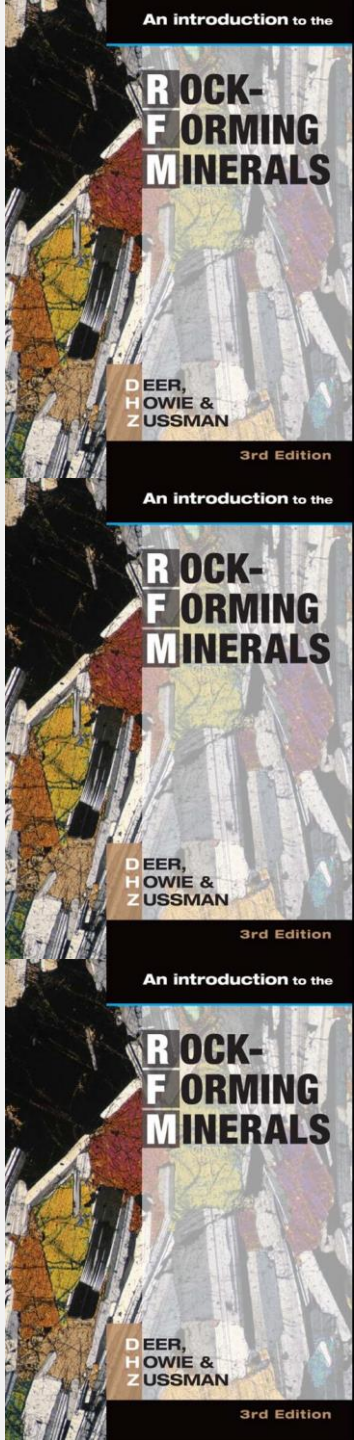







Found in:

Felsic igneous rocks
- Granite and granodiorite
- Rhyolite and dacite

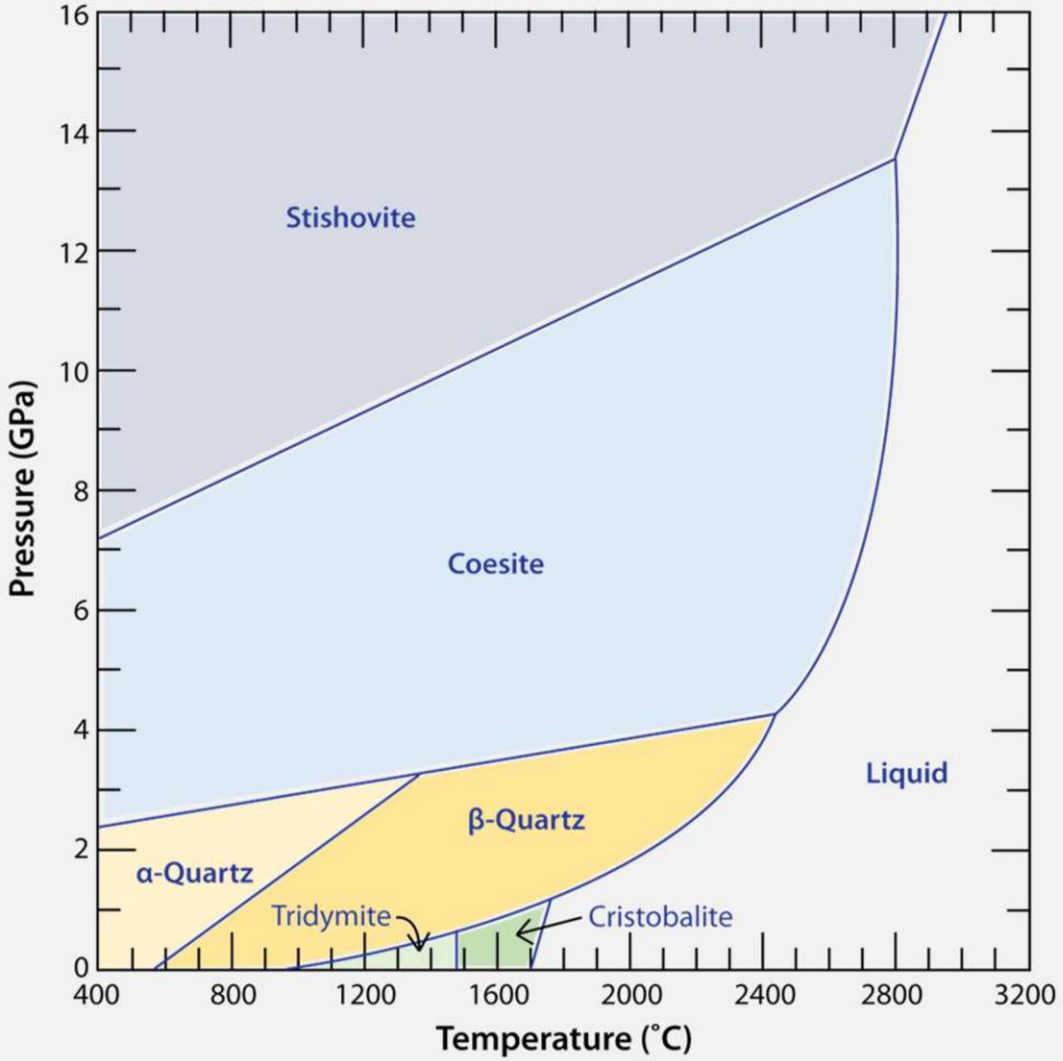





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# The $SiO_2$ polymorphs

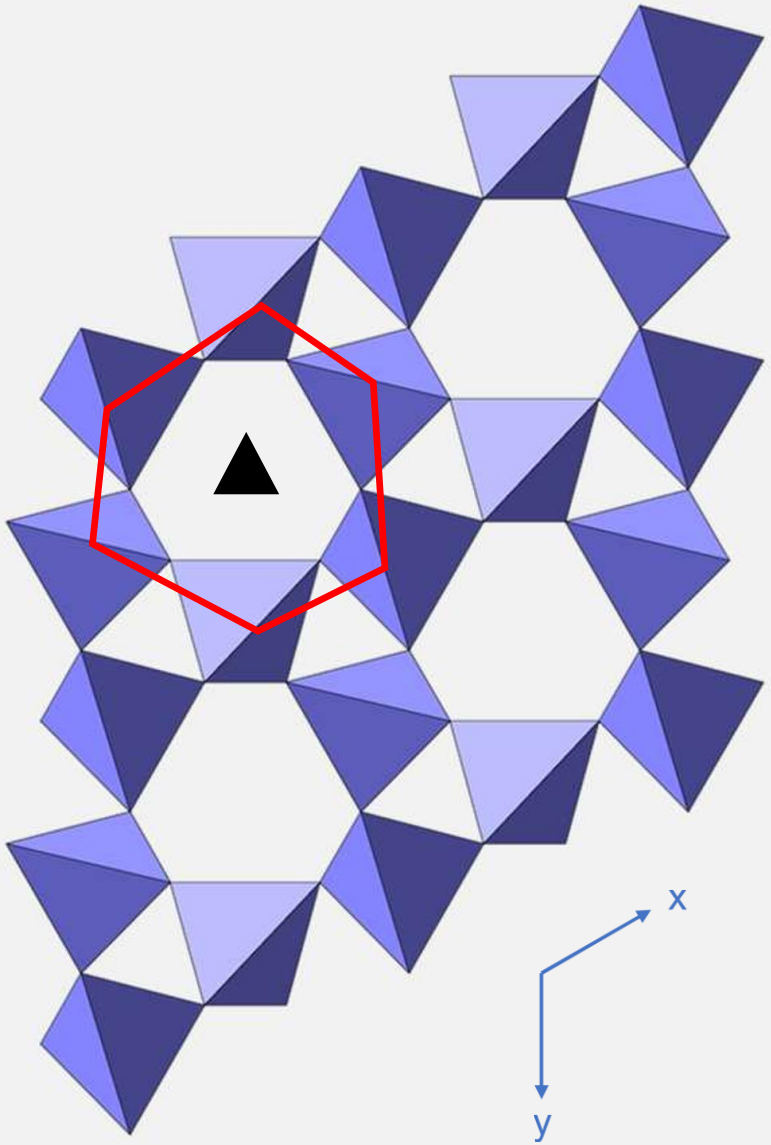
See also EART 26201 Principles of Geochemistry, Lecture 3

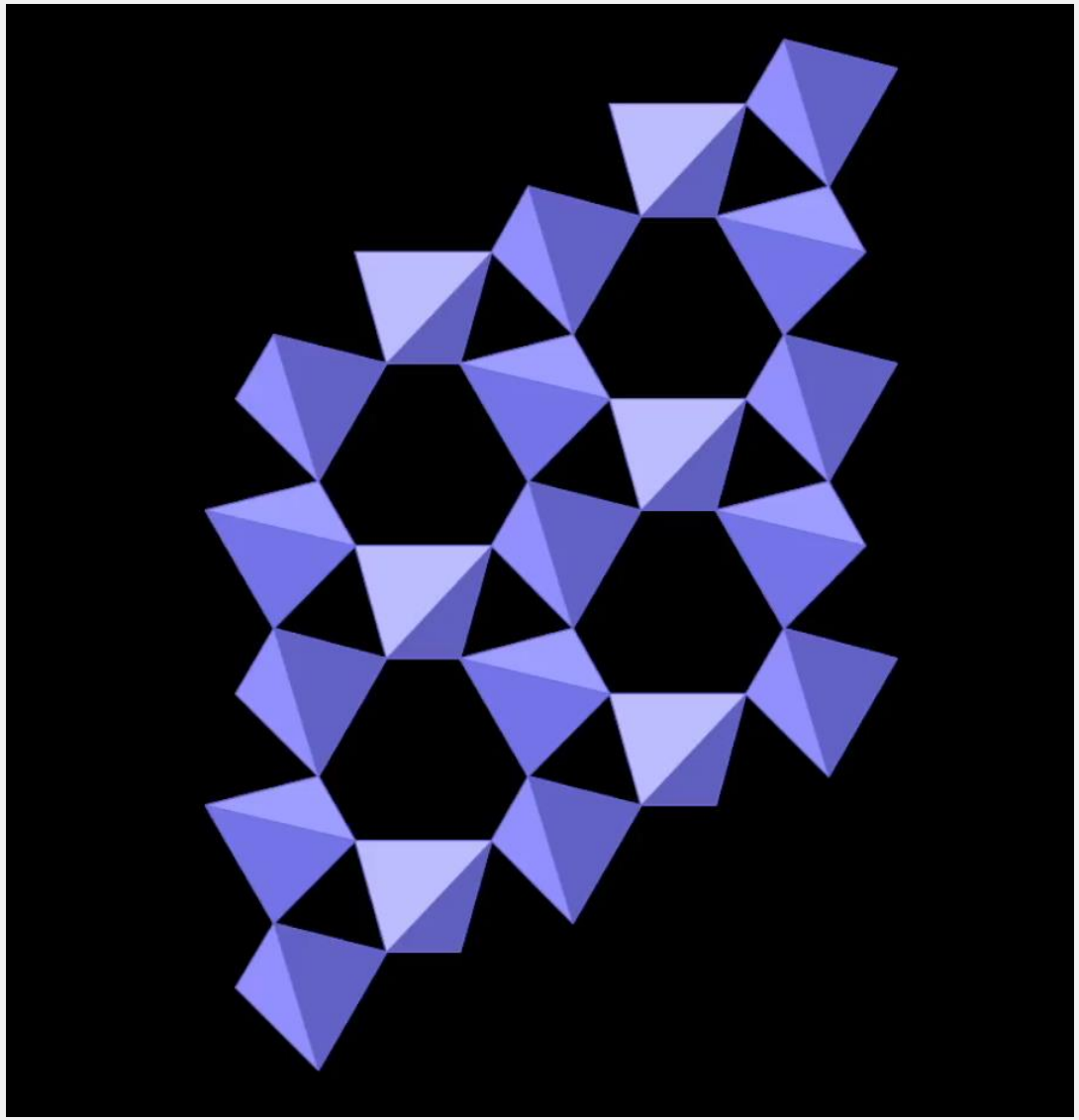



- $SiO_2$ exists as different polymorphs (same chemistry, different structure) depending on the pressure and temperature conditions
- At the Earth's surface, quartz is the stable polymorph
- Tridymite, cristobalite, coesite and stishovite can sometimes be found as metastable phases in rocks at the Earth's surface


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# Quartz

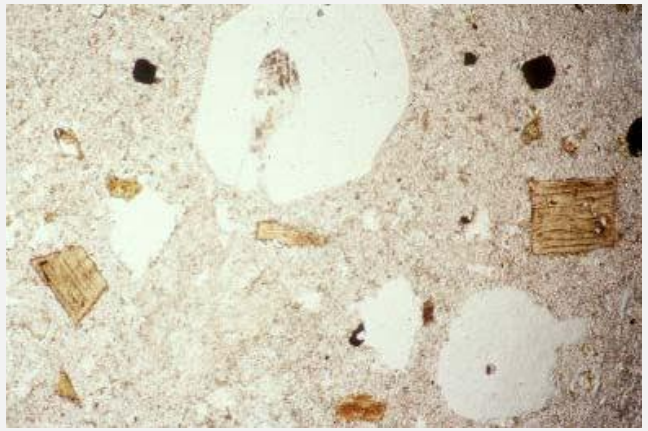


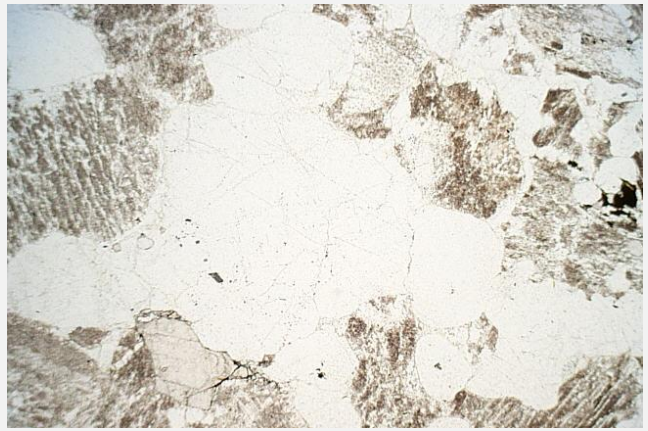




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# Quartz

Distinguish from feldspars by the absence of twinning, zoning or alteration

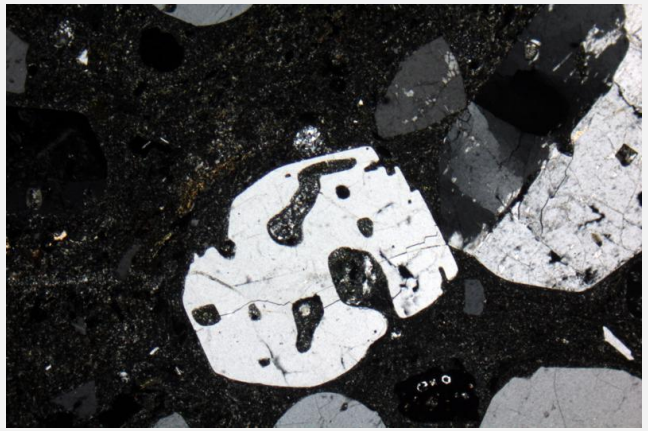


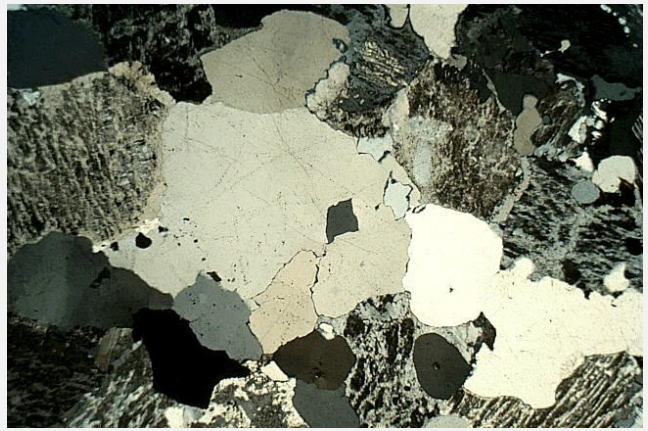


Key optical properties:

PPL

- Very low relief
- Colourless
- No cleavage
- Never undergoes alteration





XP

- Low first-order interference colours
- No twinning
- No zoning
- May show patchy or undulose extinction if the rock has been deformed



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# Thin sections for classroom session

Olivine

https://www.virtualmicroscope.org/content/s339-07-basalt Rotation 2

https://www.virtualmicroscope.org/content/olivine-gabbro-huntly

Augite

https://www.virtualmicroscope.org/content/porphyritic-olivine-basalt-mauritius

Opx vs cpx

https://www.virtualmicroscope.org/content/m24-two-pyroxene-granulite-lewisian-comple-scotland Rotations 2 and 3

Hornblende

https://www.virtualmicroscope.org/content/i16-tuff-hornblende-dacite-tuff-almeria-spain

https://www.virtualmicroscope.org/content/m01-amphibolite-lough-nagilly-donegal


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# Thin sections for classroom session

Biotite and muscovite

https://www.virtualmicroscope.org/content/biotite-granite-luxulyan

Alkali feldspar

Sanidine https://www.virtualmicroscope.org/content/obsidian-sandy-braes

Orthoclase https://www.virtualmicroscope.org/content/syenite-ben-loyal

Microcline https://www.virtualmicroscope.org/content/i03-granite-leinster-granite-ireland

Plagioclase

https://www.virtualmicroscope.org/content/basalt-giants-causeway

https://www.virtualmicroscope.org/content/diorite-magellan-straits - Rotation 1 zoning

Quartz

https://www.virtualmicroscope.org/content/granite-lundy (Rotation 1 is topaz)


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# Thin sections for classroom session

Look and draw:

Porphyritic olivine basalt https://www.virtualmicroscope.org/content/porphyritic-olivine-basalt-mauritius

Troctolite https://www.virtualmicroscope.org/content/troctolite-chromite-layer

Lundy granite https://www.virtualmicroscope.org/content/granite-lundy

Some good examples

https://muse.union.edu/hollochk/kurt-hollocher/petrology/the-almost-forgotten-art-of-hand-drawings-in-petrology/

https://www.reddit.com/r/geology/comments/a4ppsb/these_amazing_hand_drawn_diagrams_of_thin/

Some less good examples

…. from MH's undergraduate notes



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